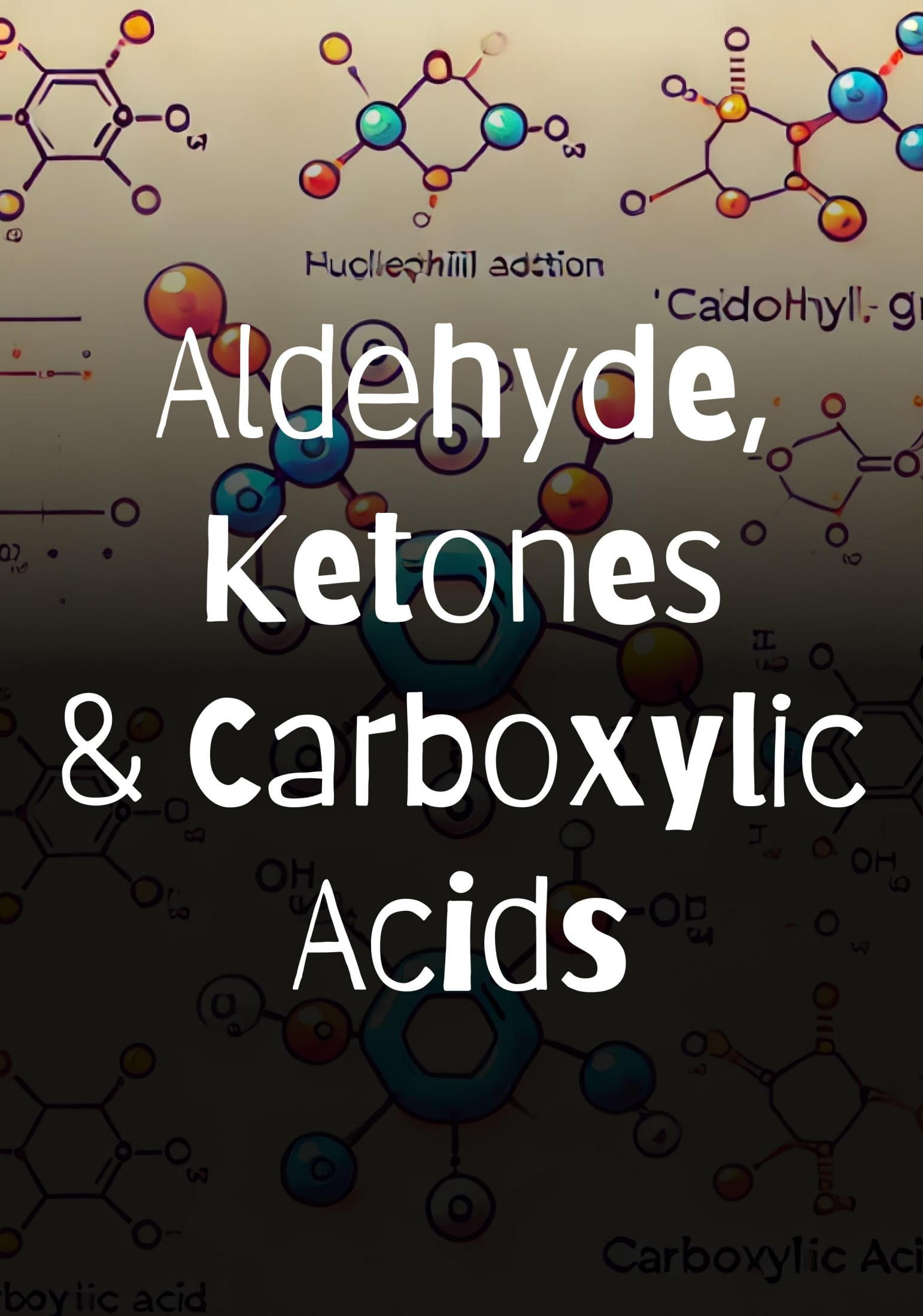

Aldehyde,
Ketones
& Carboxylic
Acids

ALDEHYDE & KETONES

PREPARATION

1. By hydration of alkynes

$$CH \equiv CH + \text{dil. } H_2SO_4 / HgSO_4 \longrightarrow CH_3-\overset{O}{\overset{\|}{C}}-H$$

$$Ph-C \equiv C-CH_3 + \text{dil. } H_2SO_4 \longrightarrow Ph-\overset{O}{\overset{\|}{C}}-CH_2-CH_3$$

2. Hydroboration & Oxidation of alkynes

$$CH \equiv CH + \begin{matrix}1.\ B_2H_6 \\ 2.\ NaOH/OH\end{matrix} \longrightarrow CH_3\ \overset{O}{\overset{\|}{C}}\ H$$

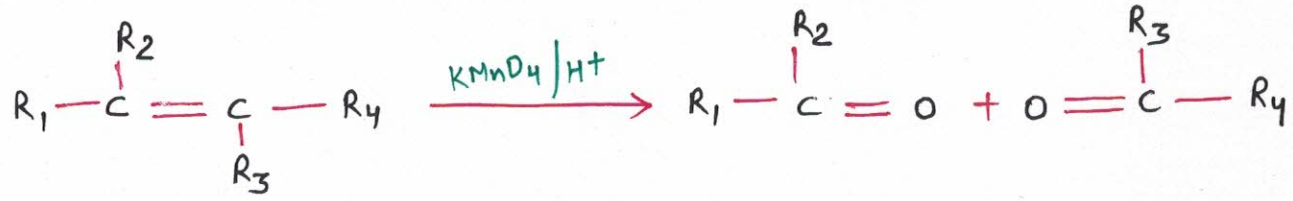
$$R-C \equiv CH + \begin{matrix}1.\ B_2H_6 \\ 2.\ NaOH\end{matrix} \longrightarrow R\ CH_2\ CH\ O$$

$$R-C \equiv CH \xrightarrow{\text{dil. } H_2SO_4} R-\overset{O}{\overset{\|}{C}}-CH_3$$

3. Ozonolysis of alkenes

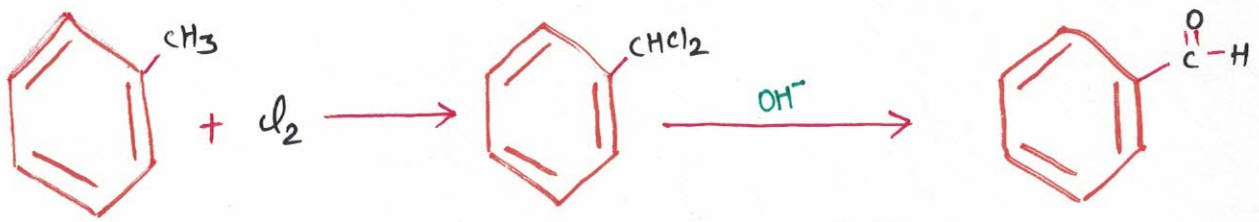
$$R-\overset{R_2}{\underset{R_3}{C}}=\underset{}{C}-R_4 \xrightarrow{O} R_1-\overset{R_2}{\underset{}{C}}=O + O=\overset{R_3}{\underset{}{C}}-R_4$$

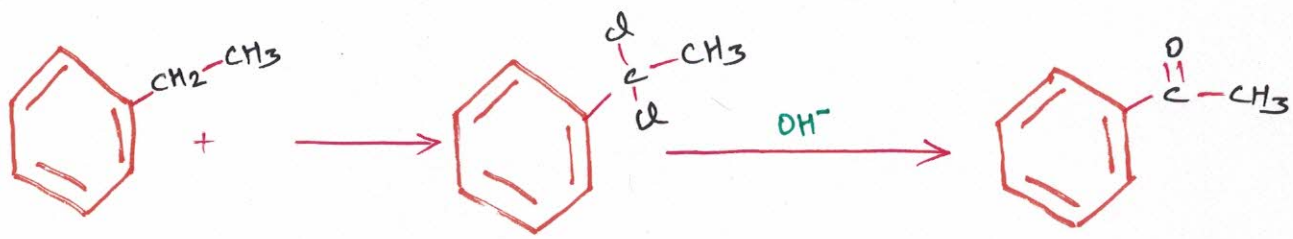
4. Oxidation cleavage of alkenes

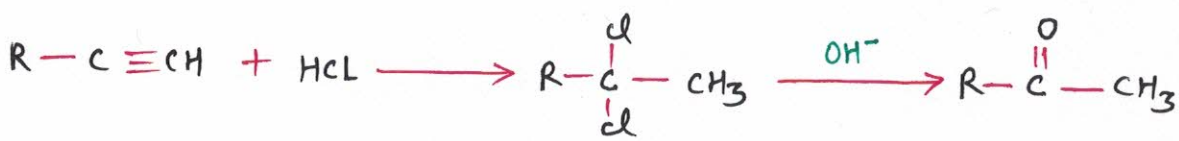


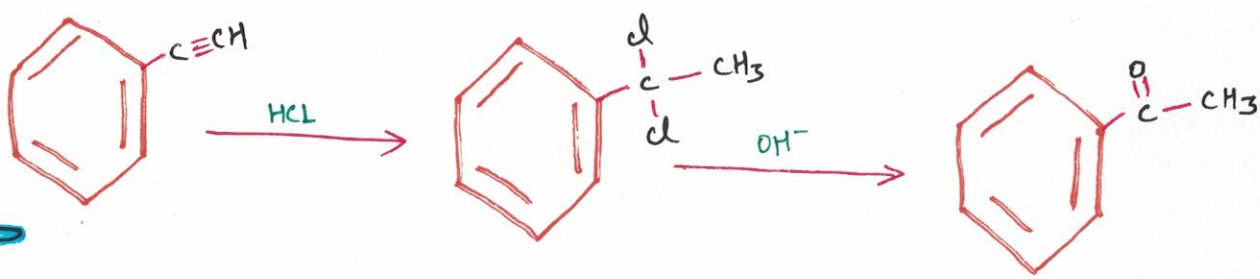
5. By hydrolysis of gemdihalides

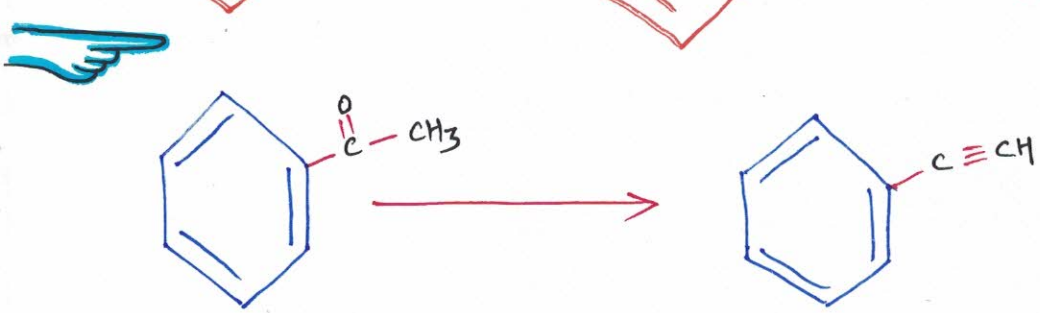


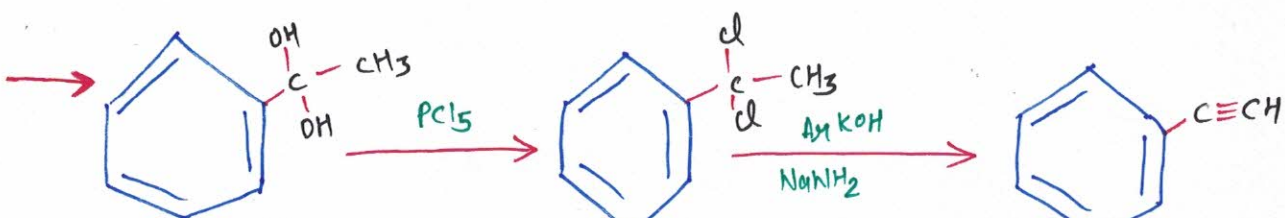






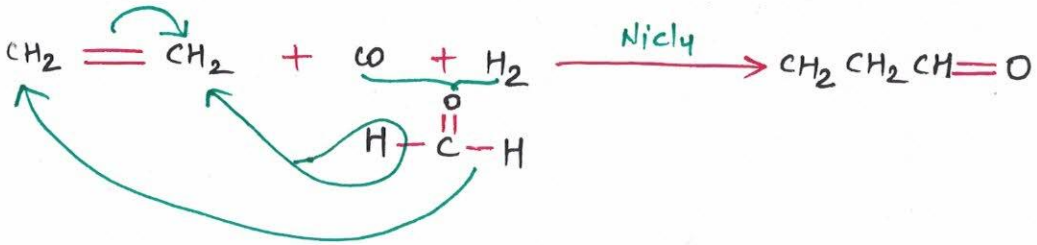






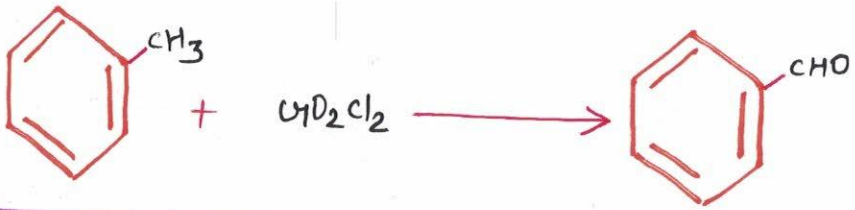
Terminal gemhalide ⟶ Aldehyde
Non Terminal gemhalide ⟶ ketone

6. OXO reaction / Hydroformylation

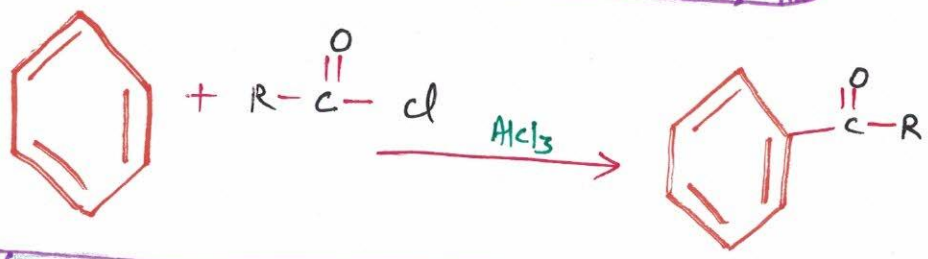




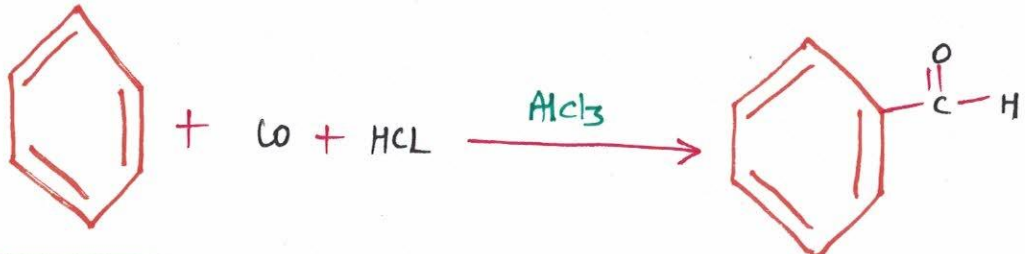
7. Etards Reaction



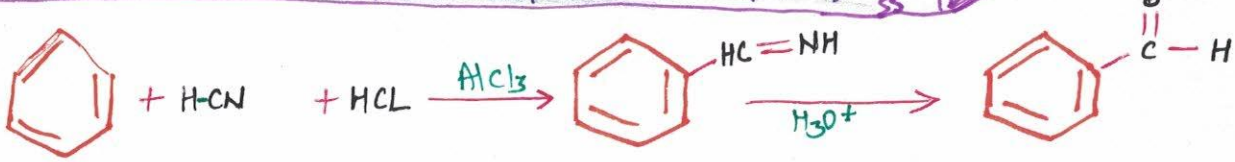
8. Friedal Crafts Acylation



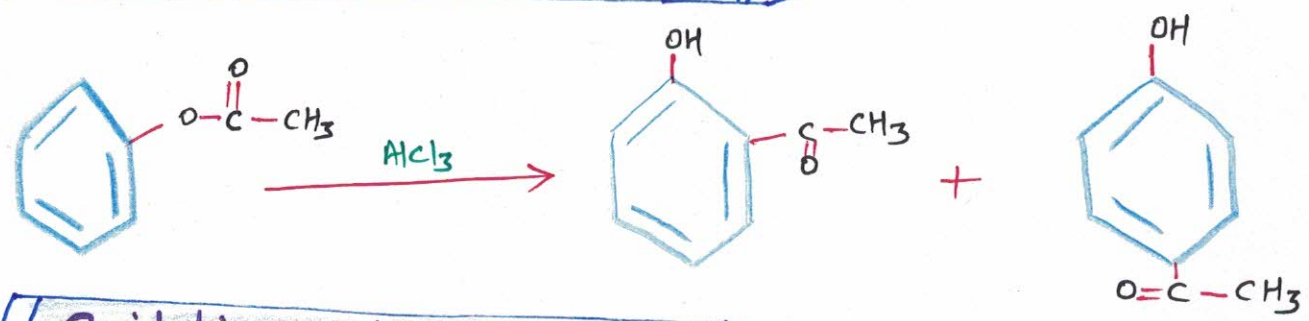
9. Gattermann Koch Aldehyde synthesis



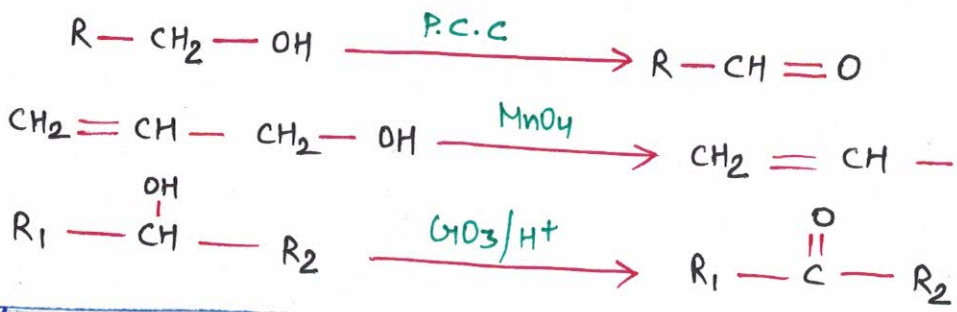
10. Gattermann Aldehyde synthesis (GAS)



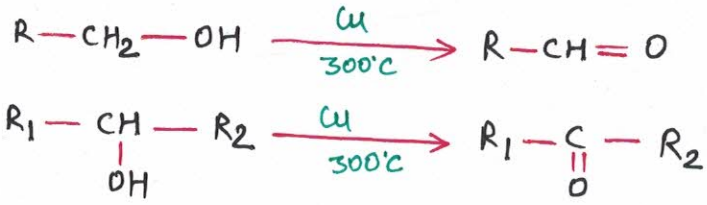
12. Reimer Teimann Reaction



13. Oxidation of alcohols



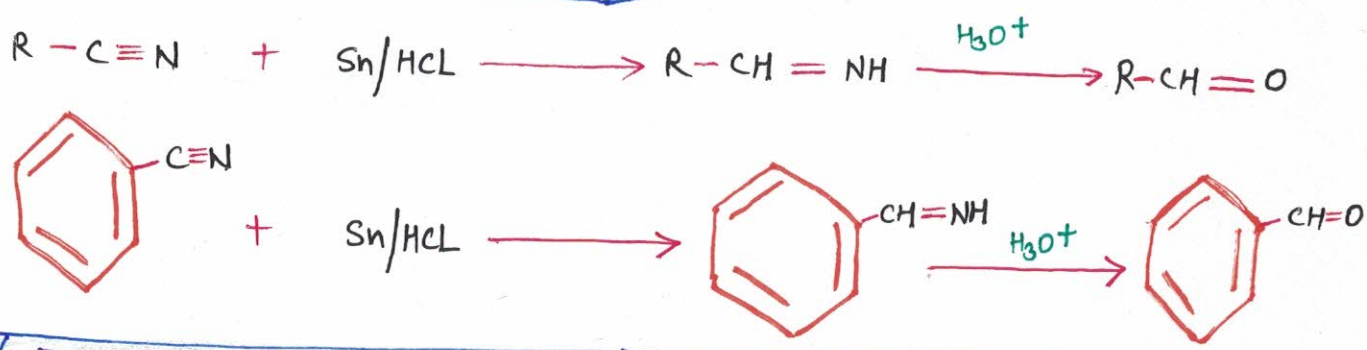
14. Dehydrogenation of alcohols



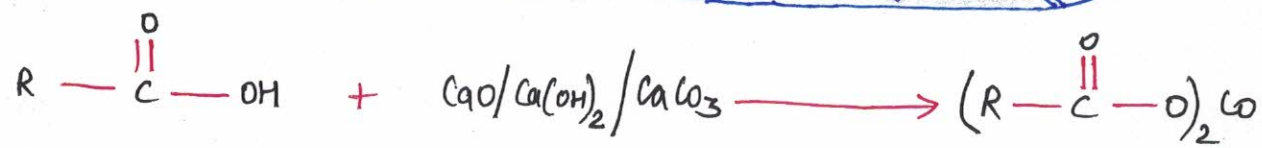
1. Pinacol – Pinacalone rearrangement
2. HIO_4 cleavage

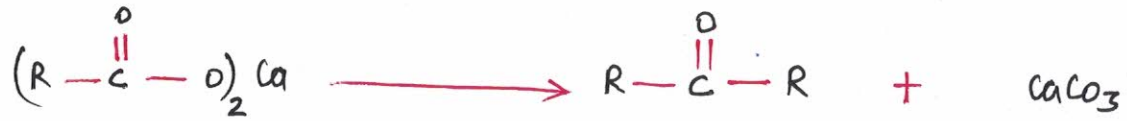
NEW REACTIONS

15. Stephen's reduction

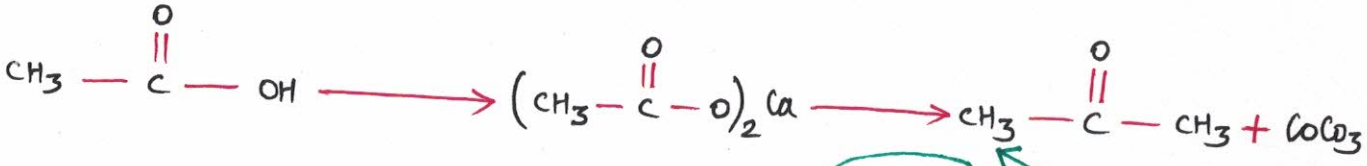


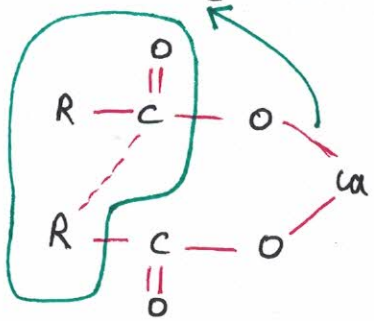
16. Dry distillation of calcium carboxylates

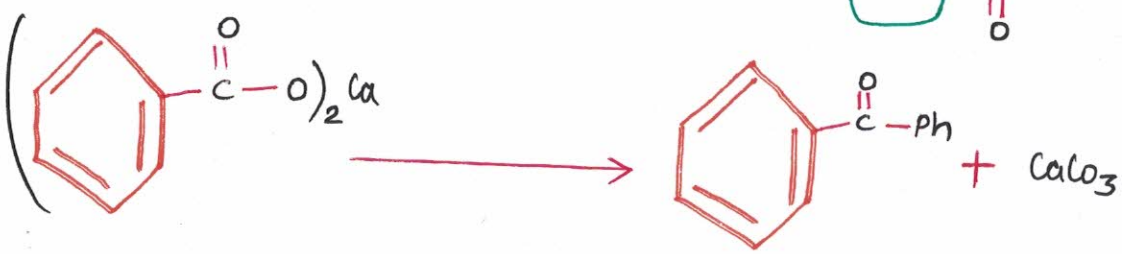


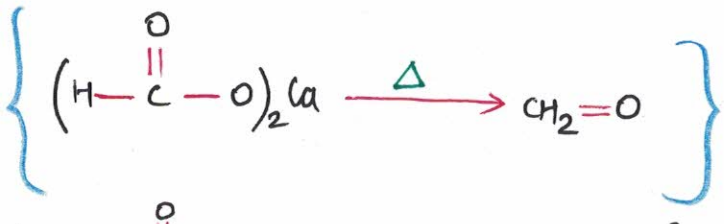


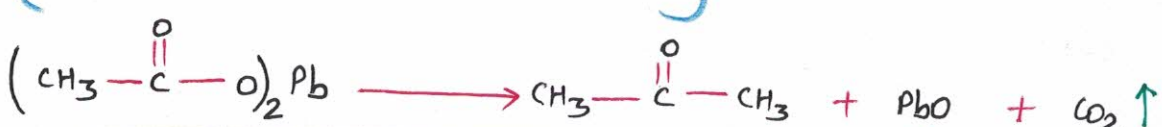
It is called dry distillation as no liquid reactant is involved in reaction.



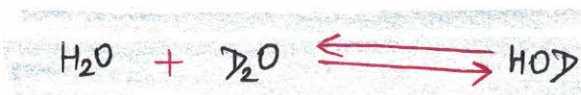


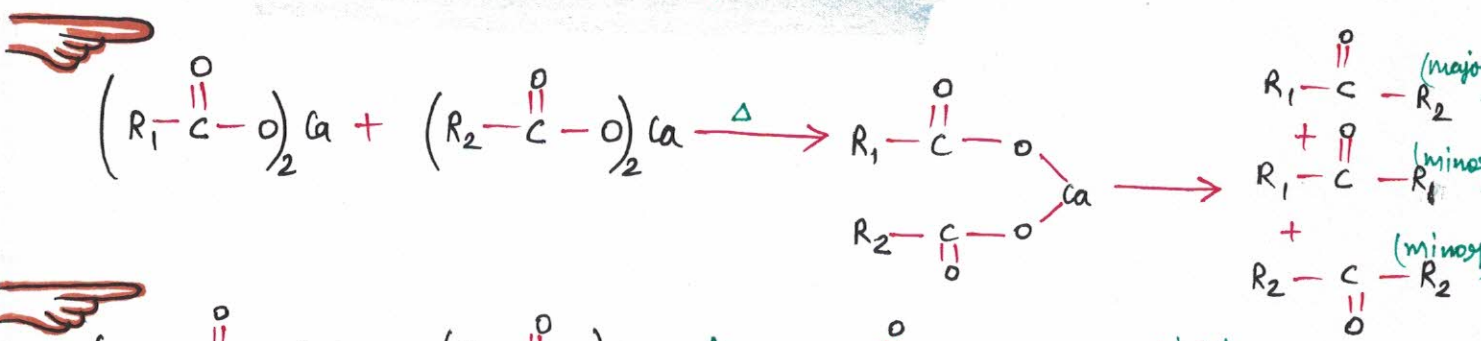


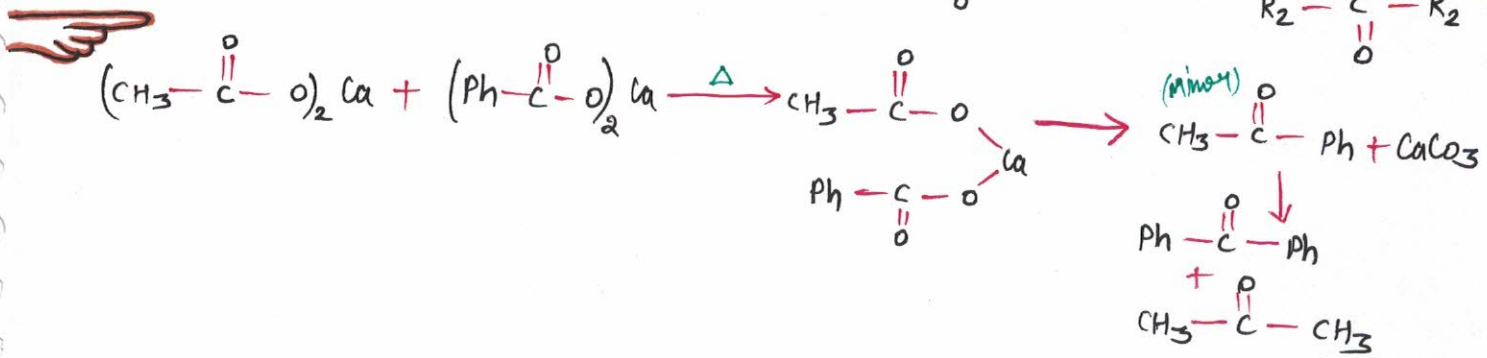


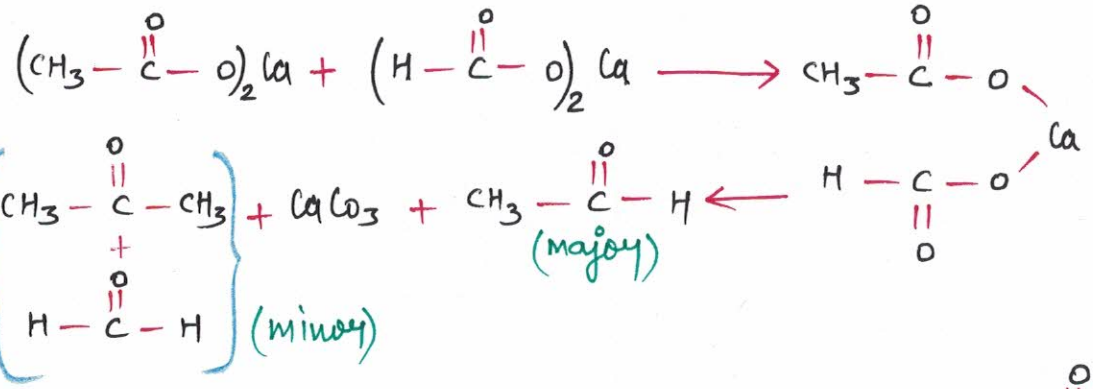


PbO_3 is not formed because $PbCO_3$ is unstable at high temp. Hence it decomposes into PbO inhereas $CaCO_3$ does not.

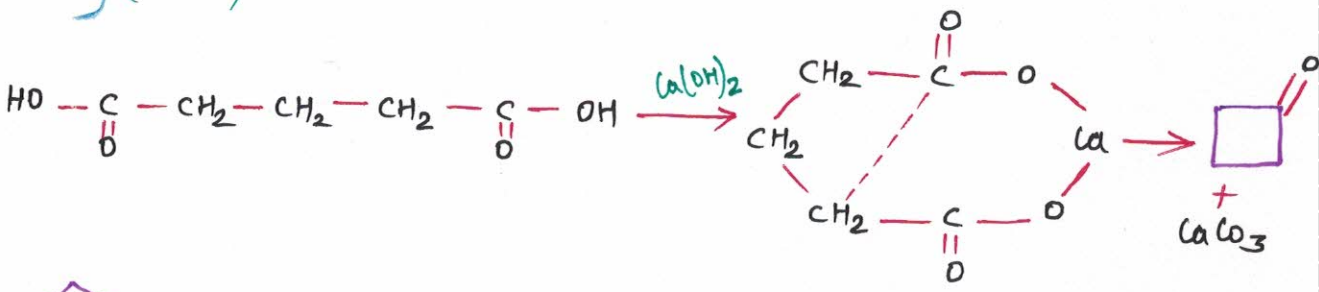




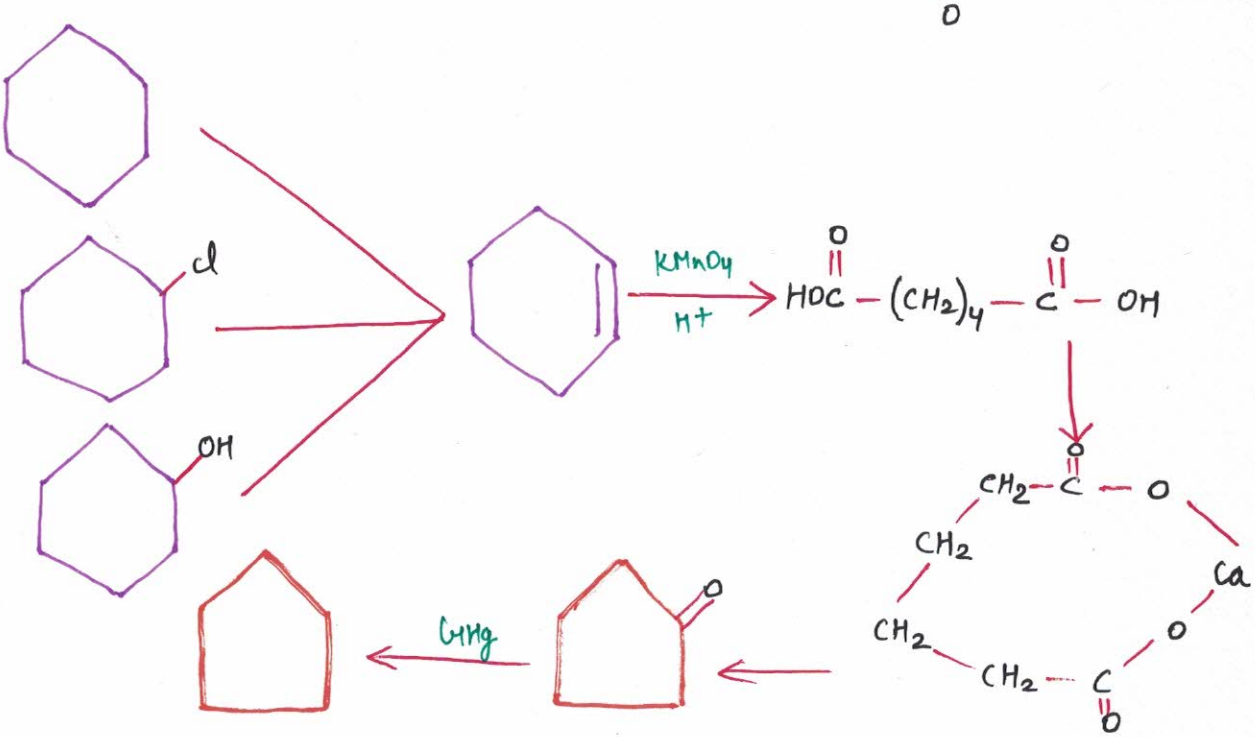




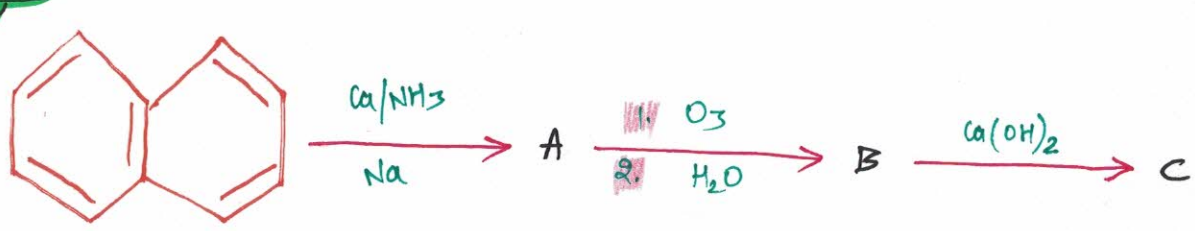


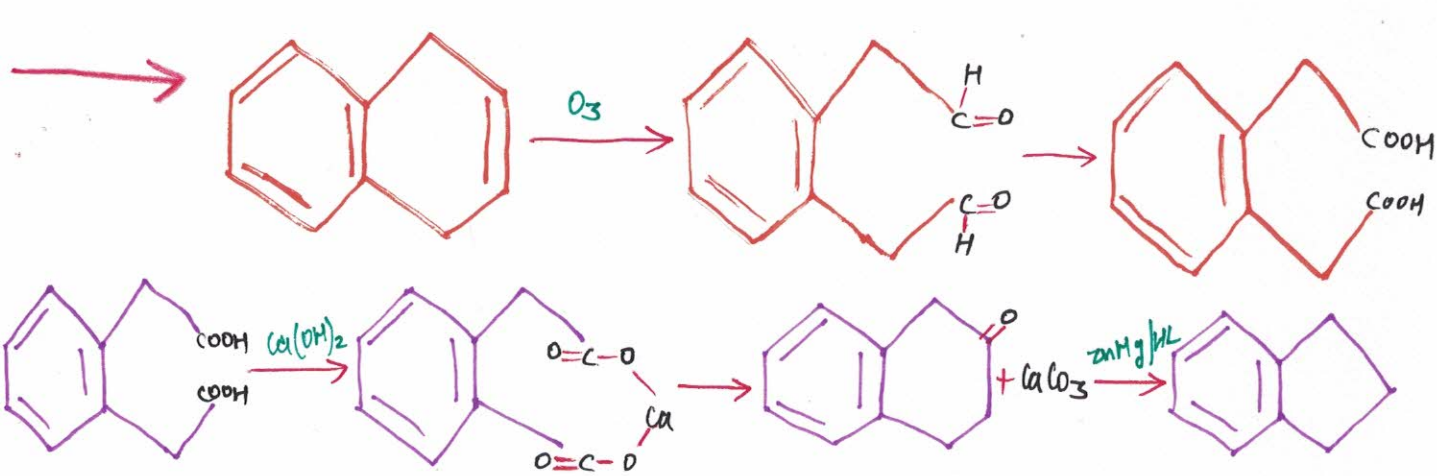




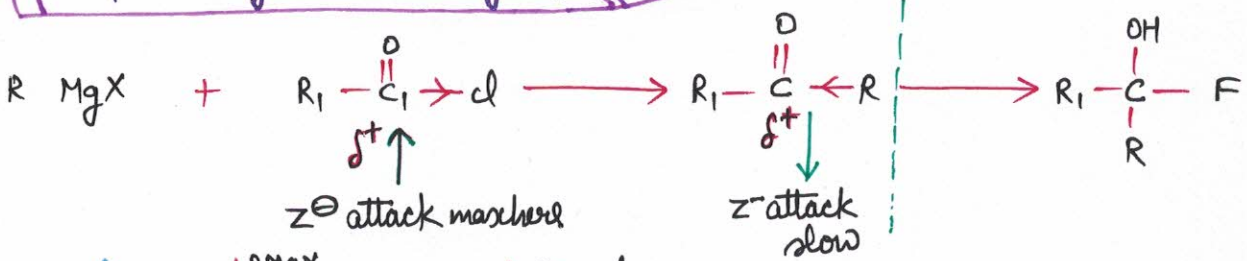


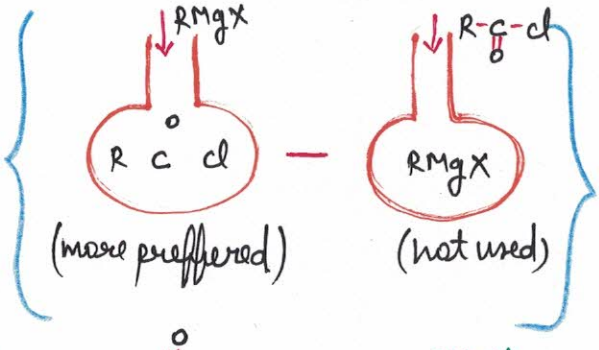




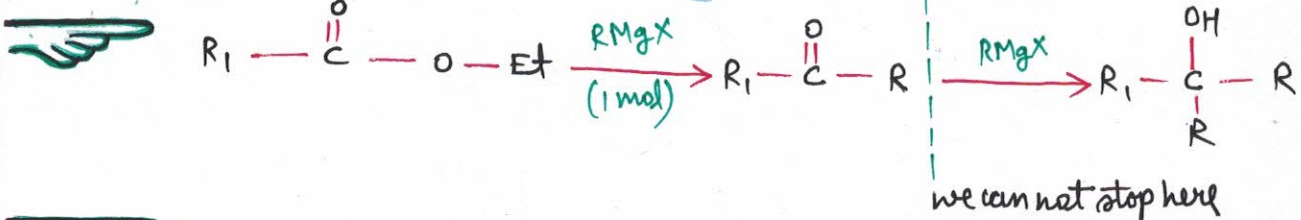


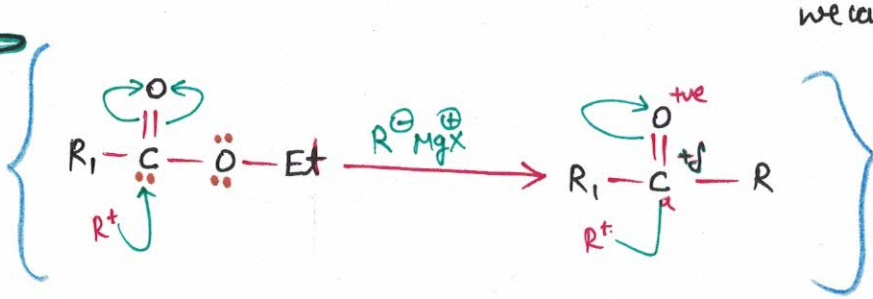
17. **By Grignard Reagent**



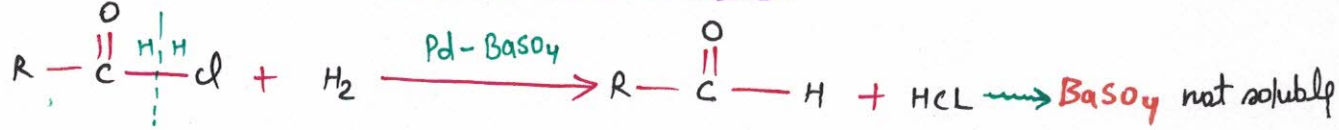


Dog Biscuit Concept.....



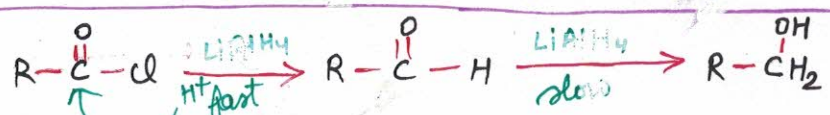


18. **By Rosenmund Reduction**

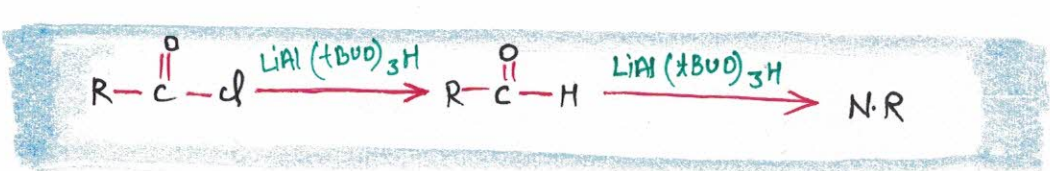


we can not use Lindlar catalyst because HCL released can react with $CaCO_3$ to form $CaCl_2 + CO_2$ and hence catalyst is destroyed. so, only Pd-$BaSO_4$ is used.

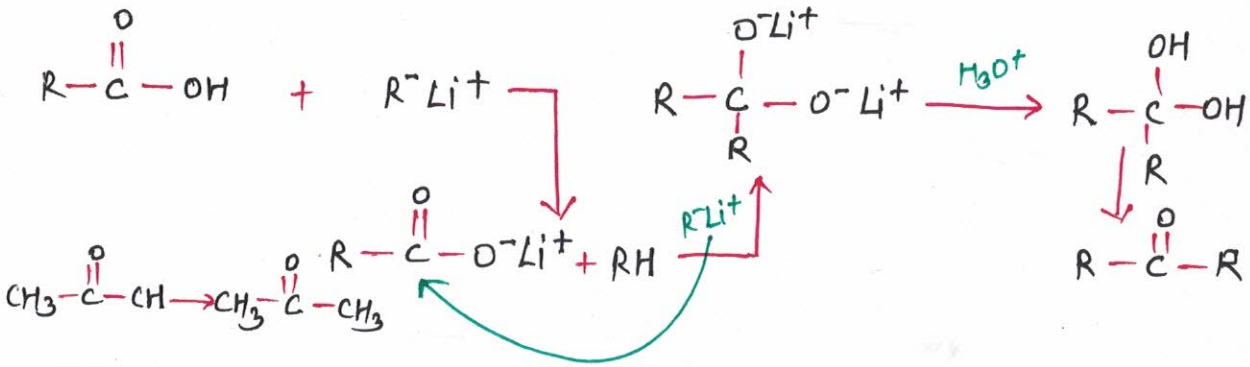
19. **By Reduction of R C(=O) Cl using $LiAl(tBuO)_3H$**

we know, 

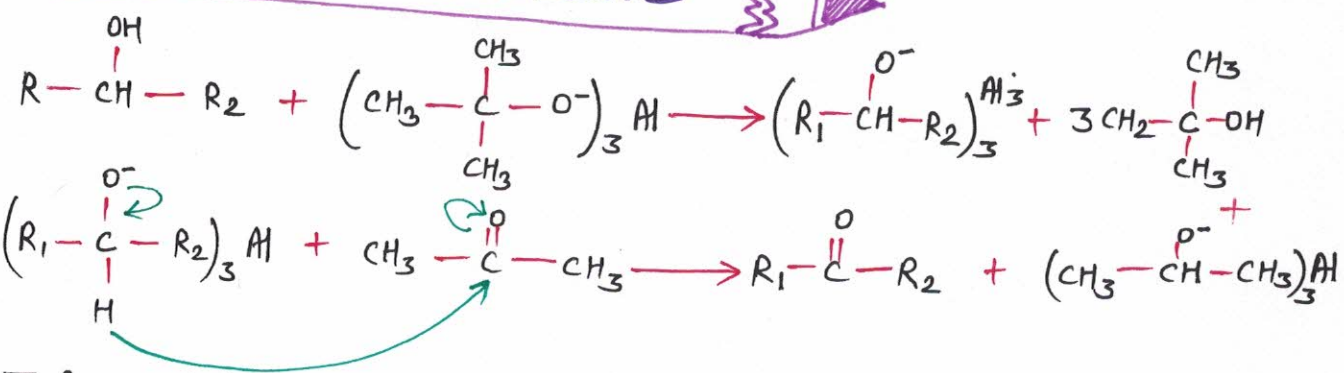
To get only aldehyde, we use $LiAl(tBuO)_3H$, Due to crowding, once we get aldehyde no further reaction takes place.



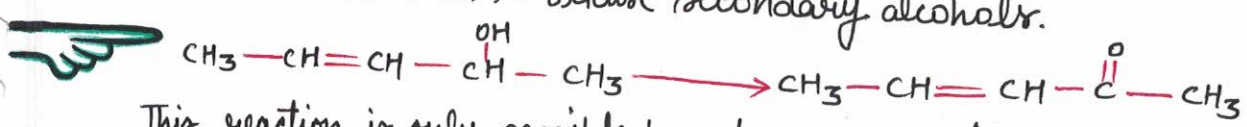
20. By reaction of R-COOH R^-Li^+



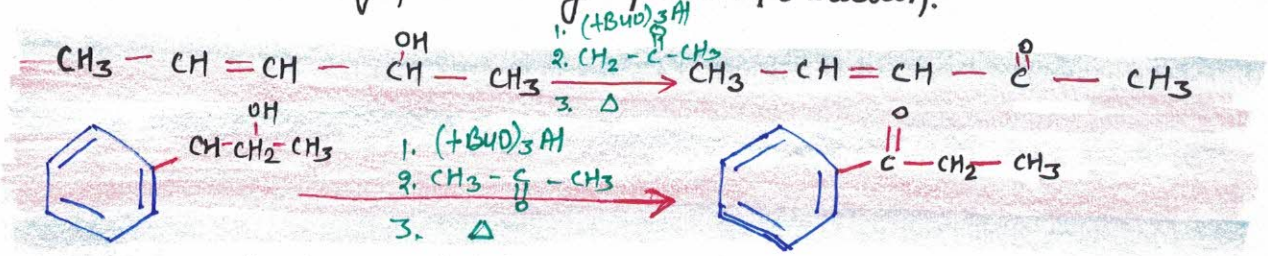
21. Oppenauer oxidation

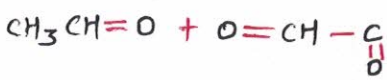
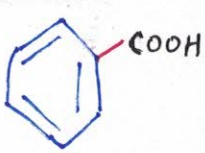


It is a selection method to oxidise secondary alcohols.



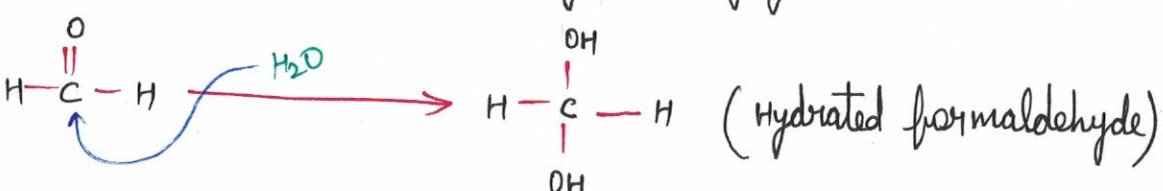
This reaction is only possible by oppenauer reaction.



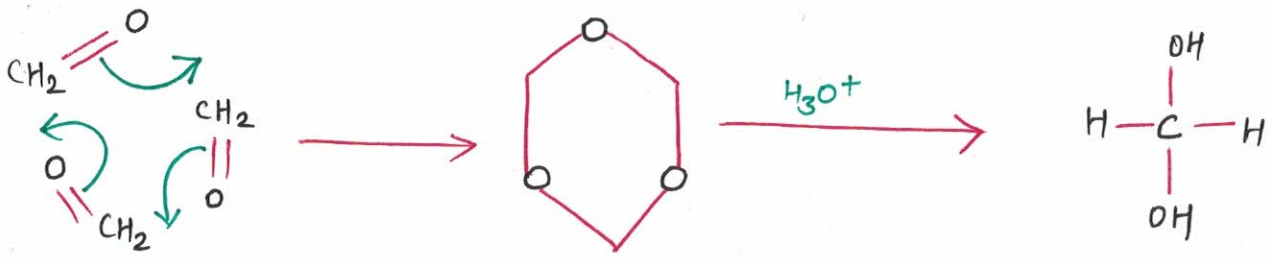
If we use $KMnO_4/H^+$, we will get $CH_3CH=O$ +  & 

PROPERTIES

 is a gas  is a liquid. If molecular wt. is very high, it exists as solid. Formaldehyde is highly soluble.



(40% solution) of formaldehyde in water to Formalin



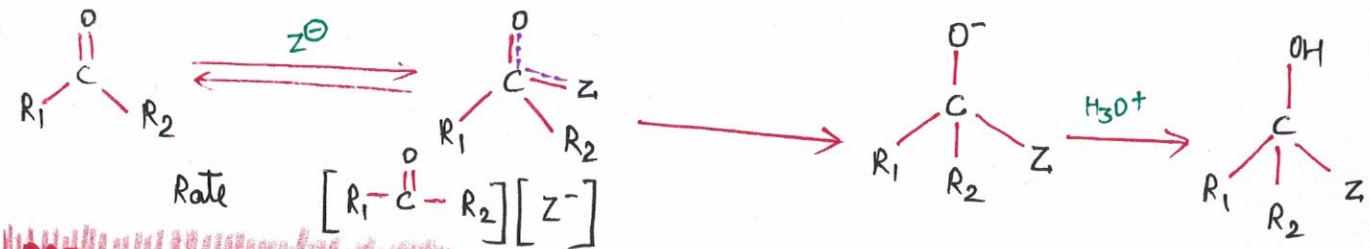
(Trioxane)



(solid Para formaldehyde)

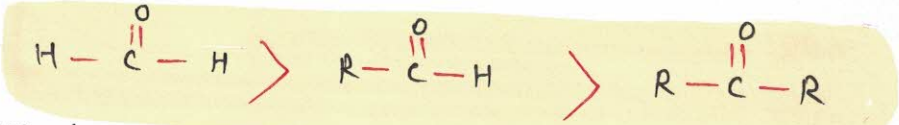
PROPERTIES OF CARBONYL COMPOUNDS

1. Nucleophilic addition

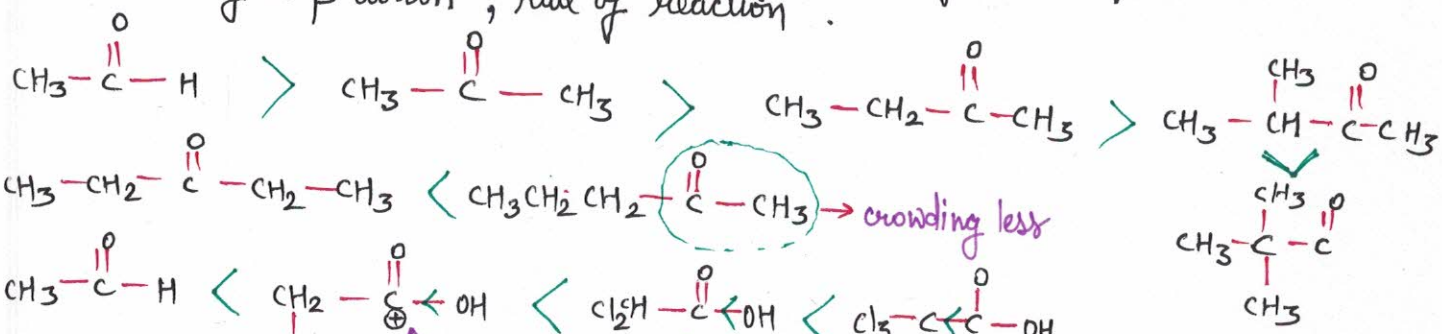


Rate $\propto [R_1-\overset{O}{\overset{\|}{C}}-R_2][Z^-]$

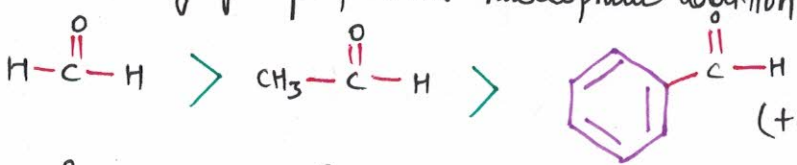
RATE OF REACTION



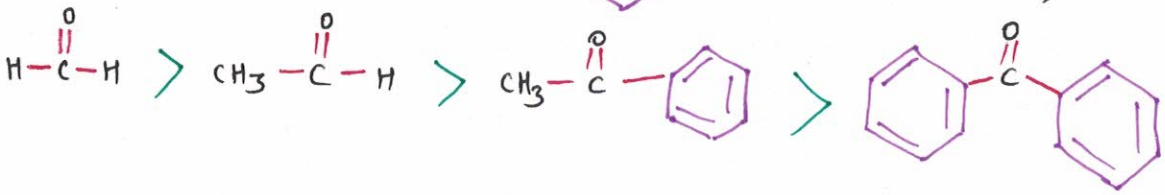
Rate is least in ketones because of crowding in T.S of ketone formed.
As branching at β carbon, rate of reaction.

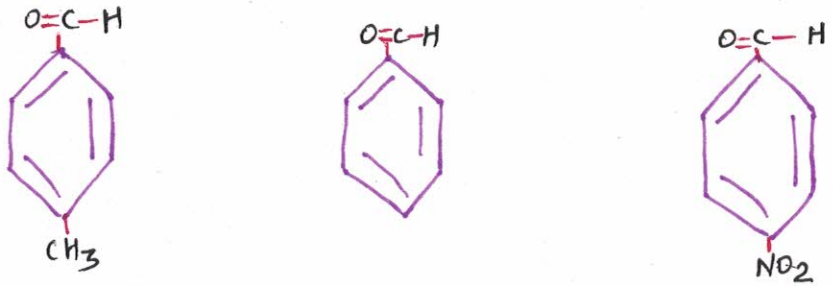


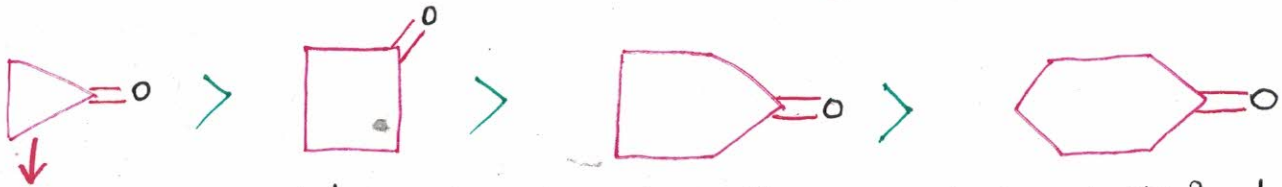
with drawing groups promote nucleophilic addition



(+M effect hence stable)



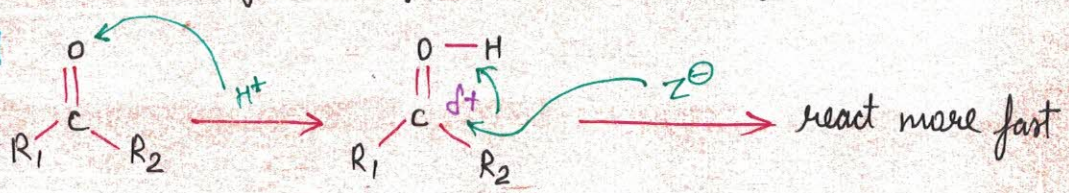




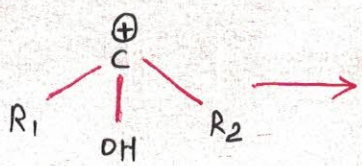
It is more because before addition it was sp^2 (60°) most strained. The moment nucleophile is added it changes to sp^3 & hence gets most relief.

Nucleophilic addition of carbonyls is catalysed by an acid.

REASON



react more fast



does not form

Every aldehyde (except HCHO) upon N.A. will give a resolvable mixture (D/L)

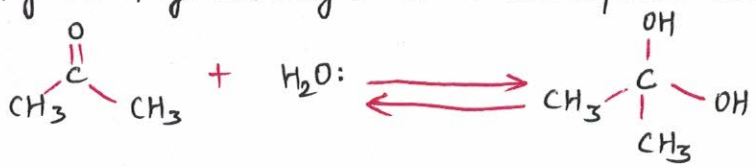
All unsymmetrical ketons upon N.A will give resolvable mixture (D/L)

HYDRATION OF CARBONYLS



(formation)

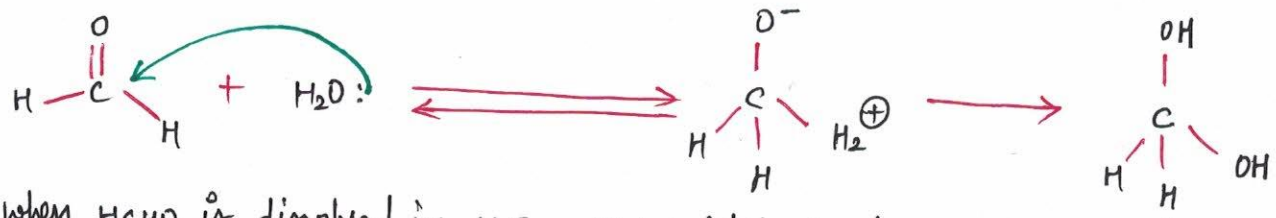
Hydration of carbonyls is a nucleophilic addition



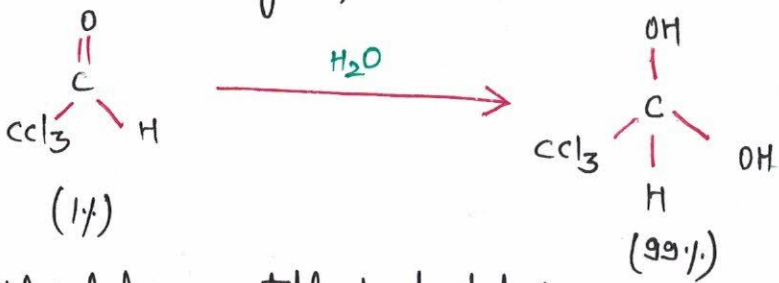
(99%) (1%)

So, higher the rate of nucleophilic addition, higher is the ability to stay in the hydrated form.

Hydrated forms exist in aqueous mix only.



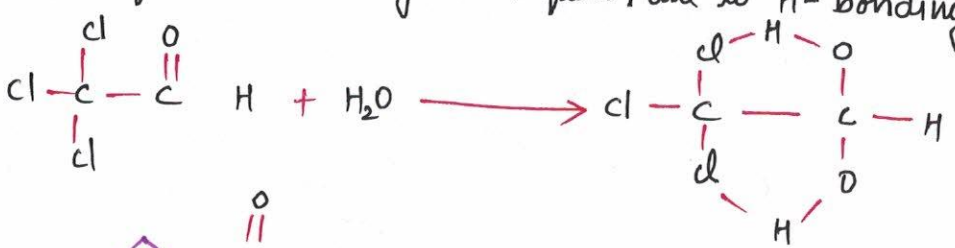
when $HCHO$ is dissolved in H_2O, 99% exist in the form of formaliin whereas CH_3CHO exist only 5%.

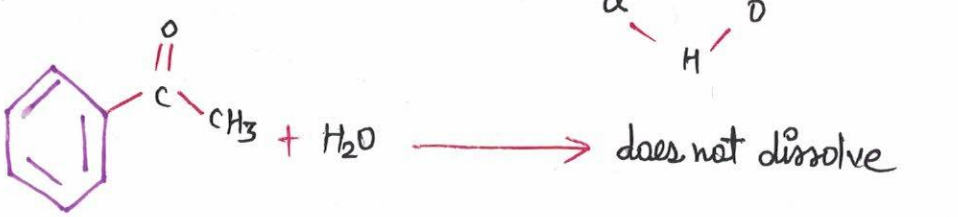


(1%)

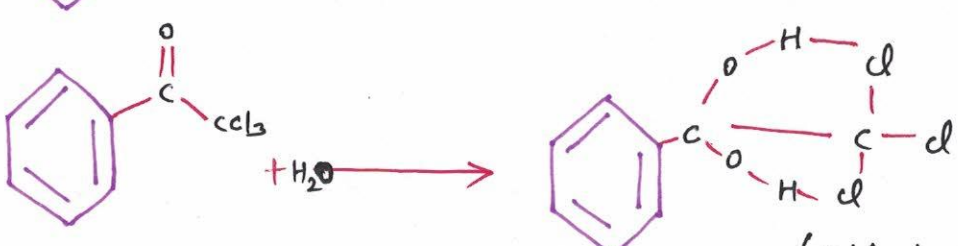
(99%)

chloral forms stable hydrated form due to H-bonding.





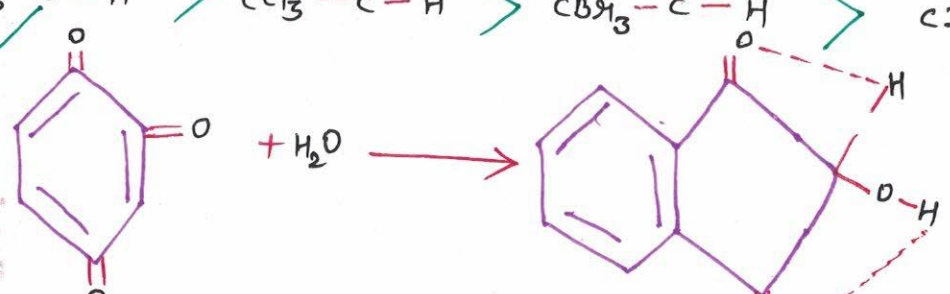
does not dissolve



(stable due to H-BOND, NH)



It will form most stable hydrate.

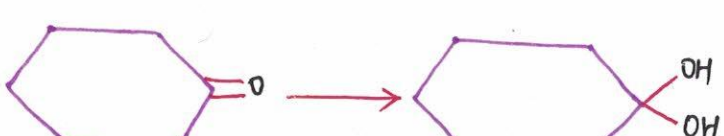


(Ninhydrine)

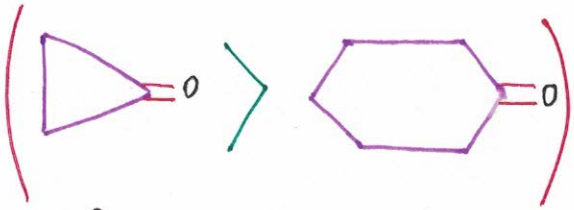
(highly stable)

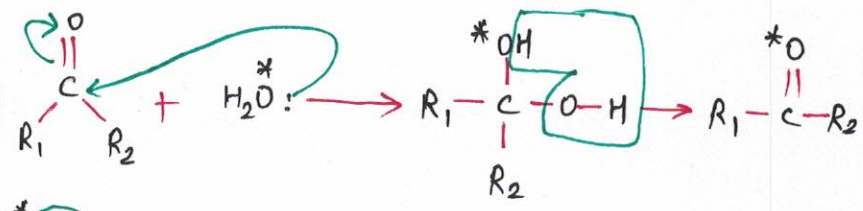
Similarly,

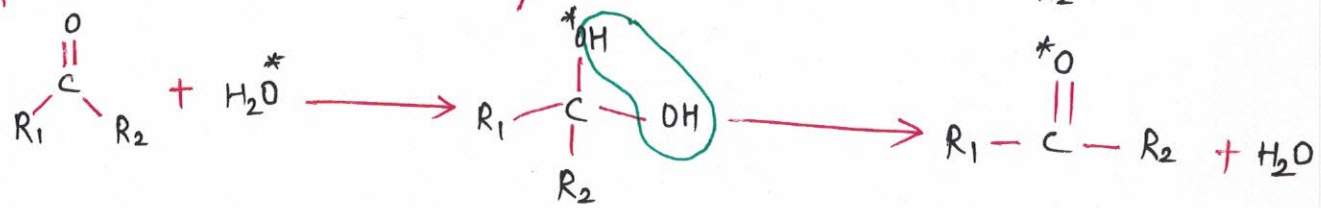




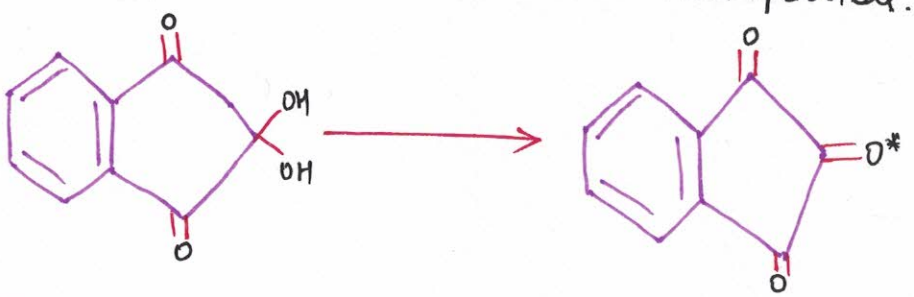
(does not form reaction moves backward.)



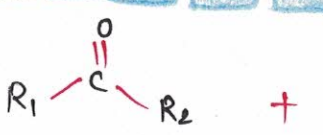


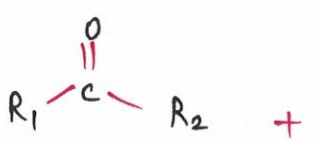


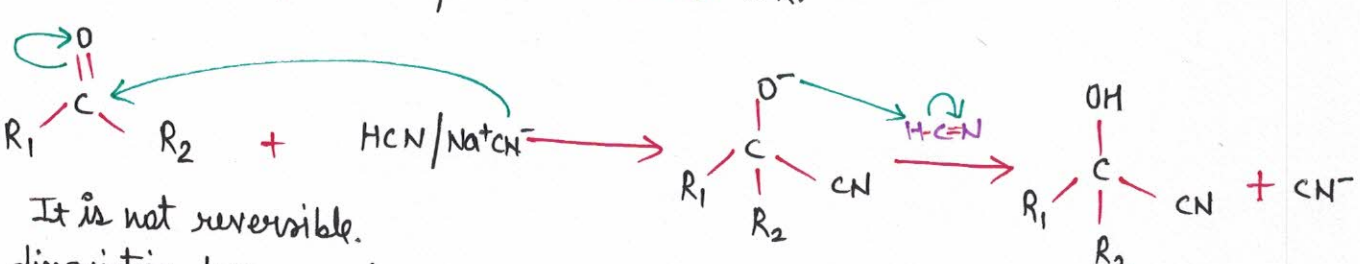
Backward & forward reactions both are possible.



REACTION WITH HCN

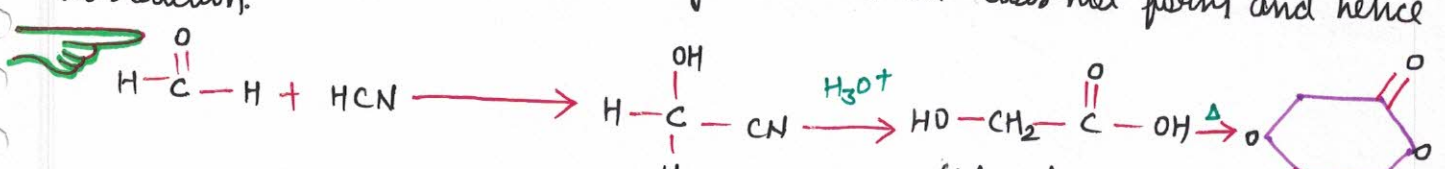






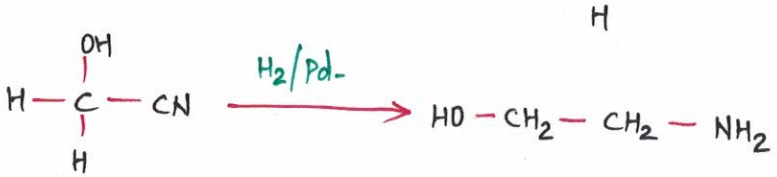
It is not reversible.

HCN dissociation decreases in presence of H^+ Hence CN^- does not form and hence no reaction.

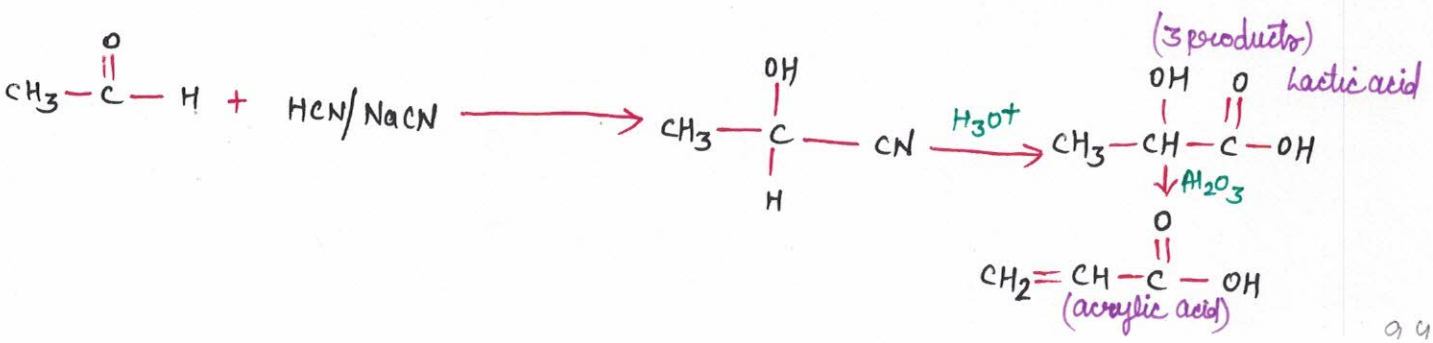


(glycolic acid)

(lactide)



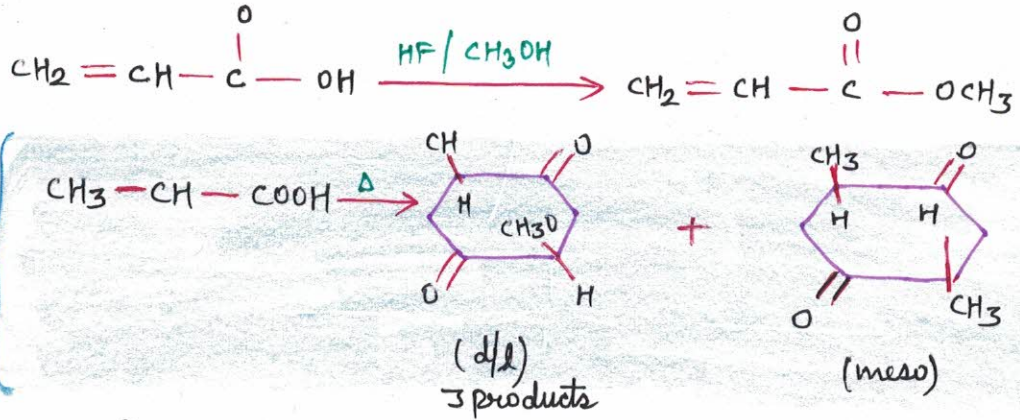
(ethenol amine)

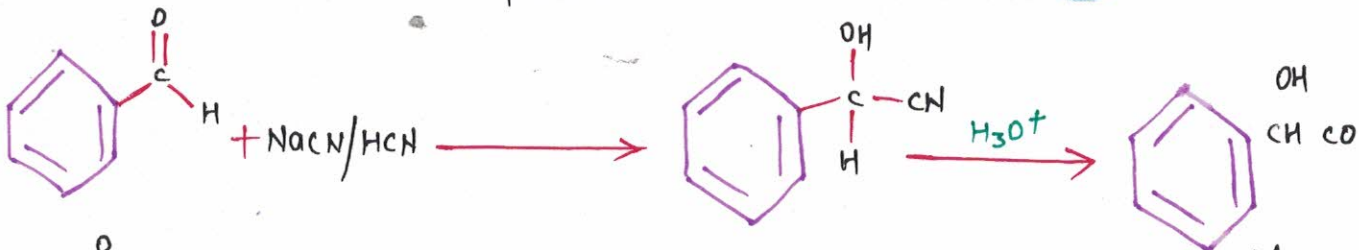


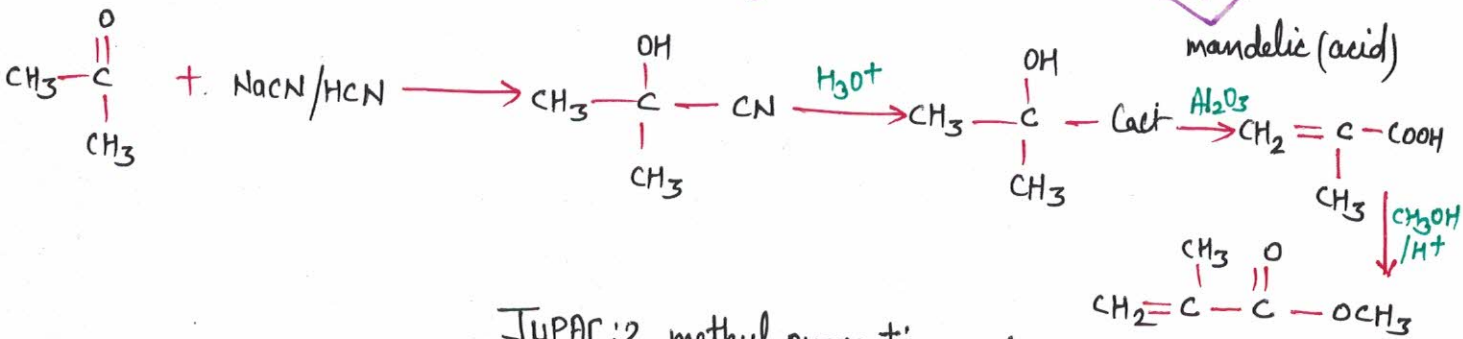
(3 products)

Lactic acid

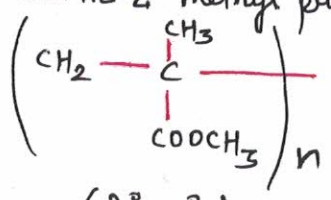
(acrylic acid)







MMA → JUPAC: 2 methyl properties

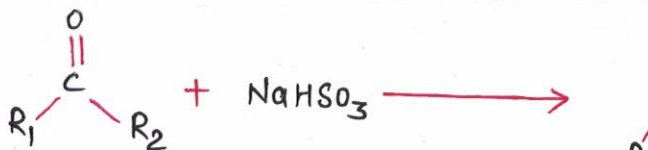


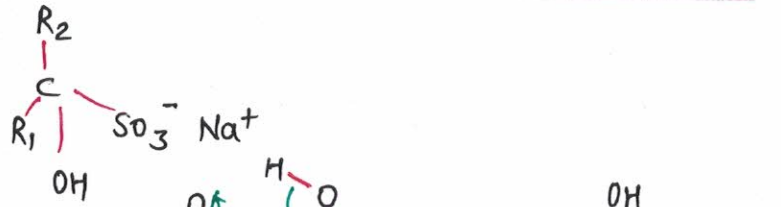
(PMMA)

(Piropix)

(MMA) (Methyl methacrylate)

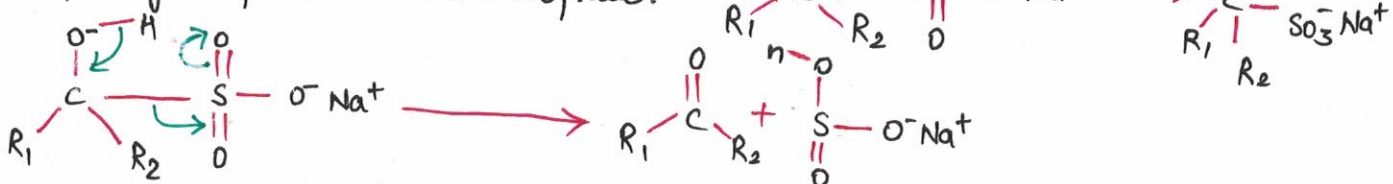
REACTION WITH $NaHSO_3$





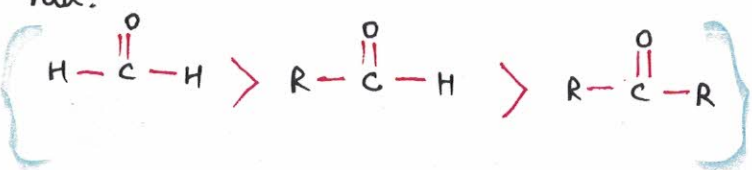
MECHANISM

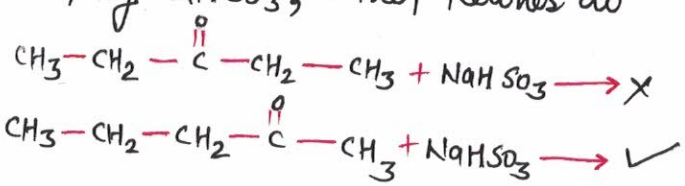
The lone pair of sulphur is the nucleophilic.



It is reversible reaction

only methyl ketons undergo nucleophilic addition by $NaHSO_3$, other ketones do not.

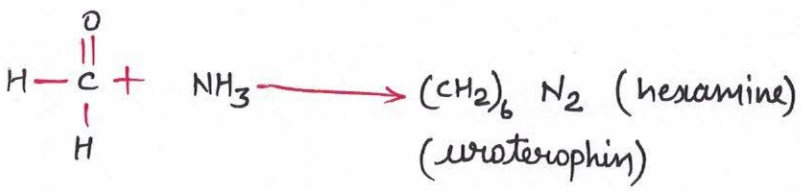


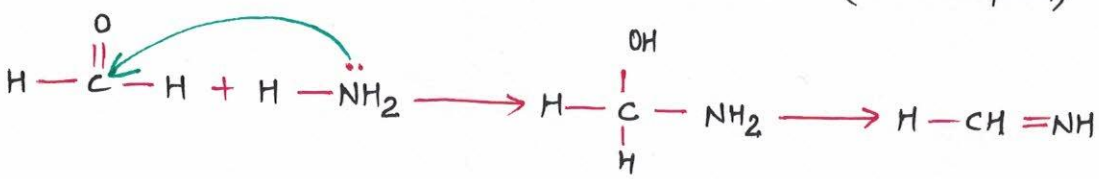


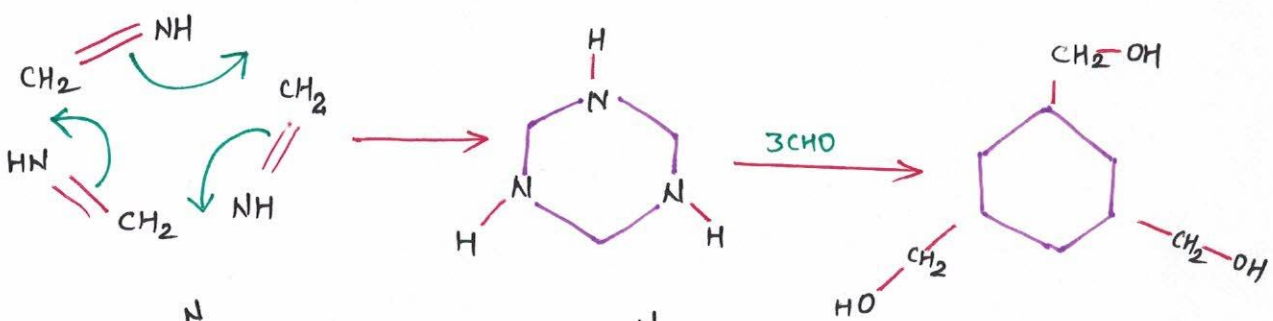
Reaction of HCN is faster than $NaHSO_3$ because C^- is better nucleophilic than sulphur.

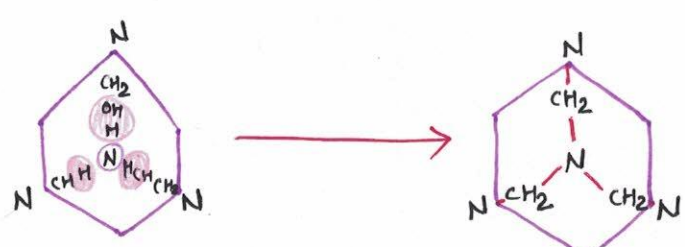
REACTION WITH NH_3 AND ITS DERIVATION

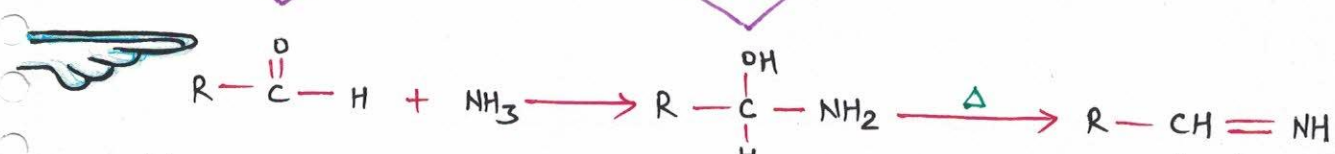
1. With NH_3





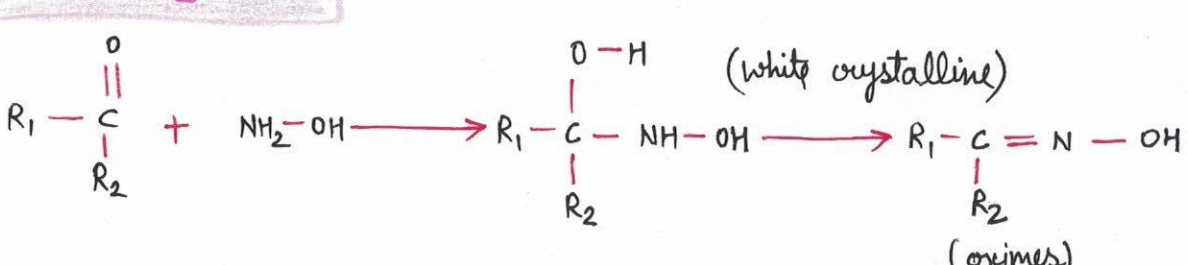




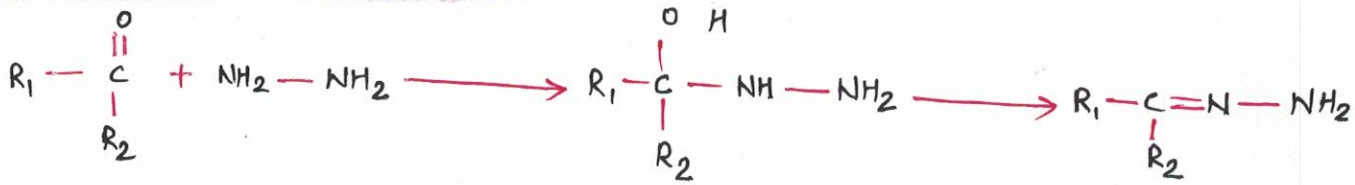


All aldehydes except HCHO will form imine on reaction with NH_3.

2. With NH_2-OH

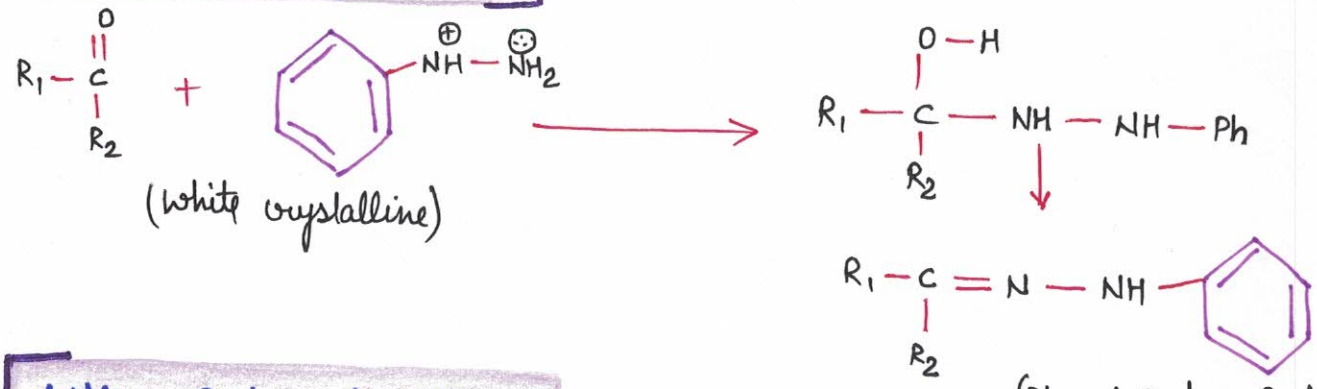


3. With NH_2-NH_2 (hydrozone)



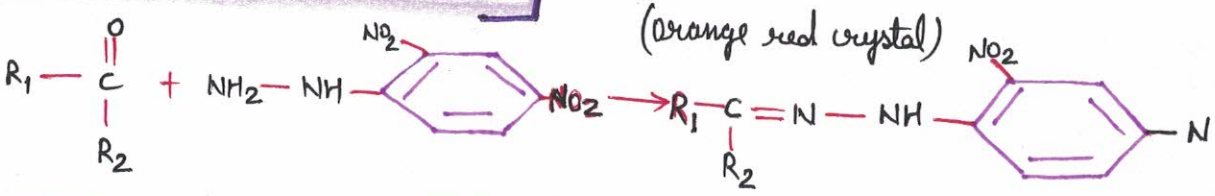
(white crystalline) (hydrozones)

4. With Phenyl hydrozone



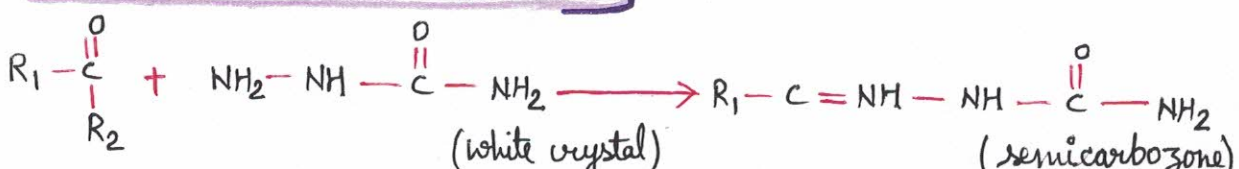
(white crystalline) (Phenyl Hydrazine)

5. With 2,4 - DNP



(orange red crystal)

6. With Semi carbozide

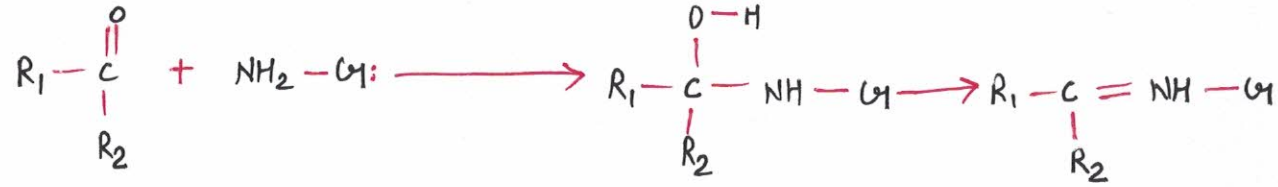


(white crystal) (semicarbozone)

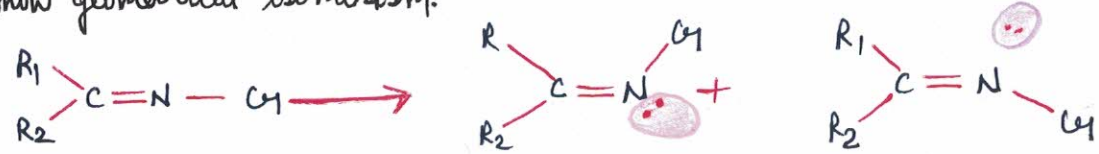
All of these reactions are test for carbonyl compounds.

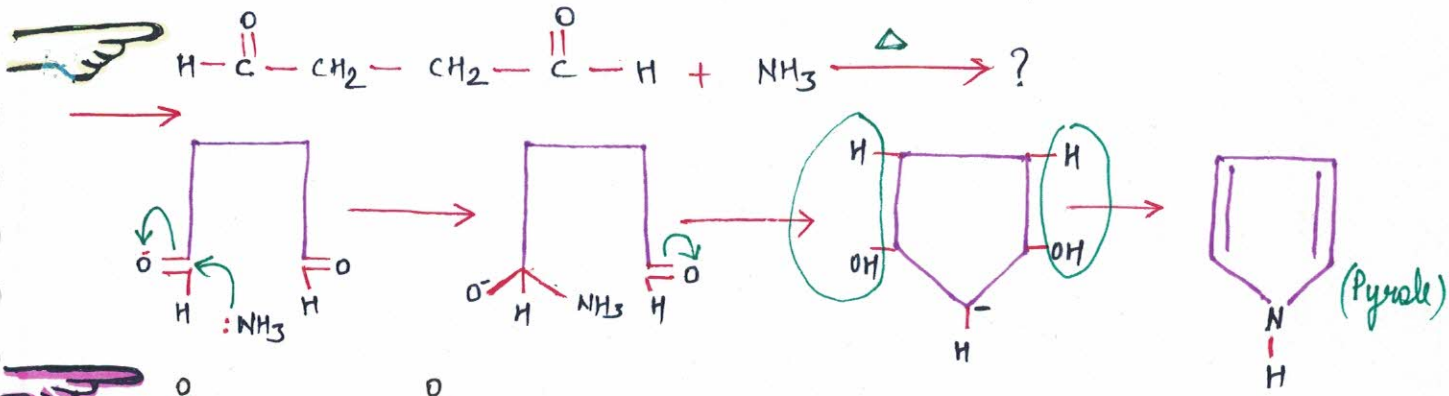
x gives oximes, hydrozones & gives a ⊕, test with 2,4 - DNP means x is a carbo-nyl compound.

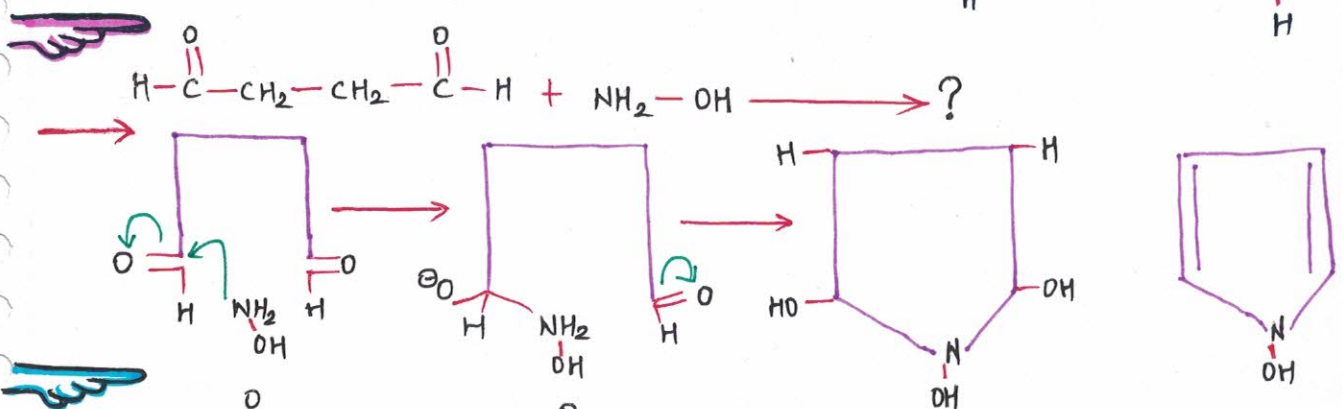
GENERAL MECHANISM

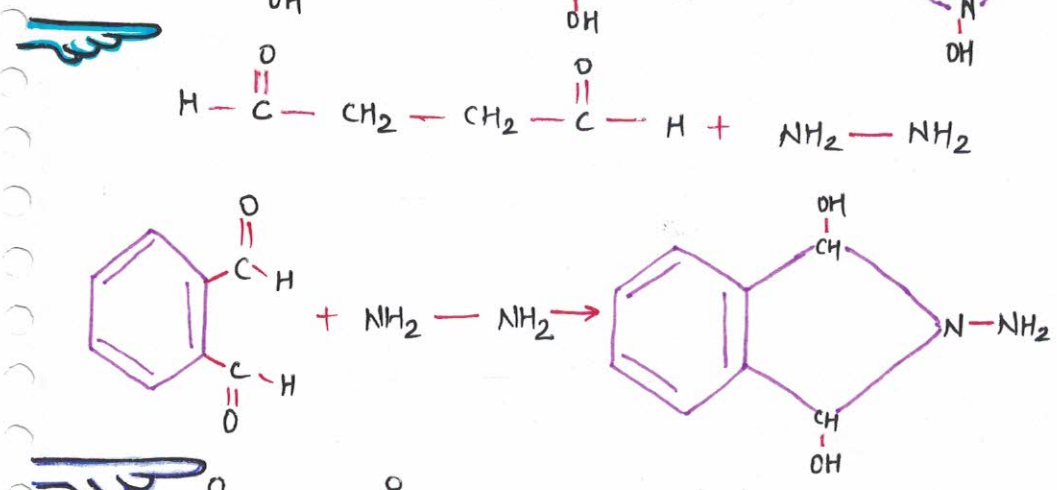


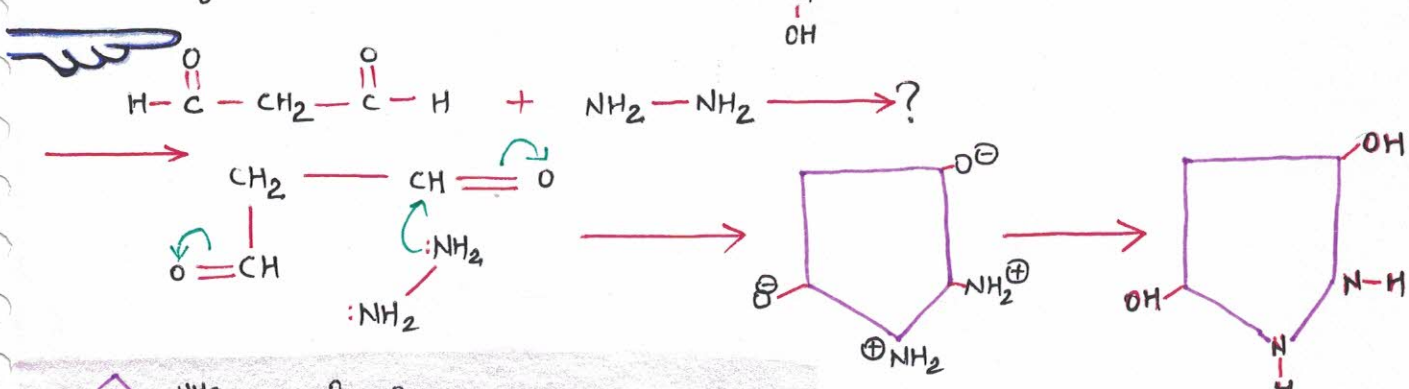
All ammonia derivations of unsymmetrical ketones and aldehydes except HCHO will show geometrical isomerism.

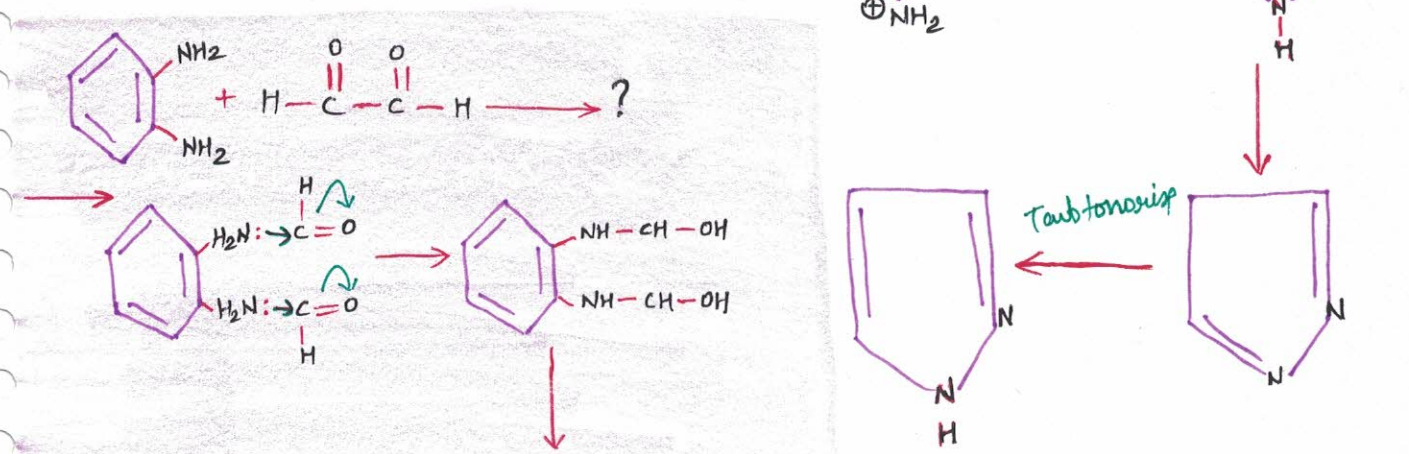



(Pyrole)







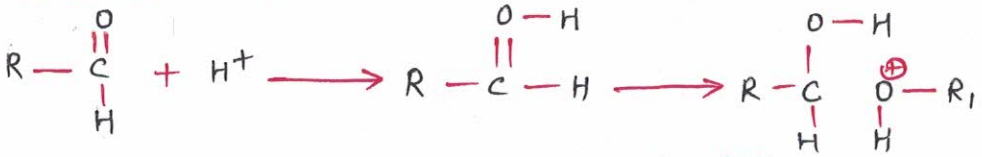

Tautomerism

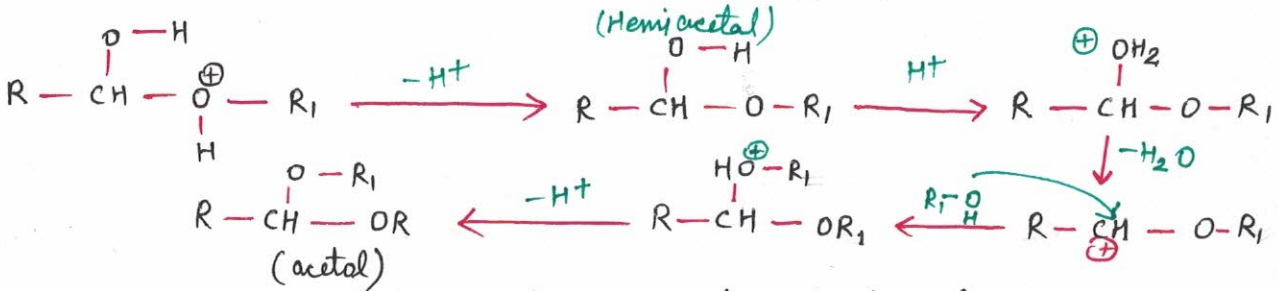
REACTION WITH ALCOHOLS



(acetals)

MECHANISM

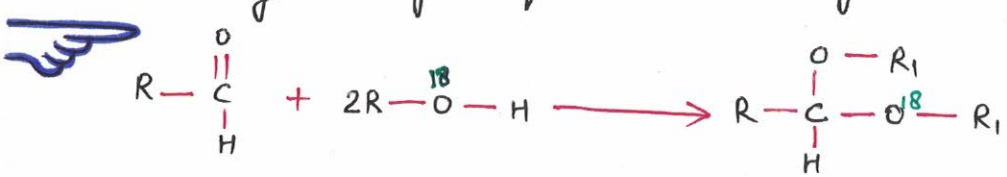


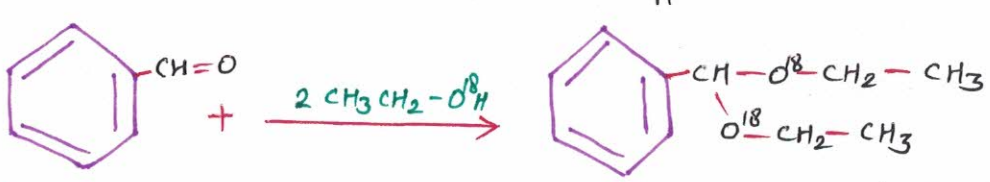


(Hemiacetal)

(acetal)

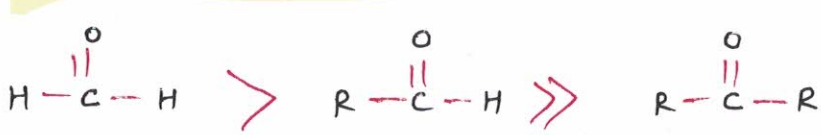
Every carbonyl requires 2 moles of alcohol.

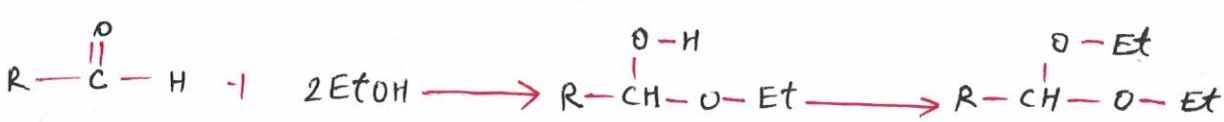


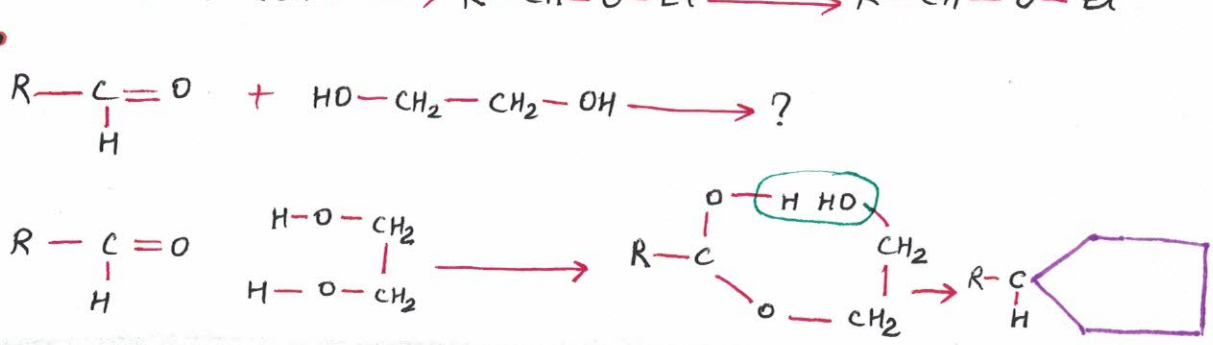


Crowding decreases rate of reaction b/w alcohol & aldehyde.

Most reactive mixture is $CH_3OH + HCHO$

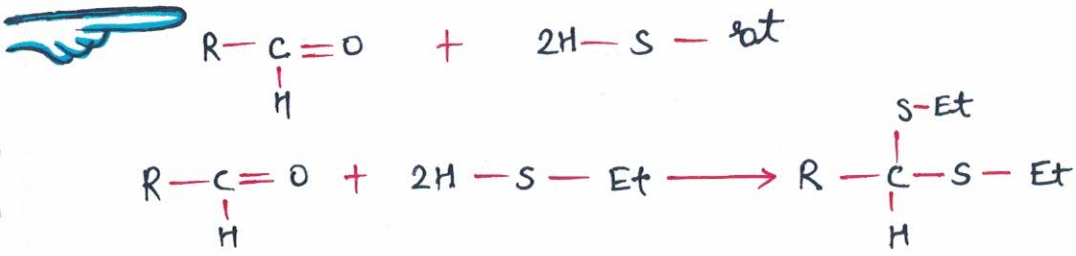




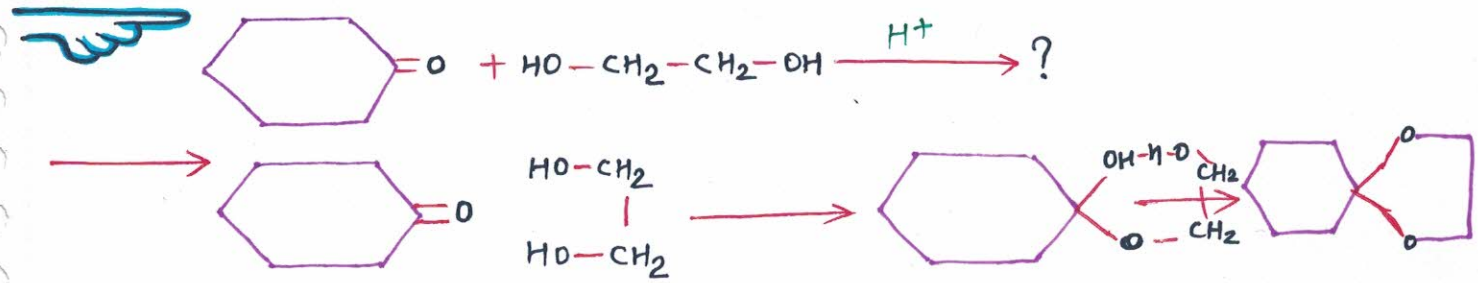


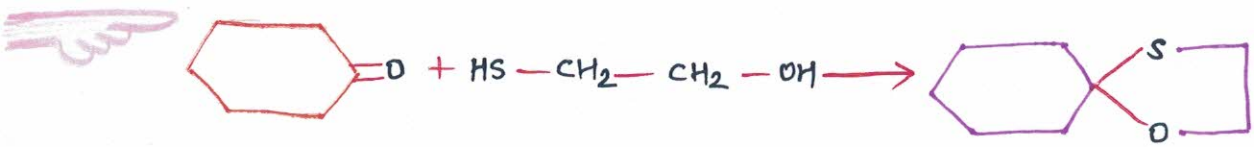
First reaction → intermolecular
2nd reaction → intramolecular
Intra > Inter (rate)

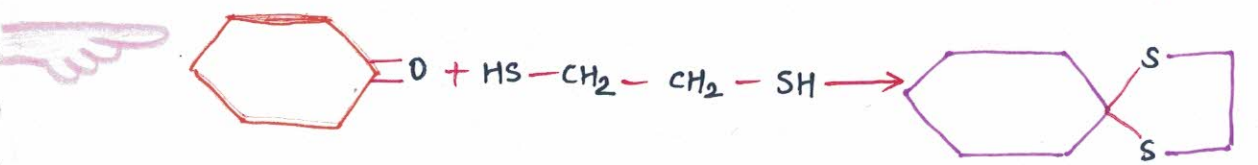
This reaction is faster than previous ones.



Thiols are better nucleophilic than alcohols this reaction is faster than alcohols.



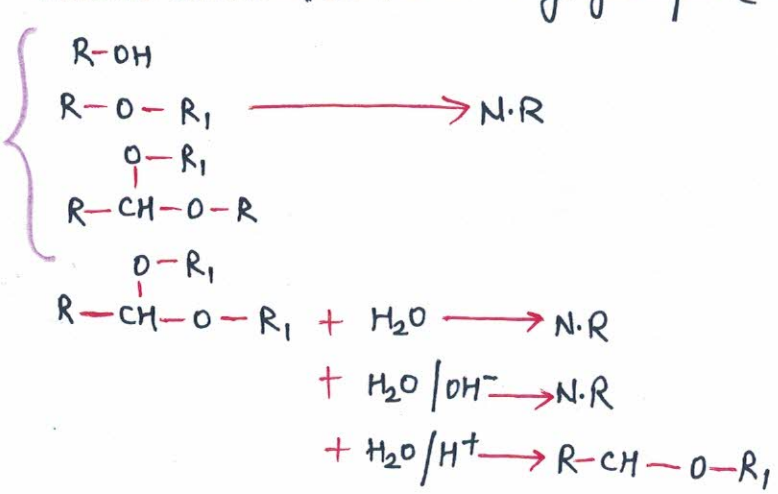


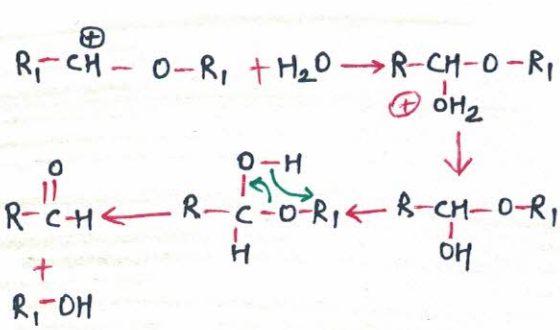


From aldehydes ⟶ acetals
From ketons ⟶ ketons

All of these reactions are reversible.

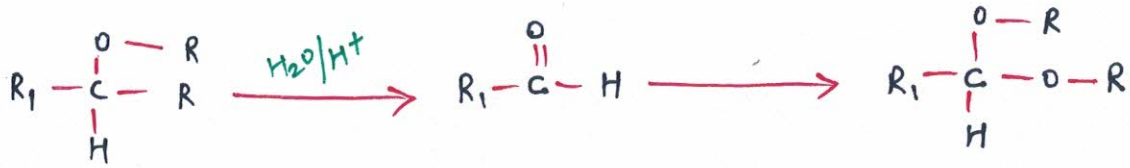
Like alcohols & ethers, acetals also do not undergo hydrolysis in basic & neutral medium as the laving groups (-O-R) is unstable.

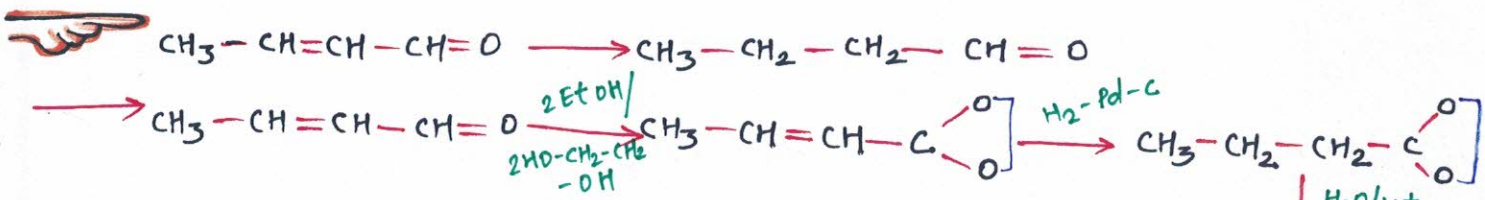


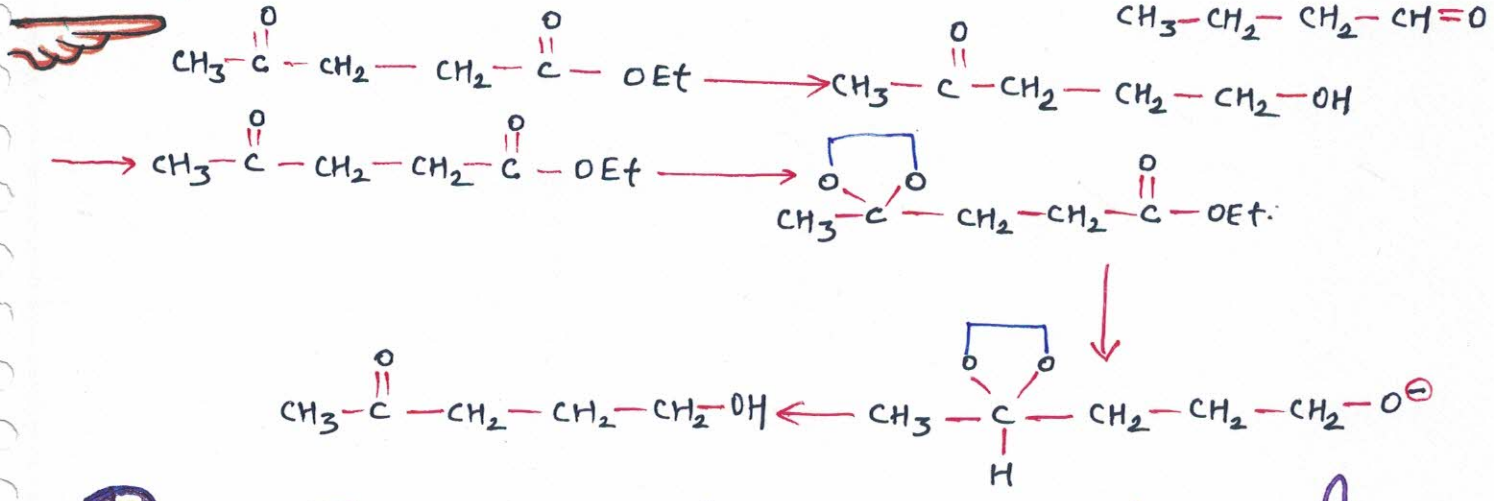


Acetals are not stable in acidic medium. they undergo hydrolysis Δ carbocation $R-\overset{\oplus}{C}H-\ddot{O}-R$ is highly stable.

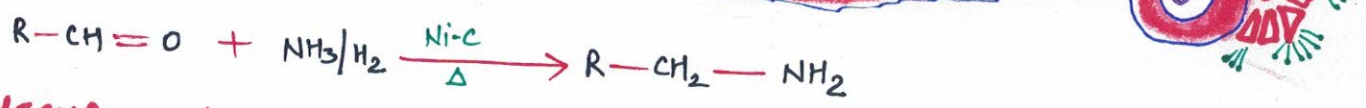




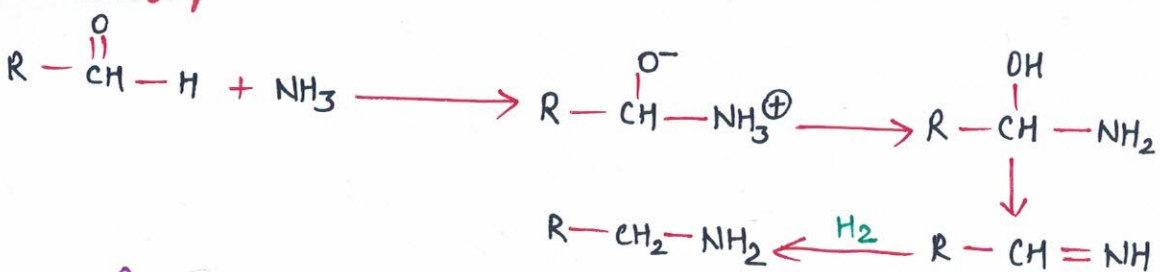


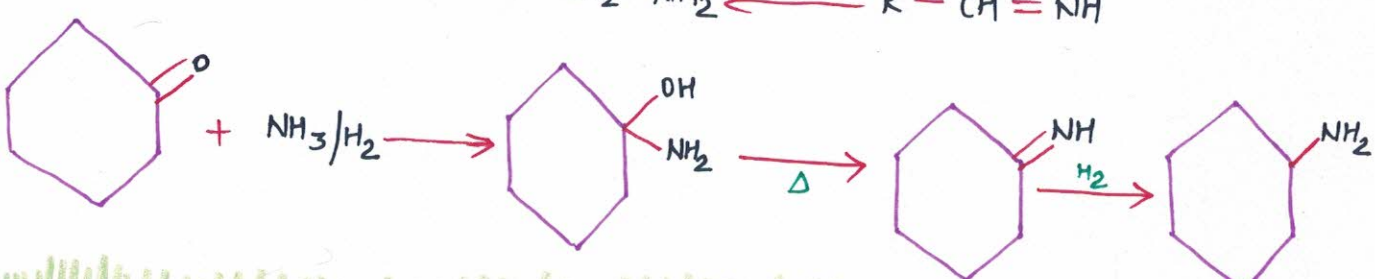


REDUCTIVE Amination of carbonyls



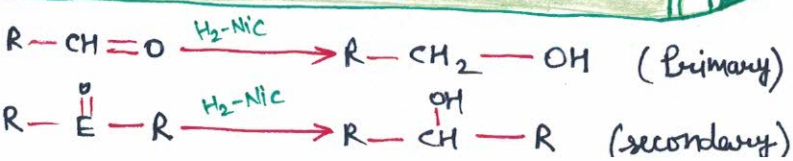
MECHANISM





REDUCTION OF CARBONYLS

① By Catalytic hydrogenation



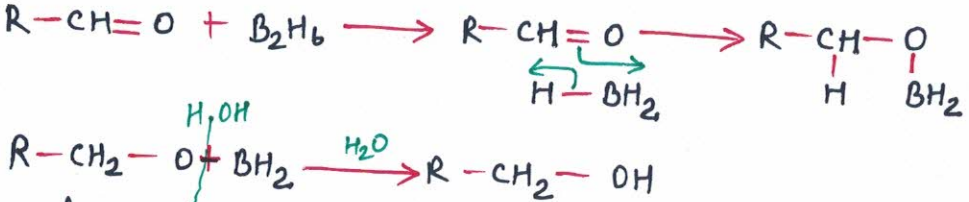
{By clemenson, Wolf Rishner, P+HI Refer to alkane chapter}

2. Reduction by $NaBH_4$/$LiBH_4$

3. Reduction by $LiAlH_4$

4. By Bouveault Blanc reaction (refer alcohols)

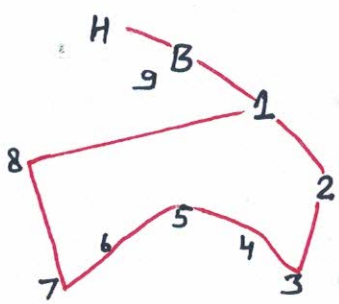
5. By hydroboration of carbonyls



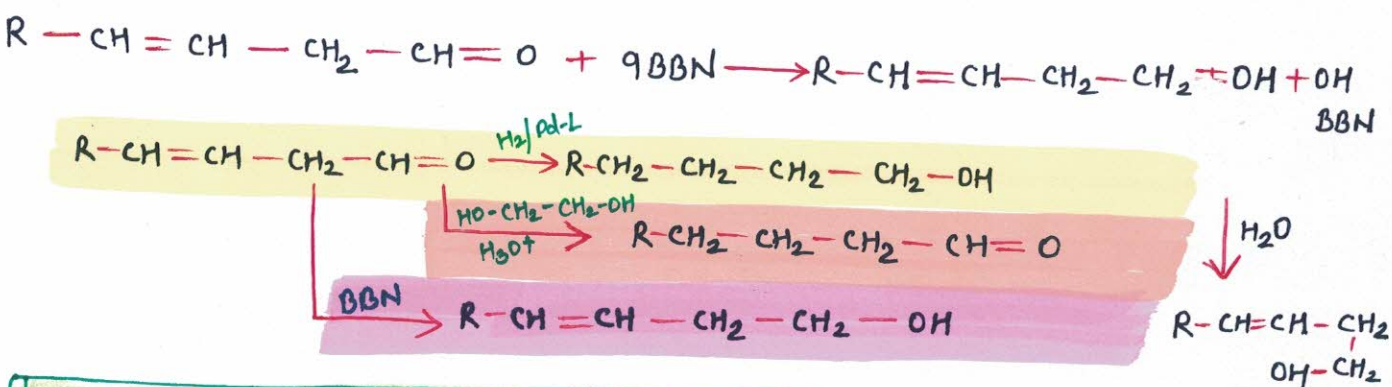
carbonyls undergo hydroboration faster than alkenes because 'O' is better than base and cyclic mechanism takes place.

6. **Reduction of 9BBN**

9 BBN reacts only with reactive carbonyl not with alkene.

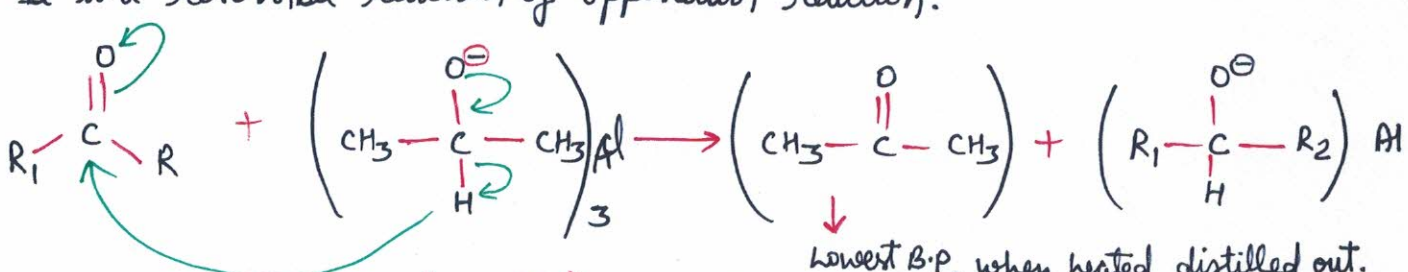


9 boro bicyclo nonane (9BBN)

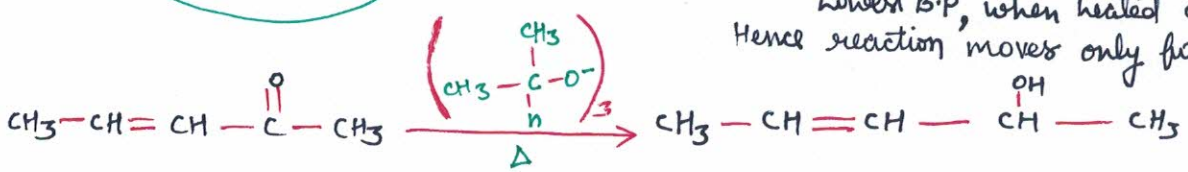


7. **Mirwein Ponndorff Verely reduction (MPV red.)**

It is a reversible reaction of oppenauer reaction.

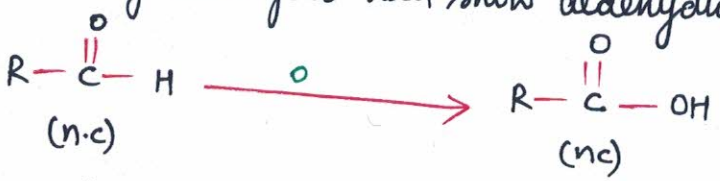


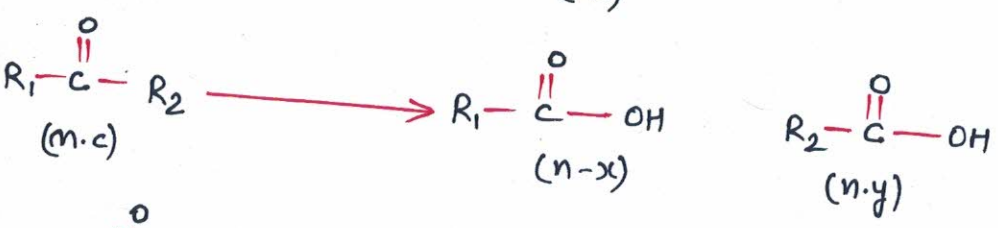
Lowest B.P, when heated distilled out. Hence reaction moves only forward.

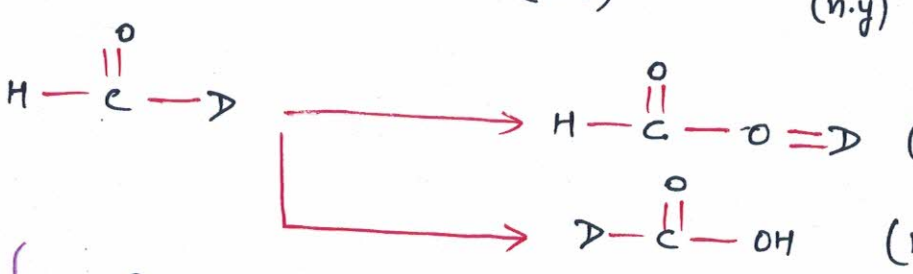


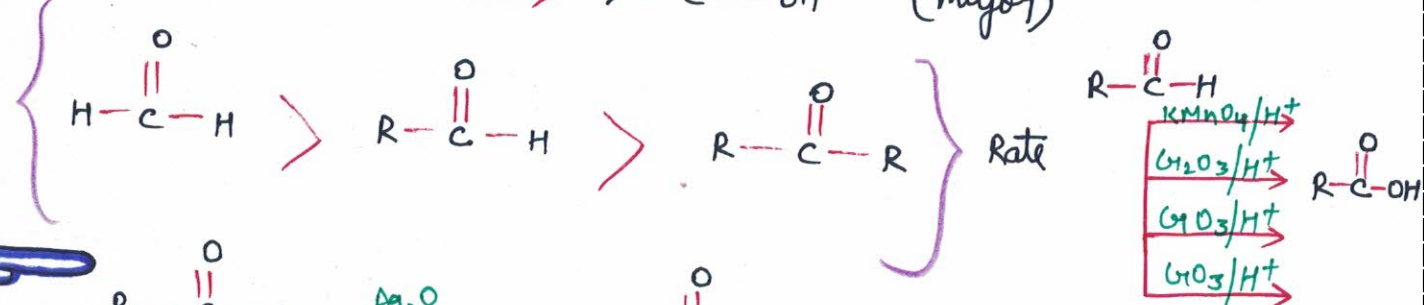
OXIDATION OF CARBONYLS

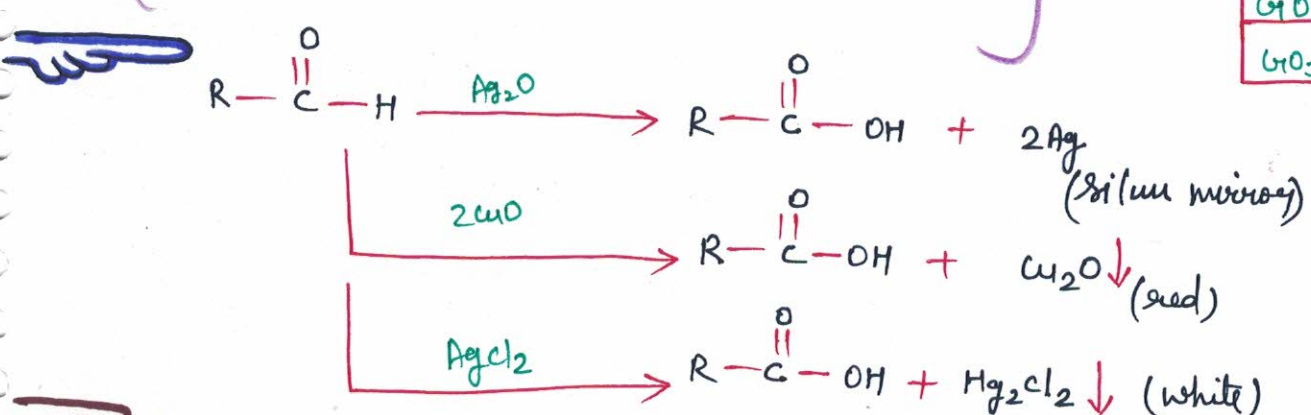
Oxidation of aldehydes will show aldehydic hydrogen isotopic effects.

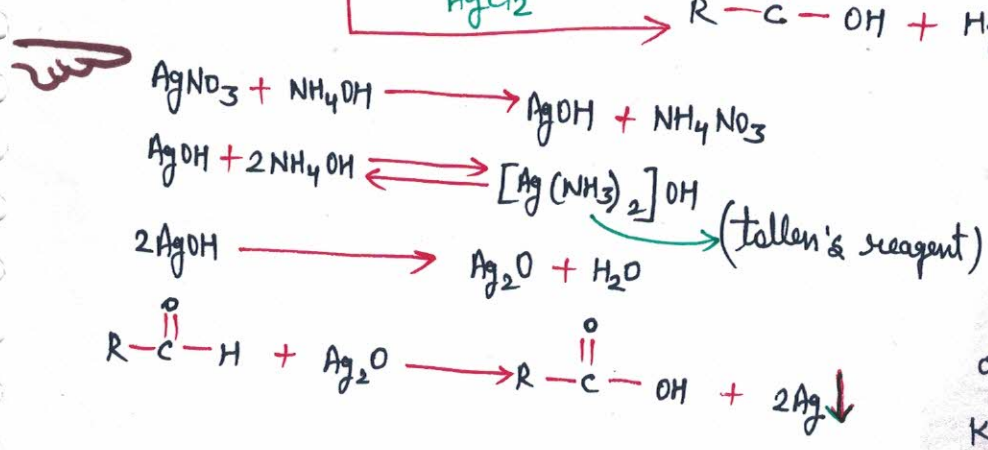












Tollen's reagent is soluble in NH_4OH so AgOH is converted into it and at the time of reaction AgOH is generated back.

only aldehydes give this test. Ketones do not undergo this reaction.

For Every carbonyl, two moles of $AgNO_3$ is required.

FAHLING'S REAGENT

$CuSO_4$ is fahling's A.

Sodium potassium Tartarate is fahling's B.

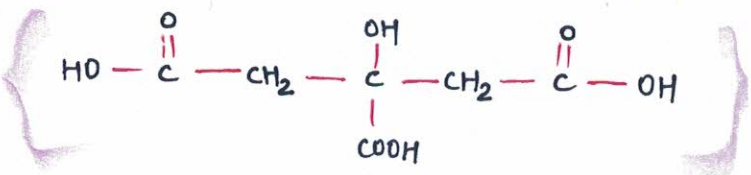
$$F.A + F.B \longrightarrow Cu(OH)_2 \xrightarrow{\Delta} CuO$$

$$R-\overset{O}{\overset{\|}{C}}-H + 2CuO \longrightarrow R-\overset{H}{\overset{|}{C}}-OH + Cu_2O\downarrow \quad \text{Red}$$

only aldehydes will give this test ketons do not give this test.

BENEDICT REAGENT

$CuSO_4$ sodium citrate $Cu(OH)_2$ CuO



citric acid

It is a tricarboxylic acid & hydroxy carbo--xylic acid.

only aldehydes give the test.

$$R-\overset{O}{\overset{\|}{C}}-H + 2CuO \longrightarrow R-\overset{O}{\overset{\|}{C}}-OH + Cu_2O\downarrow \quad \text{Red}$$

BY MERCURIC CHLORIDE ($HgCl_2$)

$$2HgCl_2 + H_2O \longrightarrow Hg_2Cl_2 + 2HCl + O$$

(white ppt)

$$R-\overset{O}{\overset{\|}{C}}-H + O \longrightarrow R-\overset{O}{\overset{\|}{C}}-OH$$

$$2HgCl_2 + R-\overset{O}{\overset{\|}{C}}-H + H_2O \longrightarrow Hg_2Cl_2 + 2HCl + R-\overset{O}{\overset{\|}{C}}-OH$$

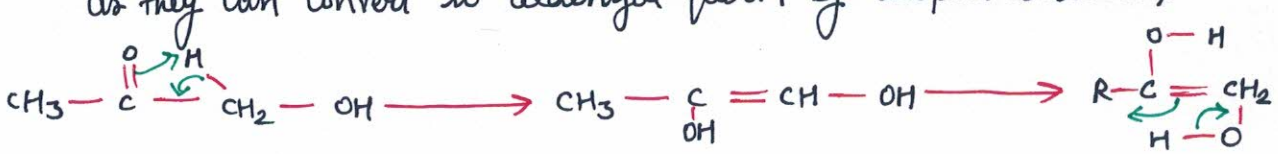
mercurous chloride (white ppt)

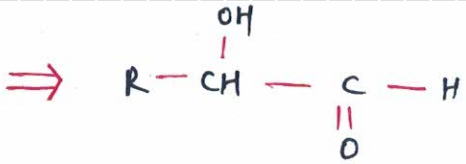
$HgCl_2$ ⟶ soluble

Hg_2Cl_2 ⟶ insoluble

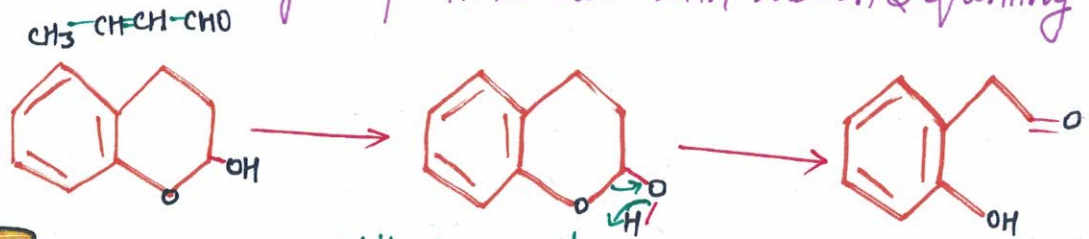
only aldehydes give this test, ketons do not.

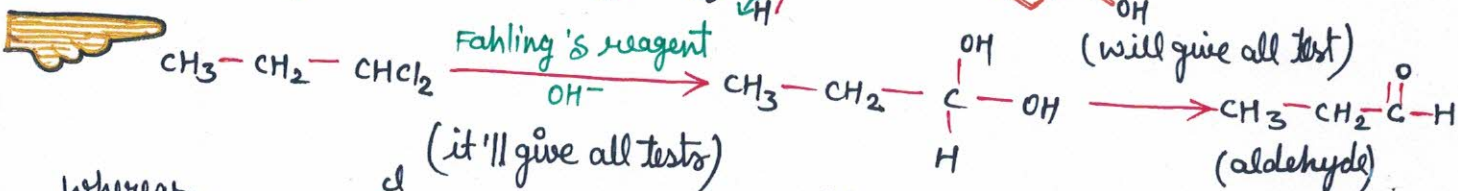
All α hydroxy ketons will give +ve test for all aldehydes test's as they can convert to aldehyde form by taytomerisation.





All hemiacetals give positive test with tollen's fahling reagent



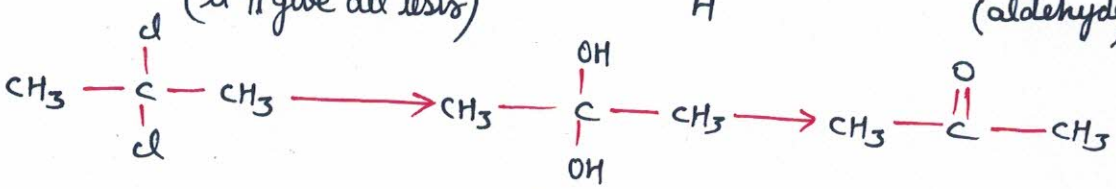


(it'll give all tests)

(will give all test)

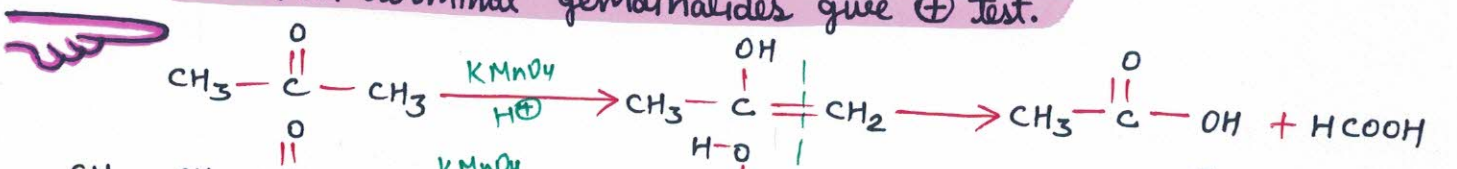
(aldehyde)

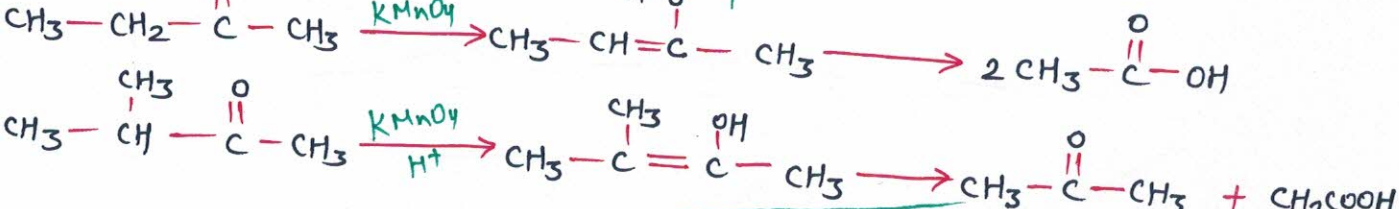
Whereas,



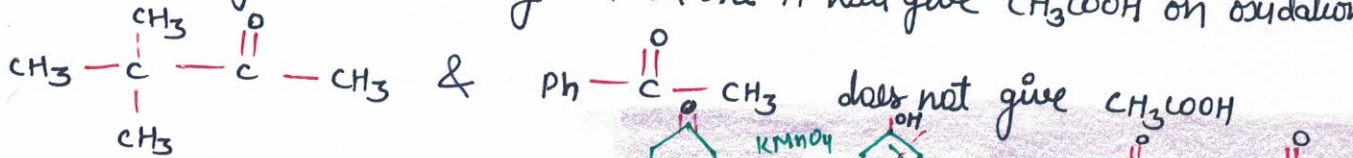
(does not give any test)

Thus, all terminal gemdihalides give ⊕ test.



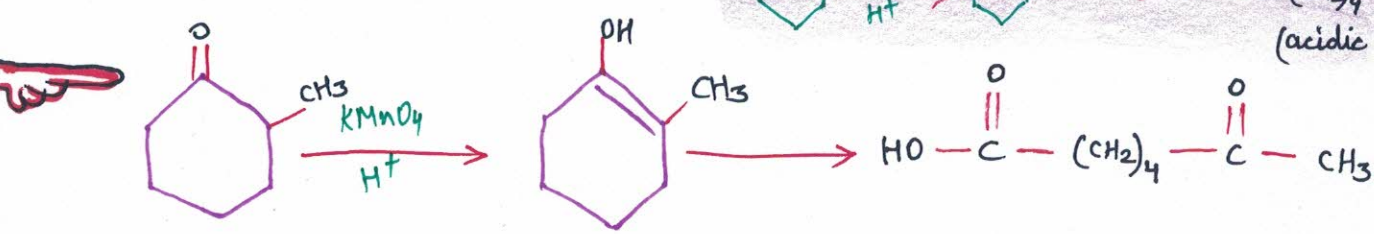


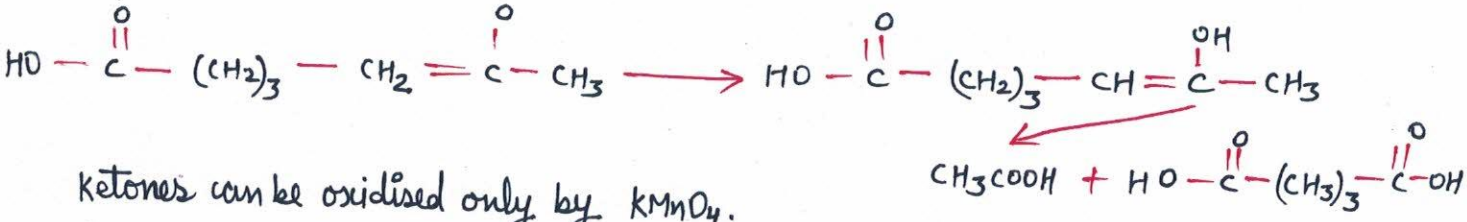
All methyl ketons having at least one "H" will give CH_3COOH on oxidation



does not give CH_3COOH

(acidic acid)

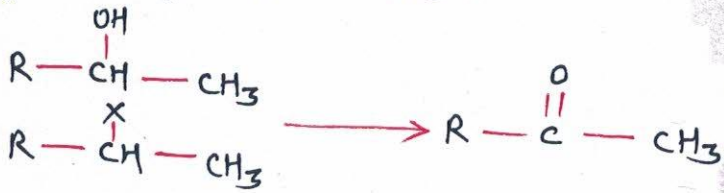




Ketones can be oxidised only by $KMnO_4$.

HALOFORM REACTION

All compounds which on reaction with X_2/OH^- can give methyl ketons can give haloform reactions.



$$R-C(=O)-CH_3 \xrightarrow{X_2} R-C(=O)-O^- + CHX_3$$

(Where in alcohols.)

$$R-C(=O)-CH_3 + F_2/OH^- \longrightarrow R-C(=O)-O^- + CHF_3$$
$$\text{"} + Cl_2/OH^- \longrightarrow \text{"} + CHCl_3$$
$$\text{"} + Br_2/OH^- \longrightarrow \text{"} + CHBr_3$$
$$\text{"} + I_2/OH^- \longrightarrow \text{"} + CHI_3 \quad (\text{yellow ppt})$$

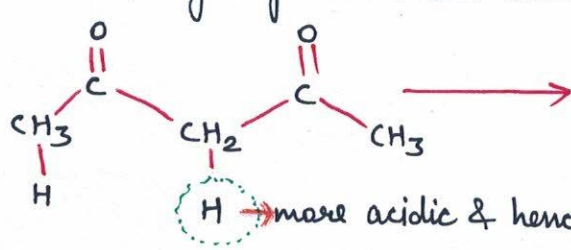
The only aldehyde which can give this reaction is acetaldehyde $CH_3-C(=O)-H$.

Only ethyl alcohol can give this test. Because both are in form of methyl ketons.

E.g. $CH_3-CH_2-C(=O)-CH_2-CH_3 \longrightarrow$ will not give this reaction.

$CH_3-CH_2-CH_2-C(=O)-CH_3 \longrightarrow$ will give this reaction.

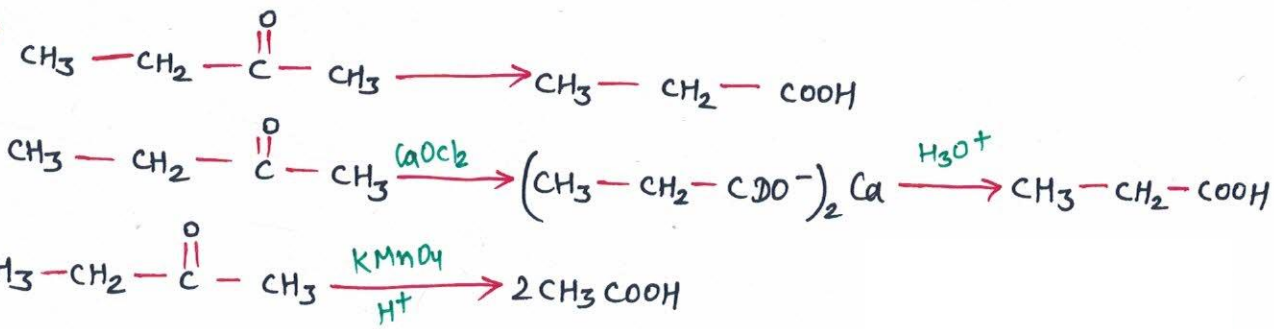
Methyl ketones having reactive methylene do not give this reaction as it's Hydrogen is more acidic than methyl hydrogen.

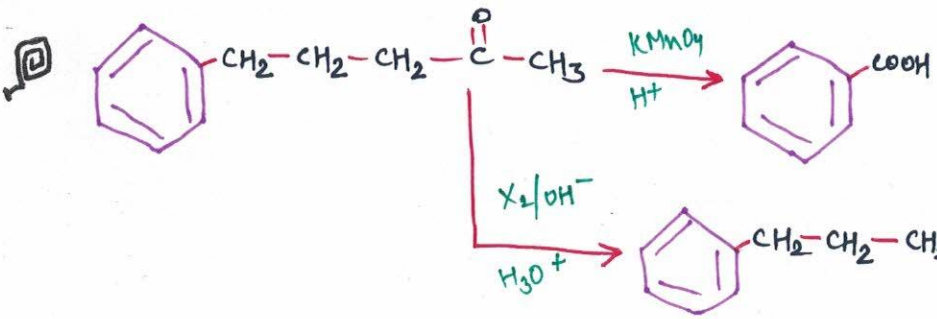


$\longrightarrow$ does not give this test.

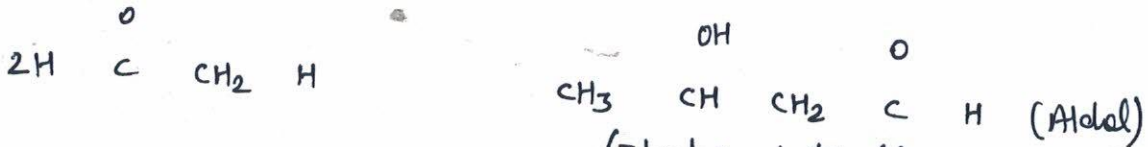
H → more acidic & hence more reactions

R > X





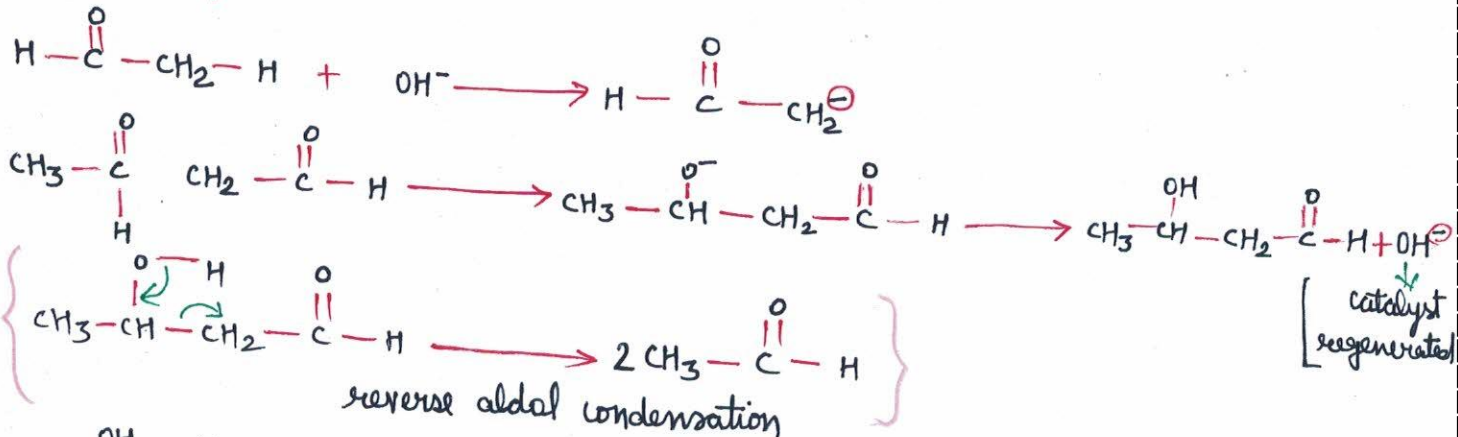
ALDOL CONDENSATION

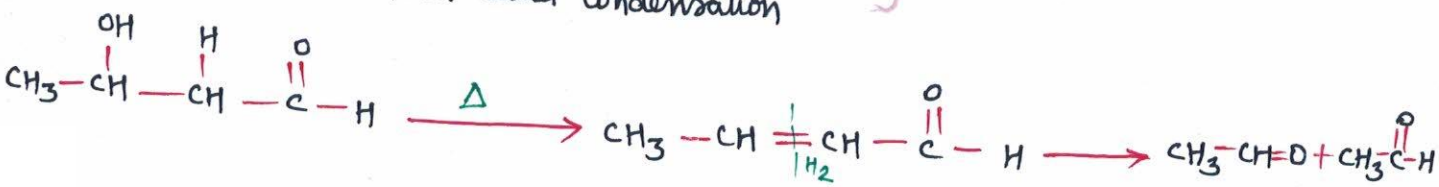


(3 hydroxybutonal)

The reactant must have "α" hydrogenand its reacted with dilute alkali which acts as catalyst.

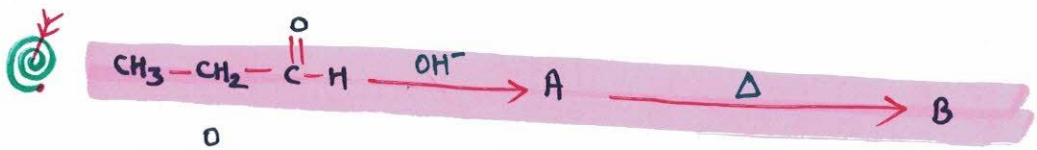
MECHANISM

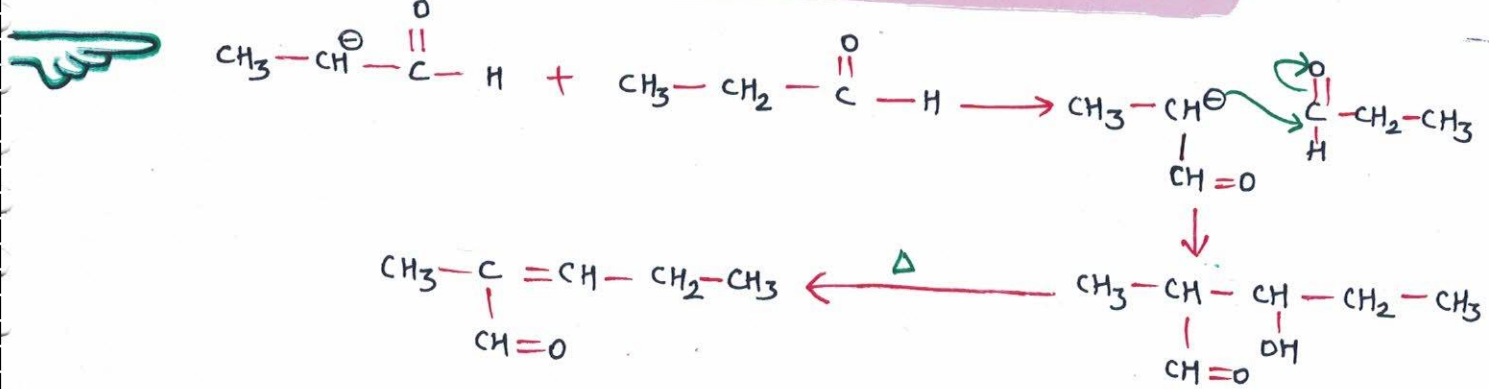


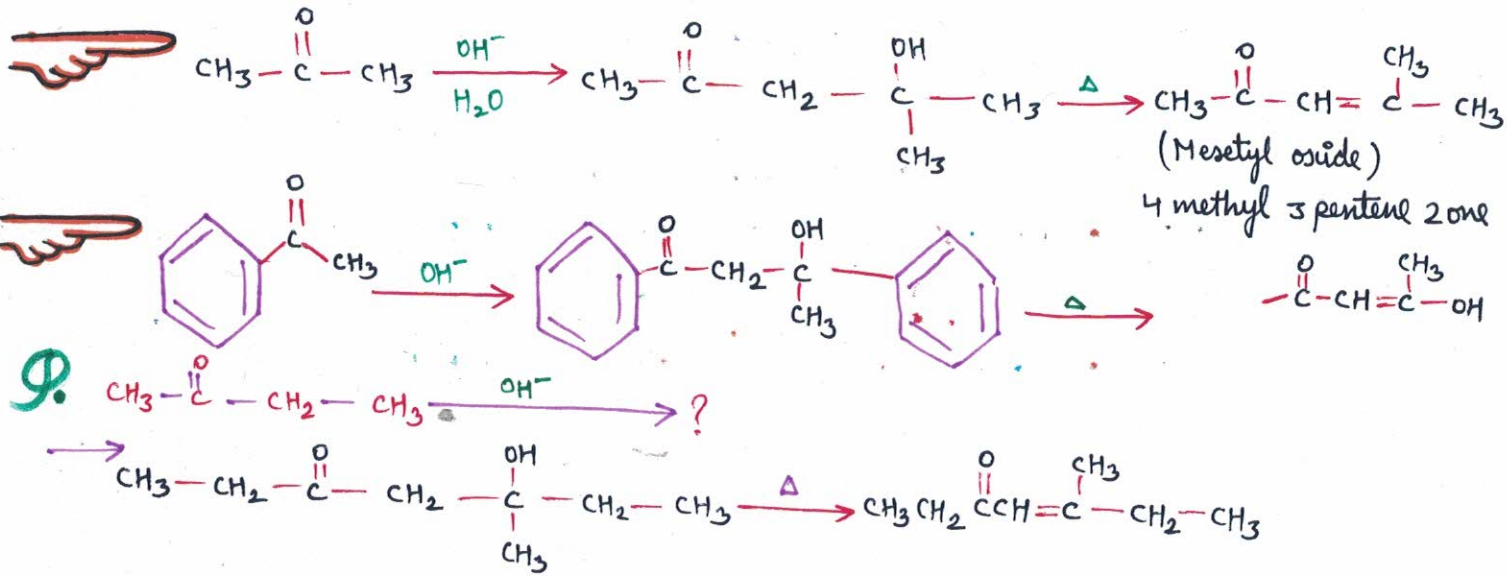


convert acetaldehyde to protonaldehyde.

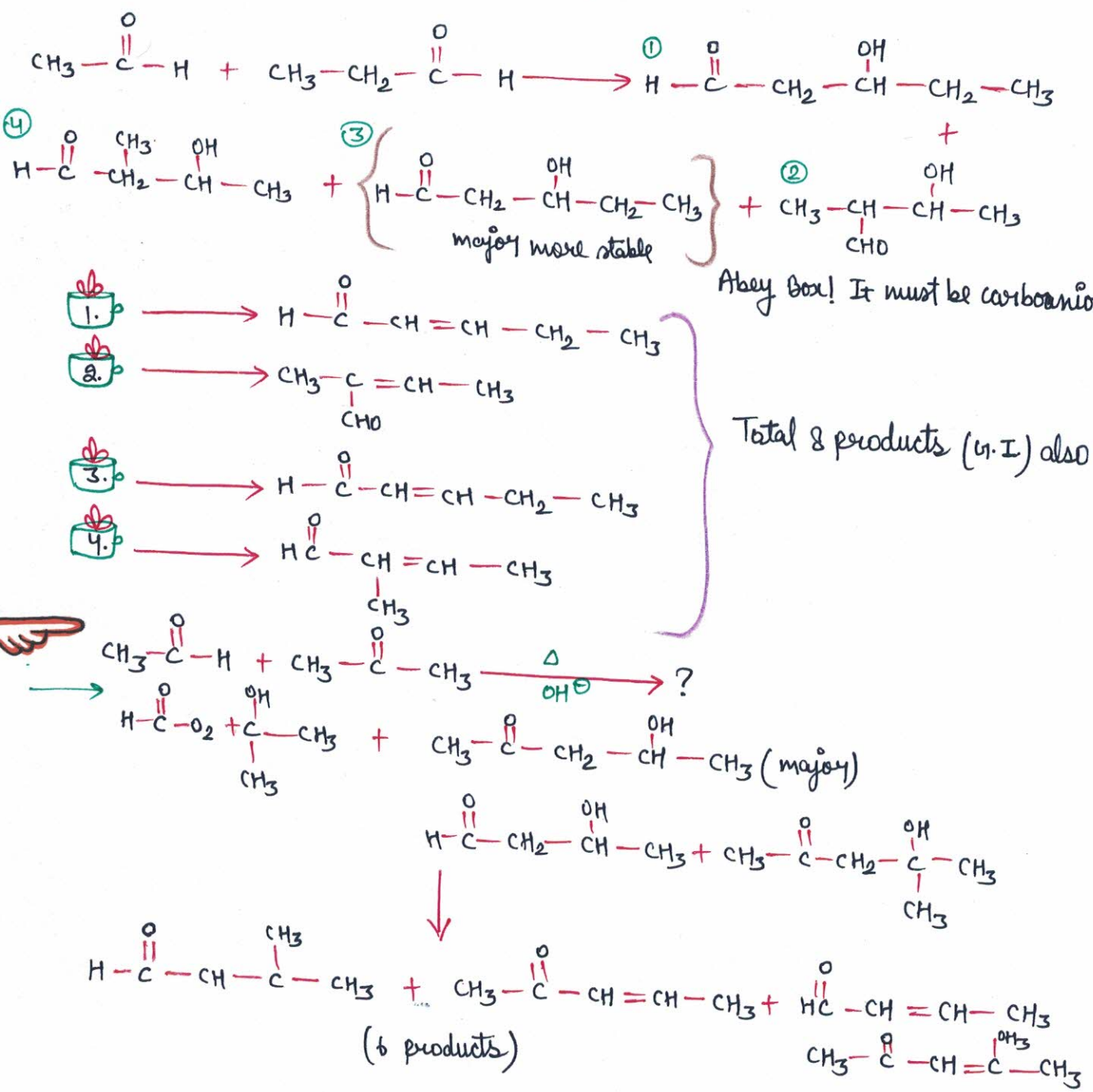
Done above.



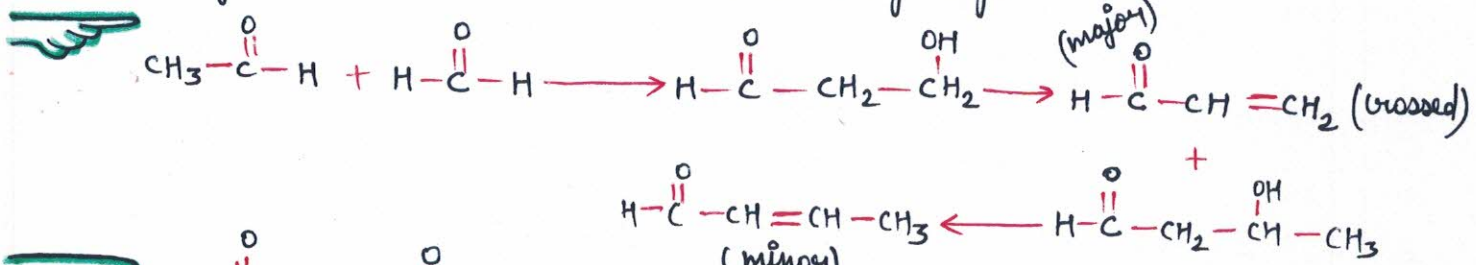


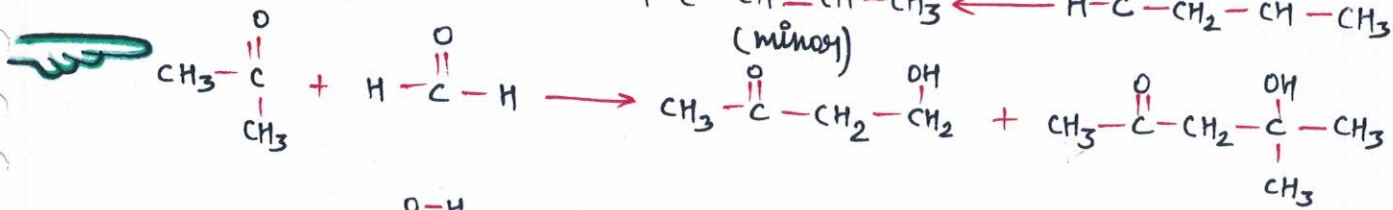


CROSSED ALDOL CONDENSATION



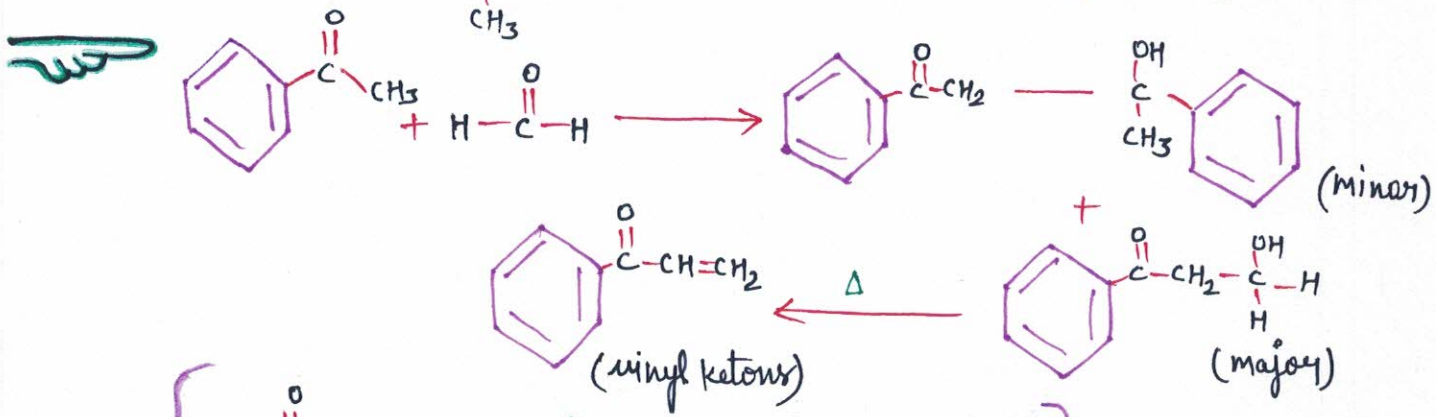
In the previous reactions, $CH_3-\underset{OH}{CH}-CH_2-\overset{O}{\overset{\|}{C}}-CH_3$ is major because, $CH_3-\overset{O}{\overset{\|}{C}}-CH_3$ will undergo nucleophilic addition and has 6 α hydrogen.





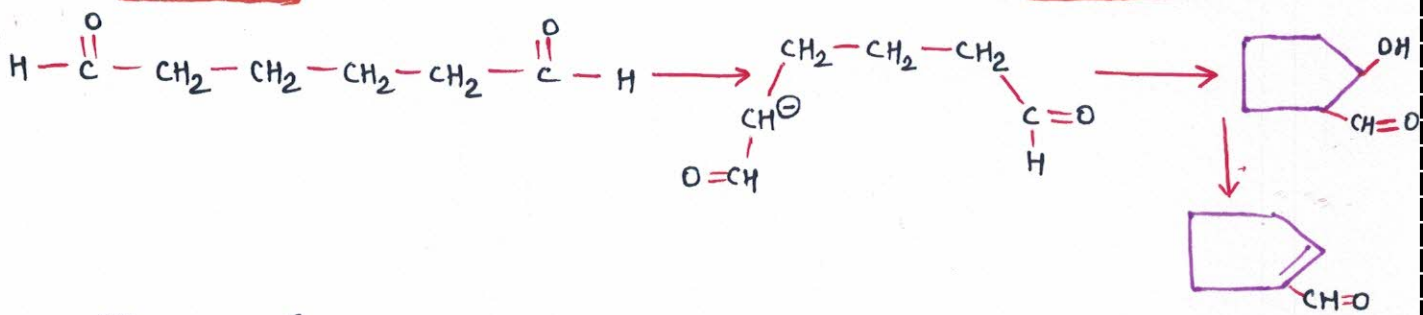
If $A+B \longrightarrow CH_3-\underset{CH_3}{\underset{|}{C}}(OH)-CH_2-\overset{O}{\overset{\|}{C}}-H$ then $A = CH_3-\overset{O}{\overset{\|}{C}}-CH_3$ – $B = H-\overset{O}{\overset{\|}{C}}-CH_3$

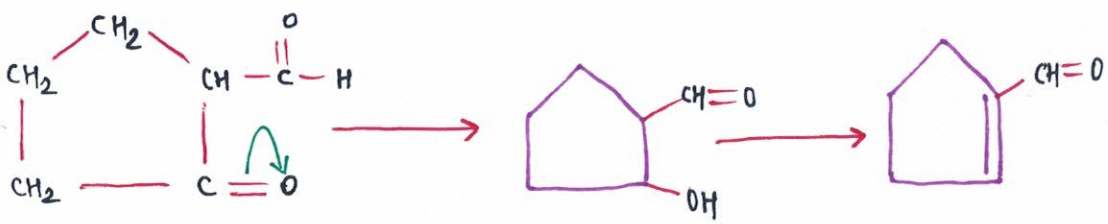
$A+B \longrightarrow CH_3-\underset{CH_3}{\underset{|}{C}}=CH-\overset{O}{\overset{\|}{C}}-H \longrightarrow CH_3-\overset{O}{\overset{\|}{C}}-CH_3 + CH_3-\overset{O}{\overset{\|}{C}}-H$

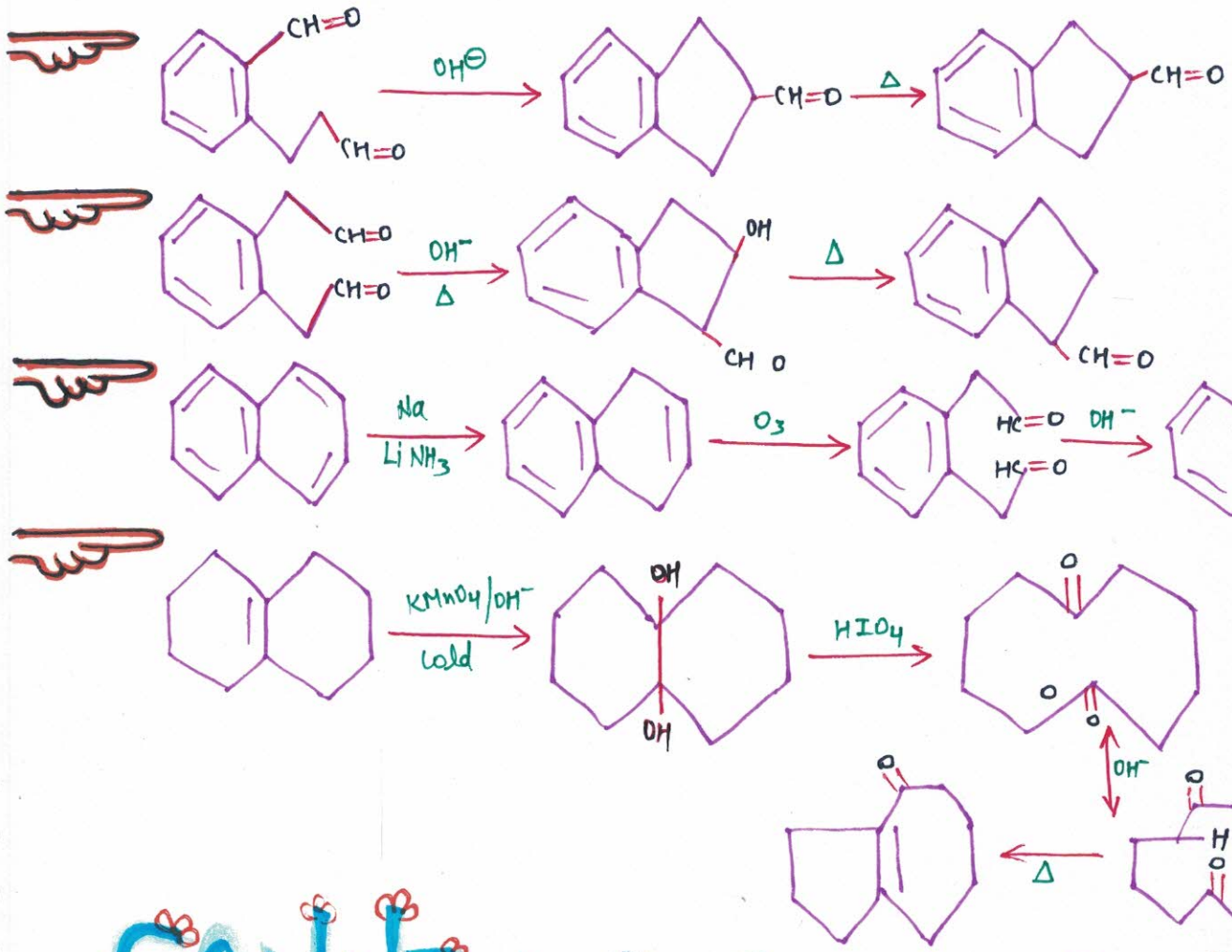


$R-\overset{O}{\overset{\|}{C}}-CH_3$ (methyl ketons) $\xrightarrow[\Delta]{HCHO/OH^-}$ $R-\overset{O}{\overset{\|}{C}}-CH=CH_2$ (vinyl ketons)

Intramolecular Aldol condensation





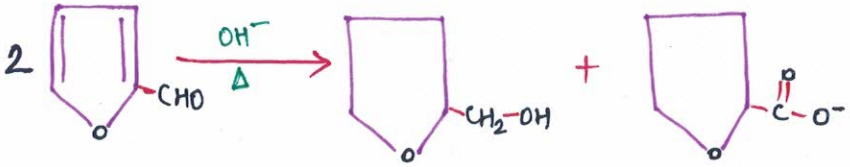


CANNIZZARRO REACTION

$$2H-\underset{\underset{H}{|}}{\overset{\overset{O}{\|}}{C}}-H + OH^- \text{ (conc.)} \xrightarrow[\Delta]{OH^-} CH_3OH + H-\overset{\overset{O}{\|}}{C}-O^-\,Na^+$$

$$2\,Ph-\overset{\overset{O}{\|}}{C}-H \xrightarrow[\Delta]{OH^-} Ph-\overset{\overset{O}{\|}}{C}-O^- + Ph-CH_2-OH$$

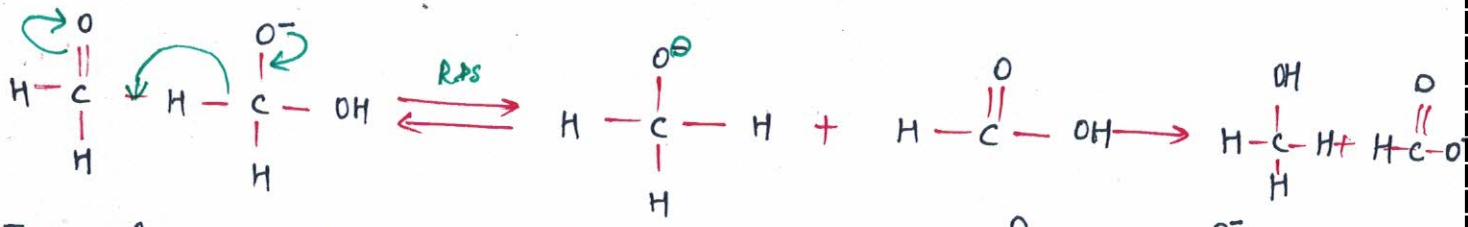
$$2\,CH_3-\underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}}-\overset{\overset{O}{\|}}{C}-H \xrightarrow[\Delta]{OH^-} CH_3-\underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}}-CH_2-OH + CH_3-\underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}}-\overset{\overset{O}{\|}}{C}-O^-$$



MECHANISM

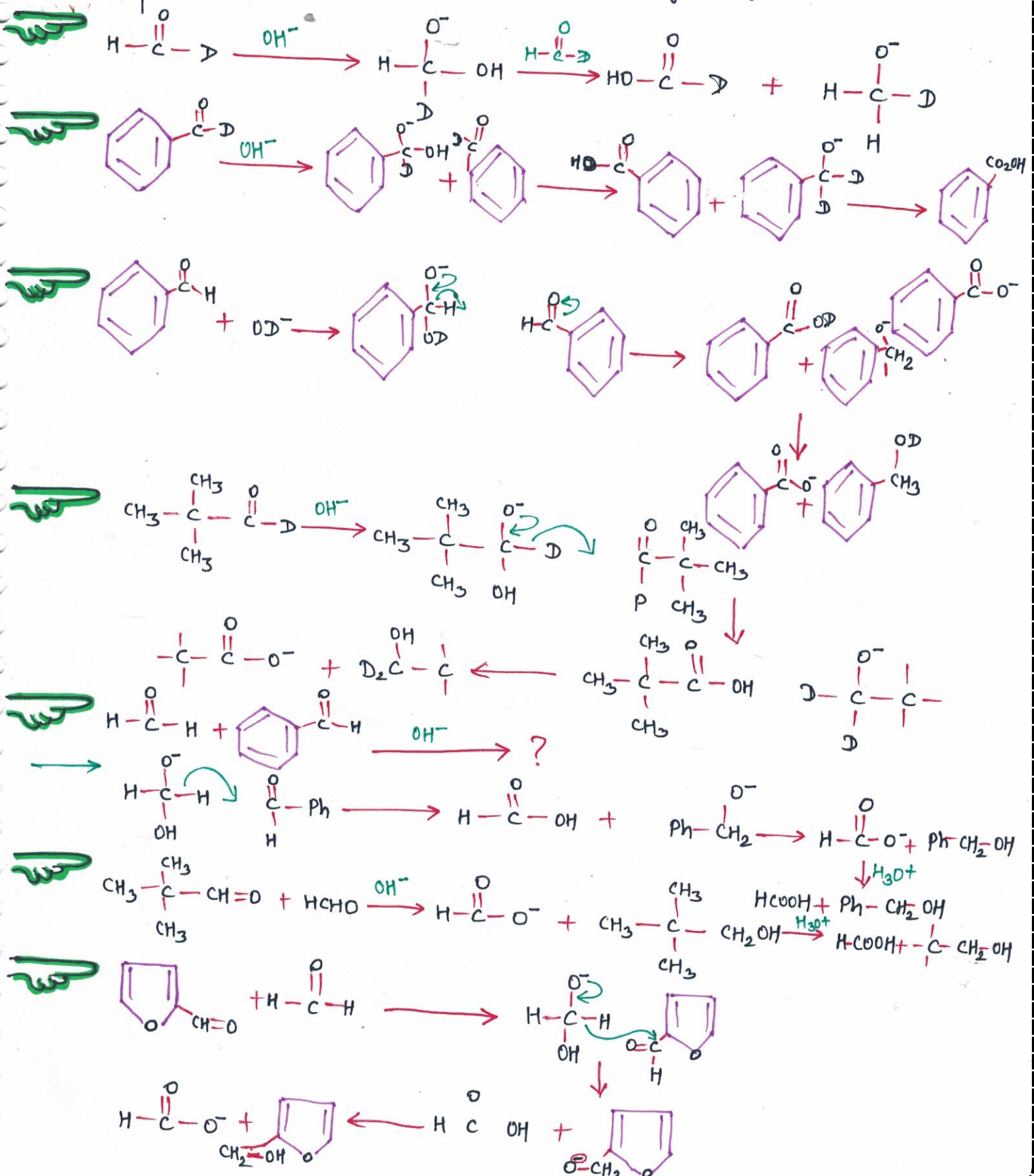
It should have no 'α" hydrogen.



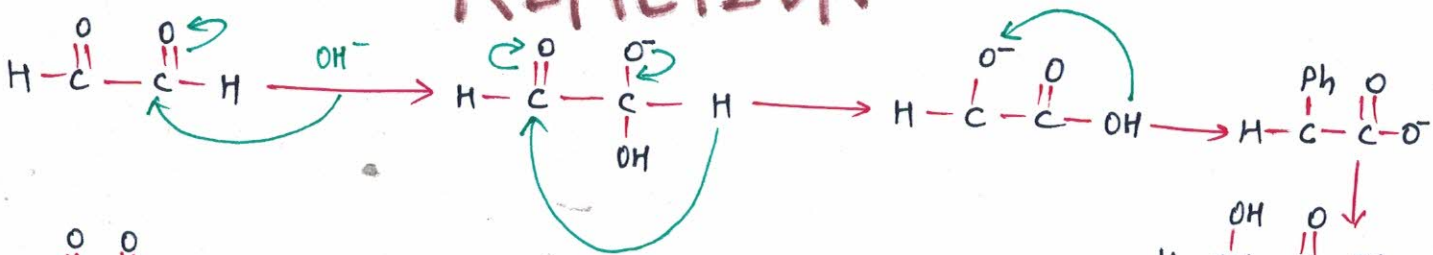


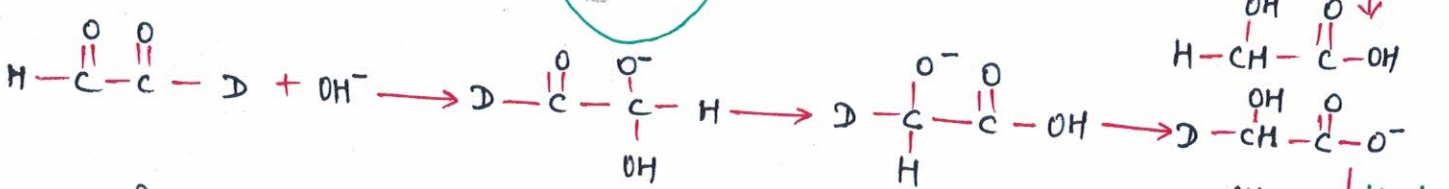
It cannizzaro reaction, OH^- is a reactant. Rate $\propto [H-\overset{O}{\overset{\|}{C}}-H][H-\underset{H}{\overset{O^-}{\overset{|}{C}}}-OH]$

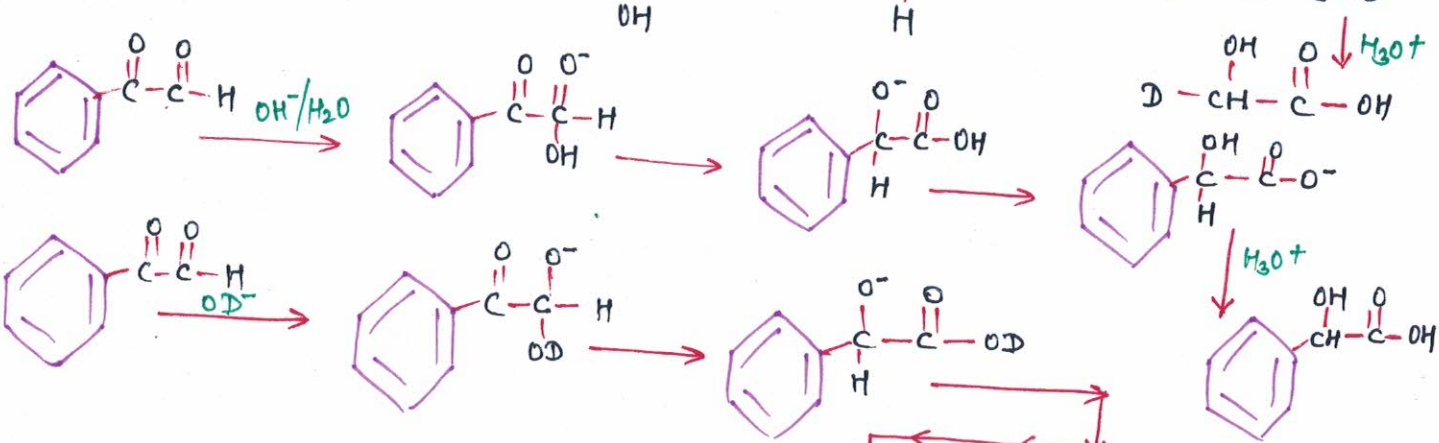
It is 3rd order kinetics. It occurs at high conc^n & high temp. aldol occurs at mild temp.

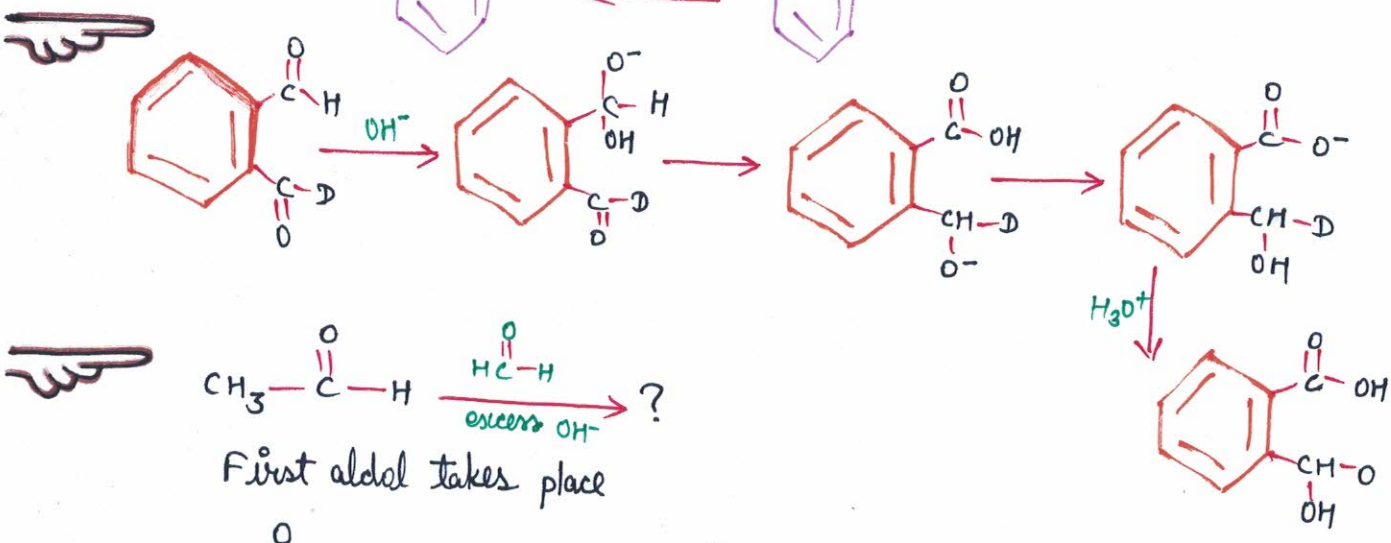


INTRAMOLECULAR CANNIZARO REACTION

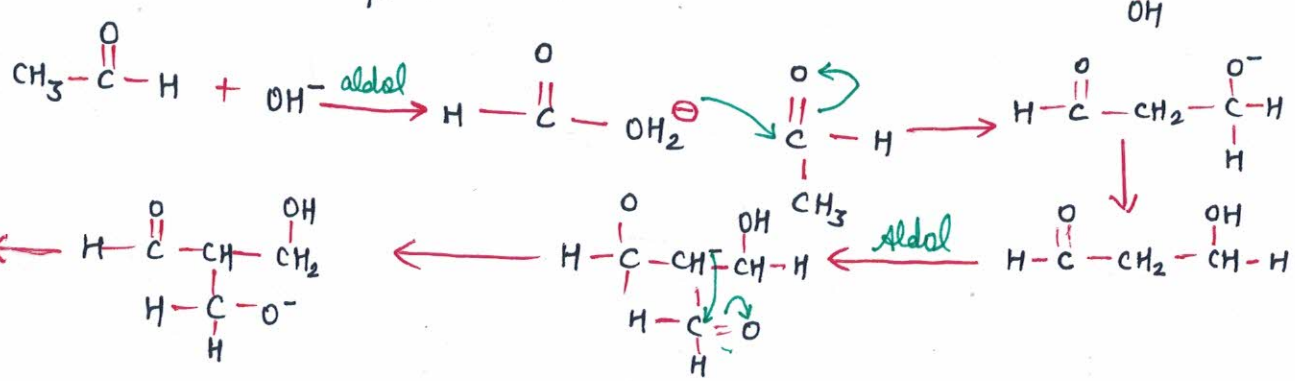


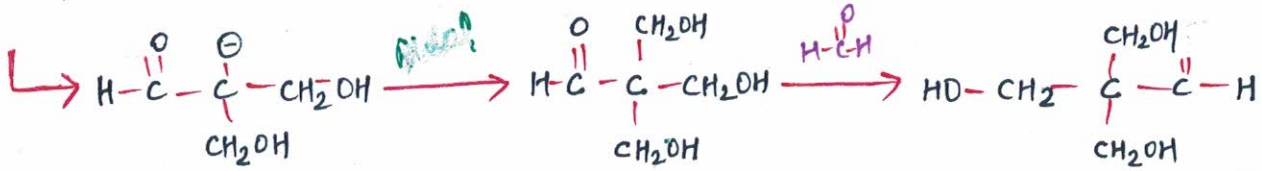


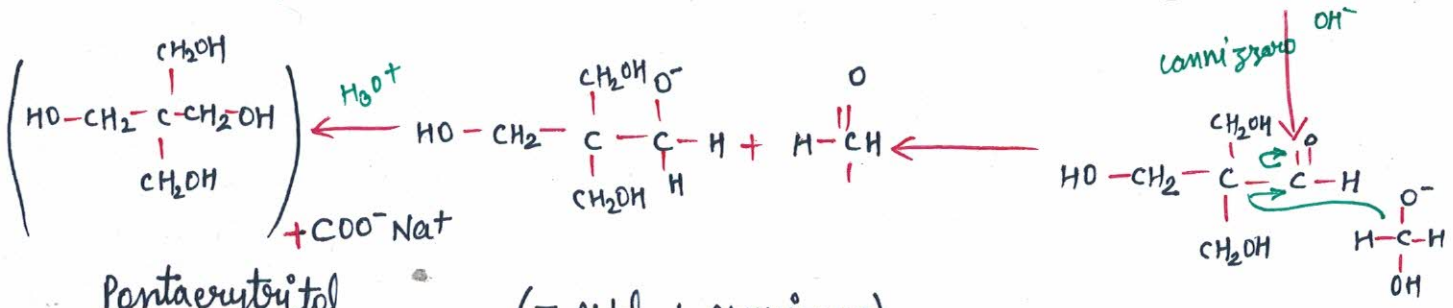




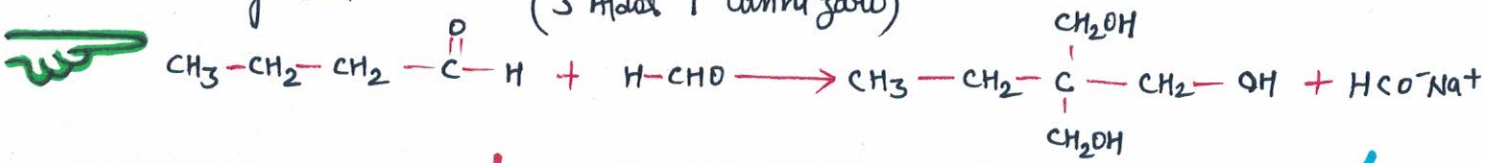
First aldol takes place



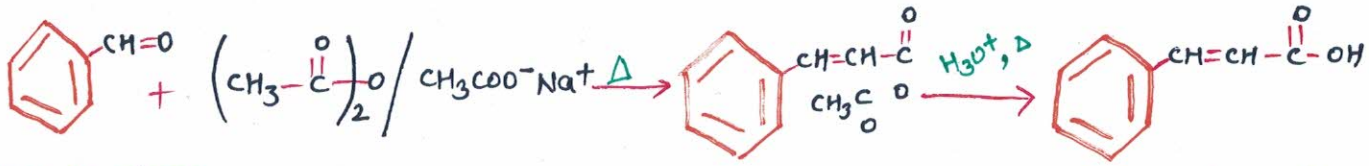




Pentaerythritol (3 Aldol 1 cannizaro)

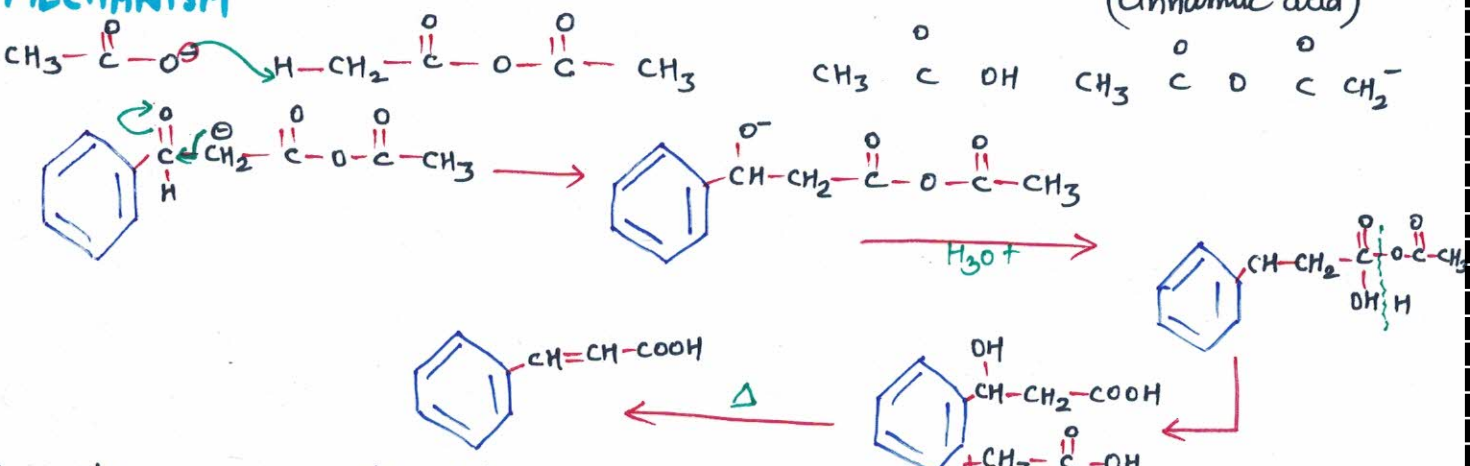


PERKIN CONDENSATION

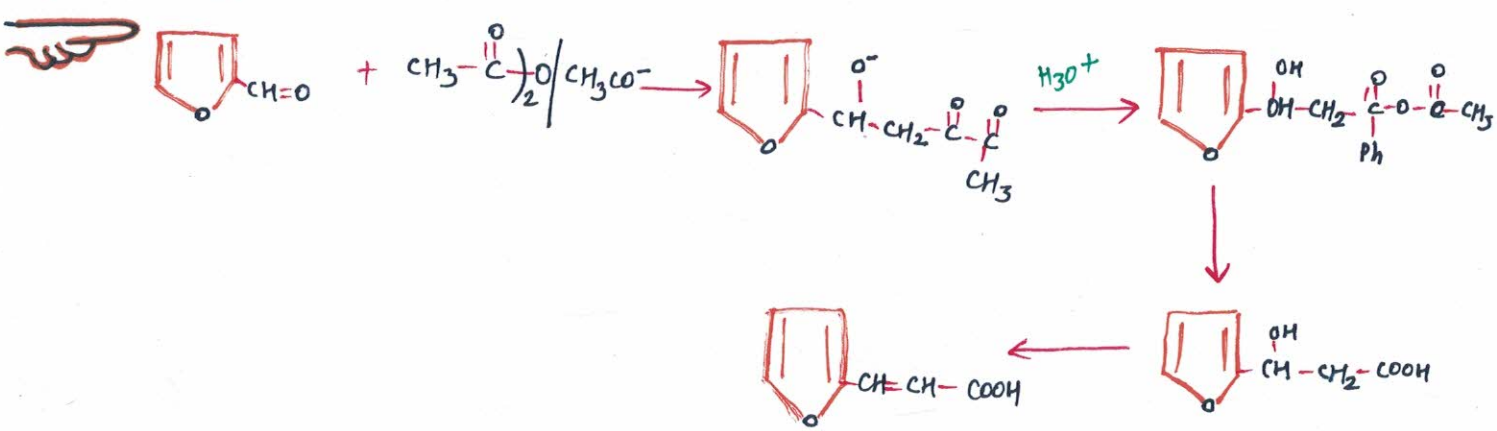


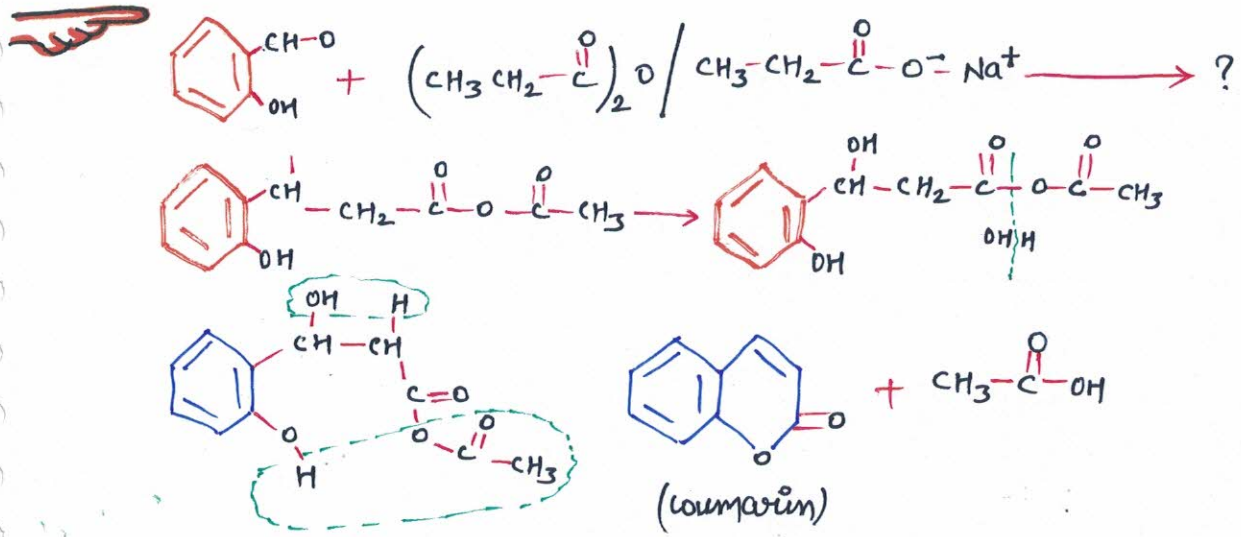
(cinnamic acid)

MECHANISM

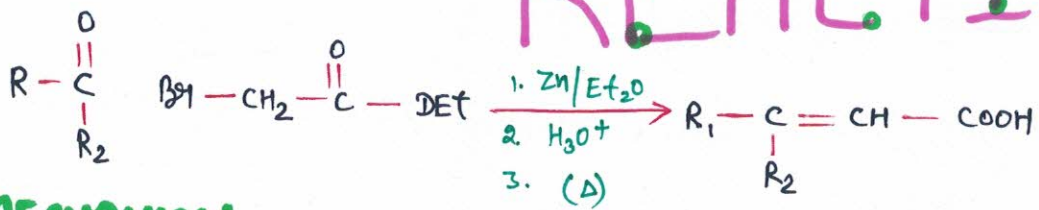


Preparation of A,B unsaturated acid by reaction of aromatic aldehyde with $(CH_3-\overset{O}{\overset{\|}{C}})_2O$. is Perkin's condensation.

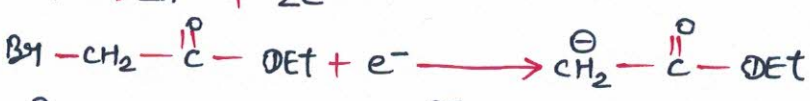


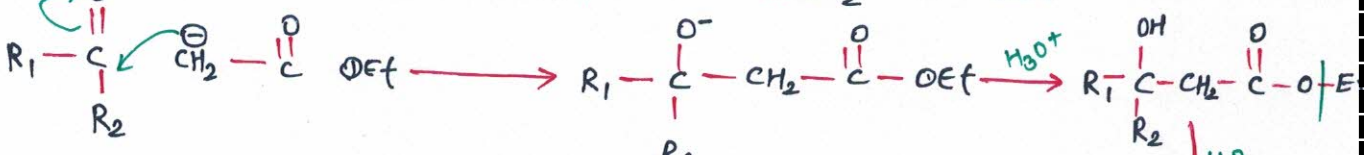


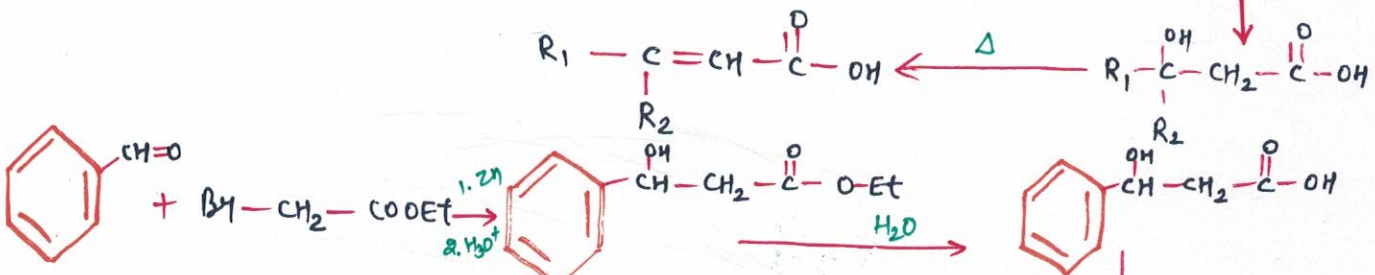
REFORMATING REACTION

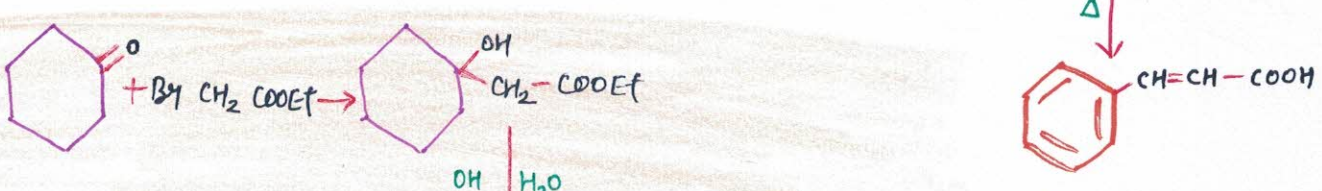


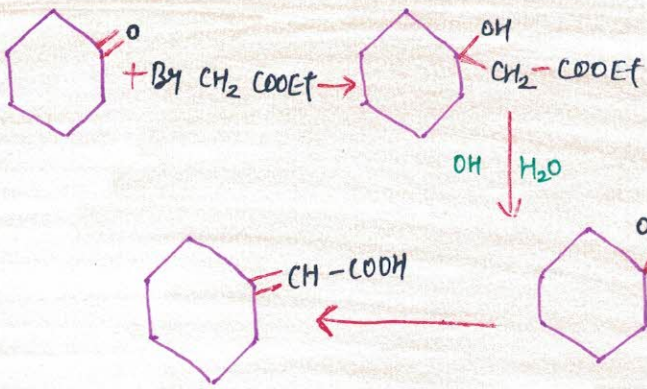
MECHANISM -

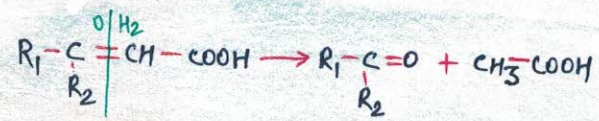




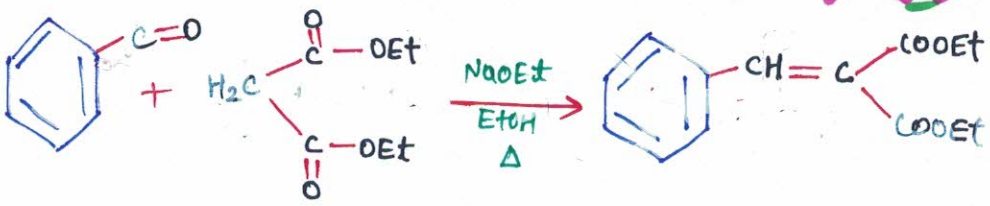




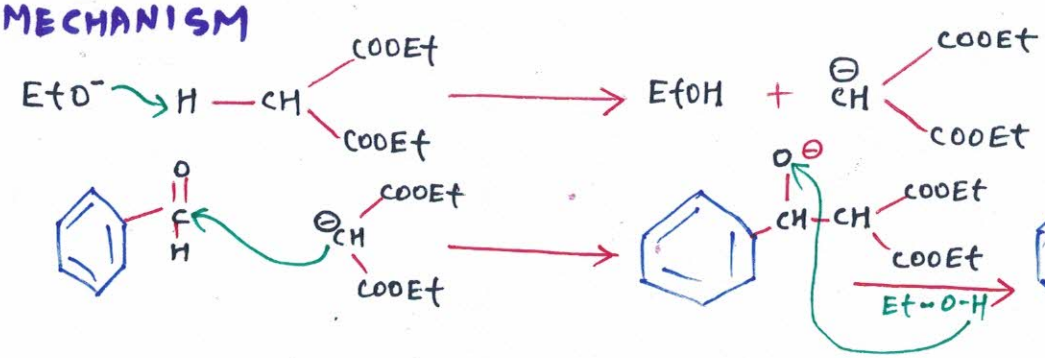




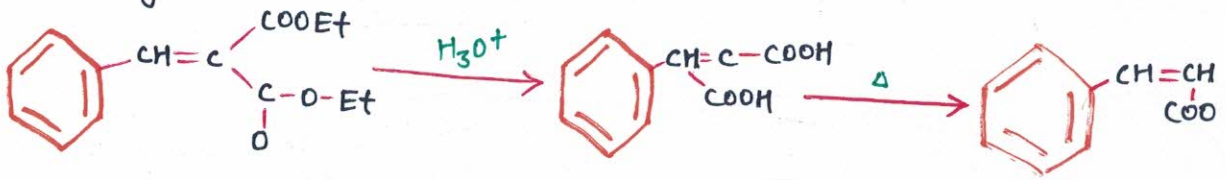
KNOWWONGAL REACTION



MECHANISM

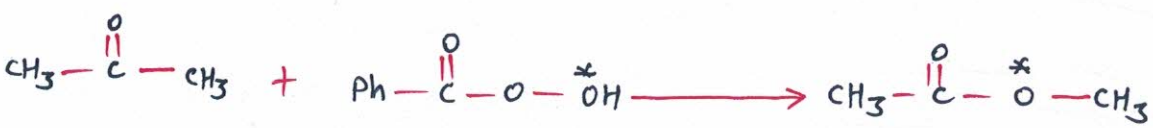


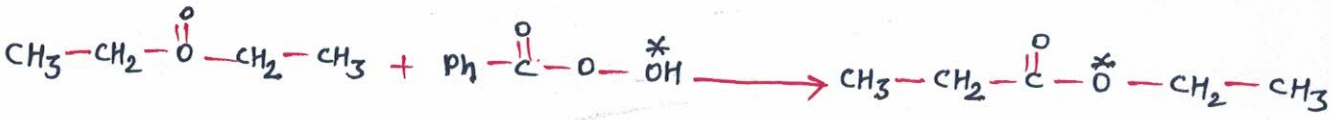
Reaction of reactive methylene group will an aromatic aldehyde is knowwongal Reaction.



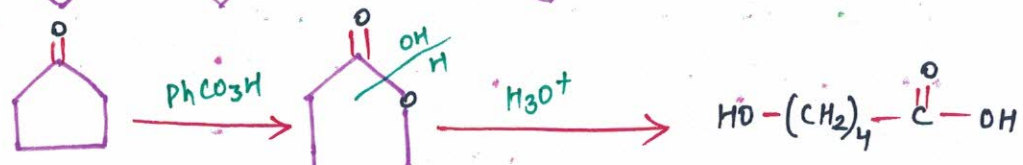
BAYER VILLIGOR REACTION (OXIDATION)

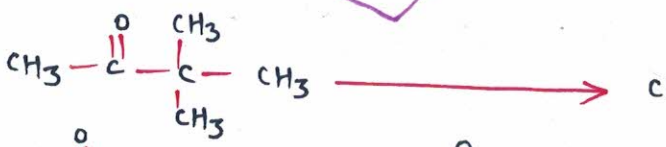


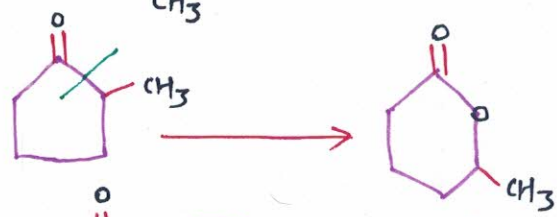






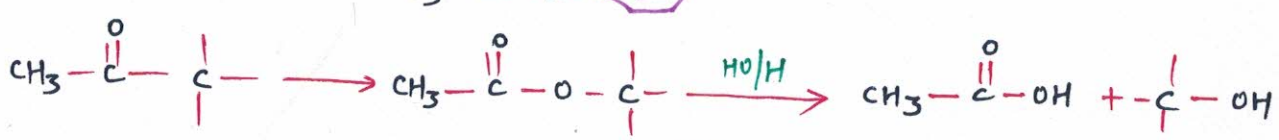






O in introduced near the more bulky group





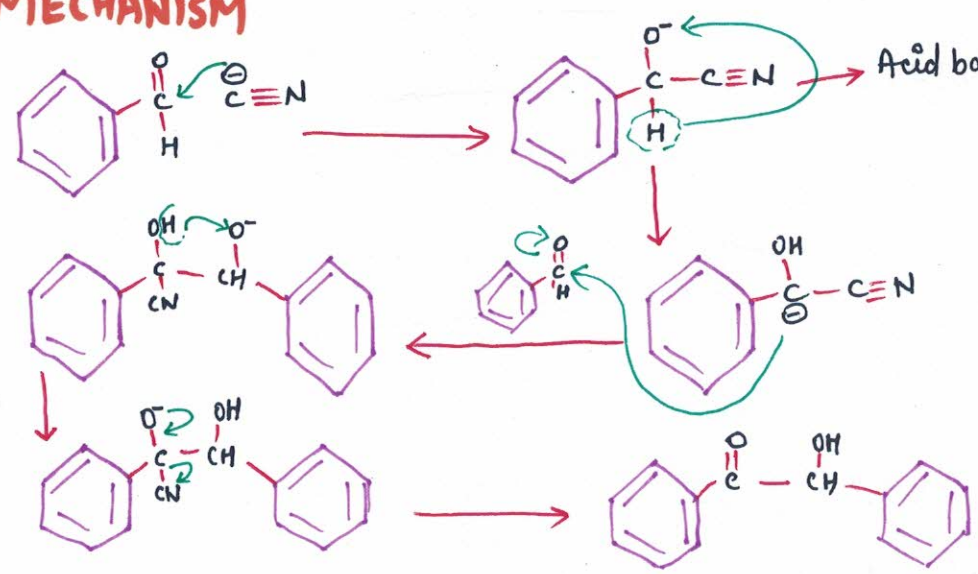
Benzoin Condensation



(Benzoin)

only aromatic aldehydes undergo it.

MECHANISM



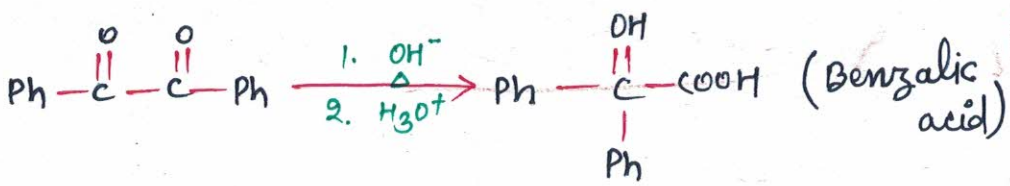
In aliphatic aldehydes, as KCN is basic salt and in $CH_3-\underset{CN}{\overset{O^-}{C}}-H$, H cannot migrate as it is not acidic, so instead of benzoic aldol condensation takes place.

$H-\overset{O}{C}-CH_3$ KCN $H-\overset{O}{C}-CH_2^-$



Benzil

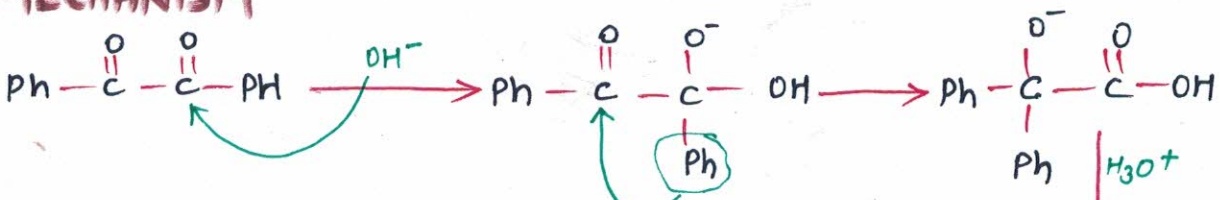
Benzil - Benzilic acid rearrangement

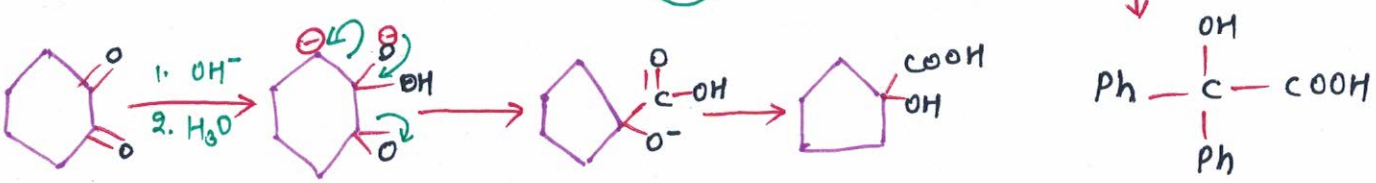


(Benzalic acid)

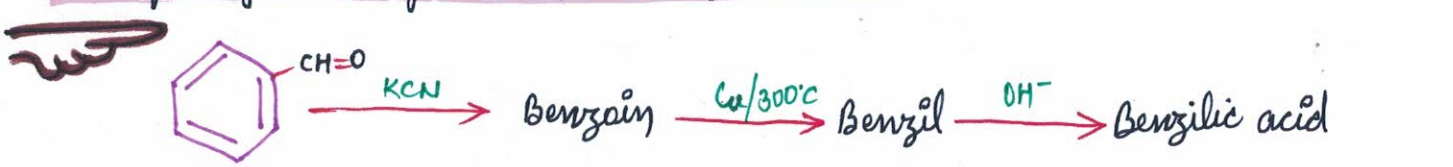
(2, Hydroxy, 2,2 diphenyl ethanoic acid)

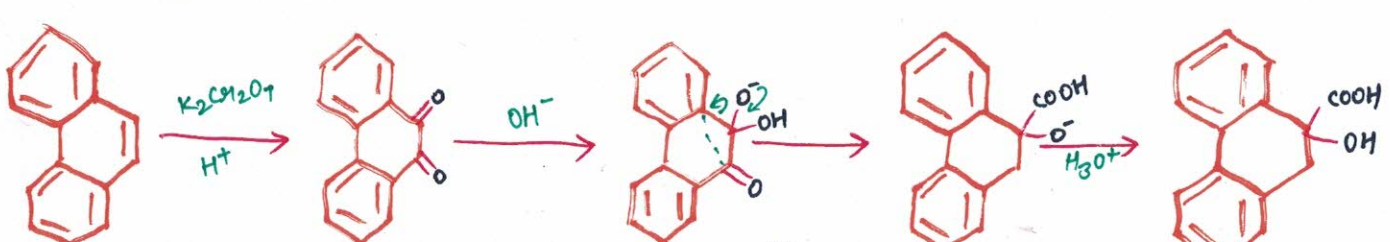
MECHANISM





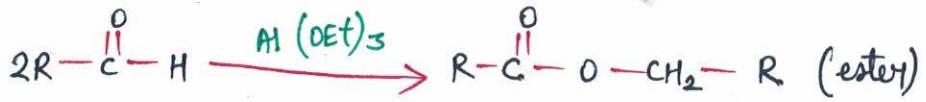
α hydroxy carboxylic acids are formed.





only α diketones undergo this reaction

Tischenko Reaction

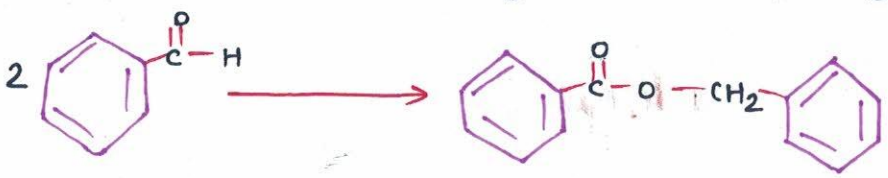


It is disproportion reaction.

As now $Al(OEt)_3$ is amphoteric, even if α hydrogen is present, aldol condensation will not take place.

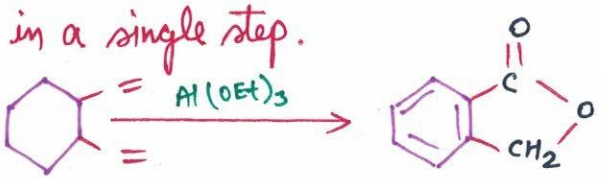
So, each & every aldehyde will give this reaction.

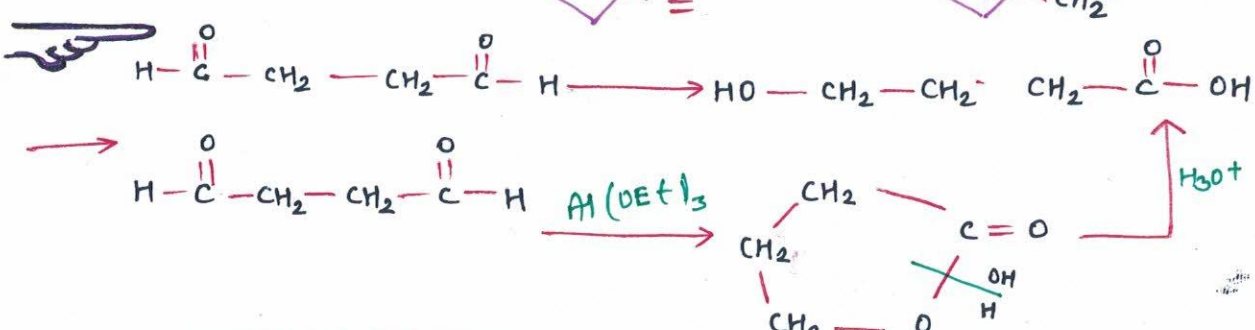


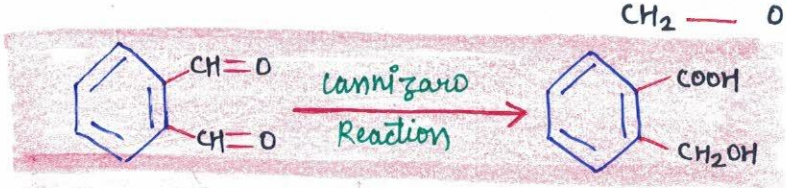


Q. Convert aldehyde into ester in a single step.

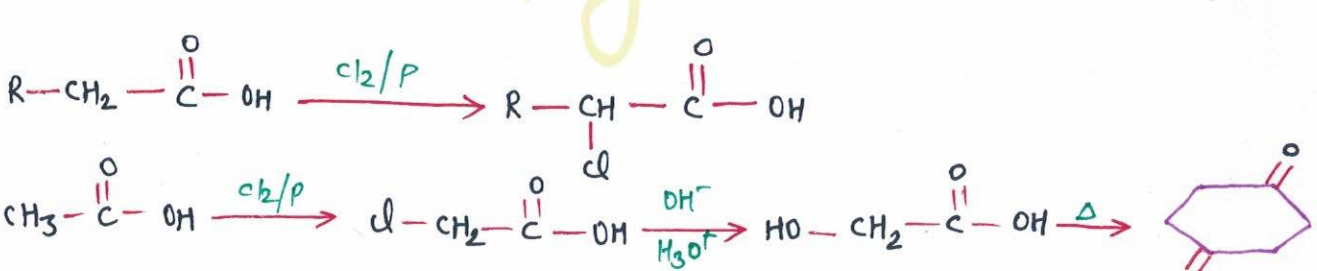
→ Tischenko Reaction

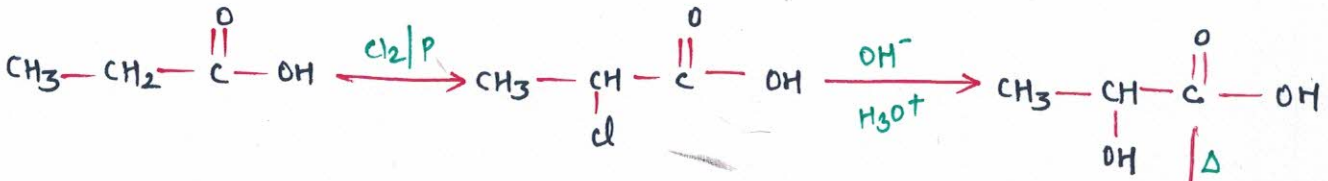






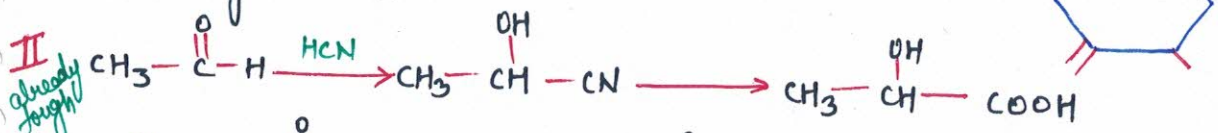
HELL VOLHARD Zilinisky Reaction (HVZ)

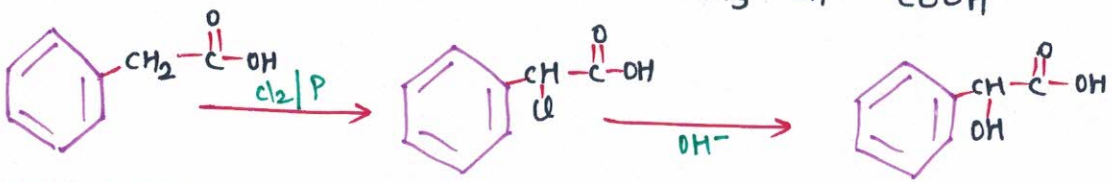




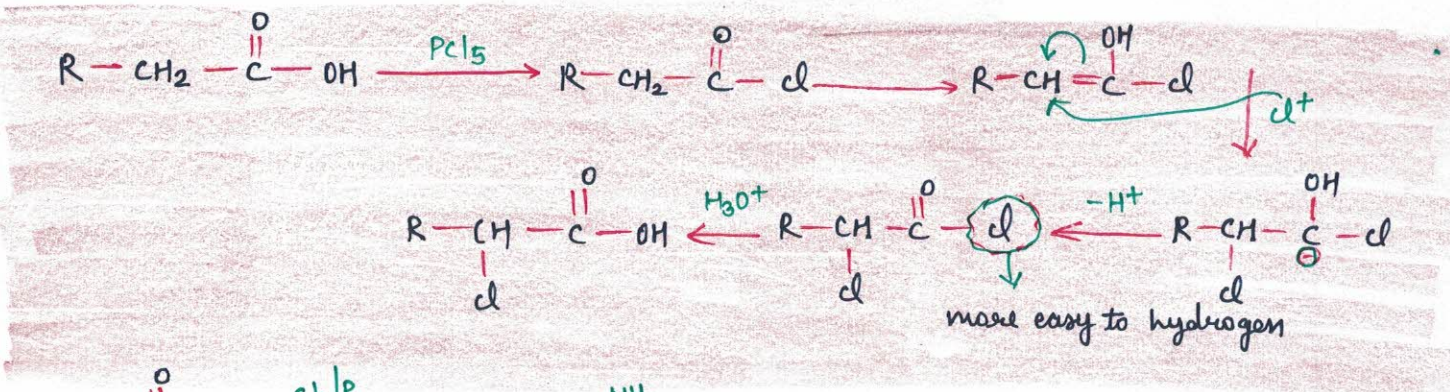
It is a method to prepare α hydroxy carboxylic acids.

II already taught





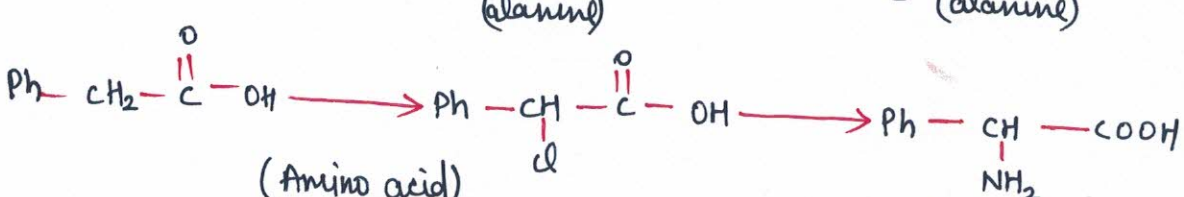
MECHANISM







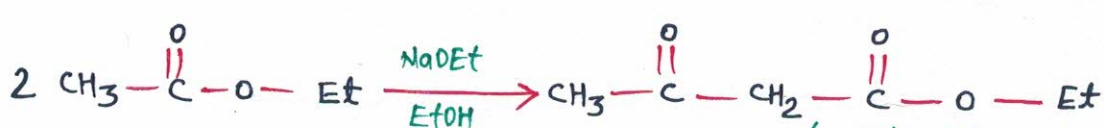
(Amino acid) (alanine) (alanine)



(Amino acid) (Phenyl glycine)

By this method we can prepare amino acids.

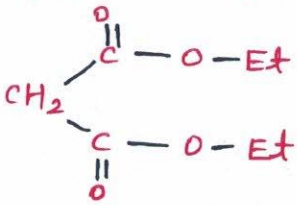
CLAISEN'S CONDENSATION

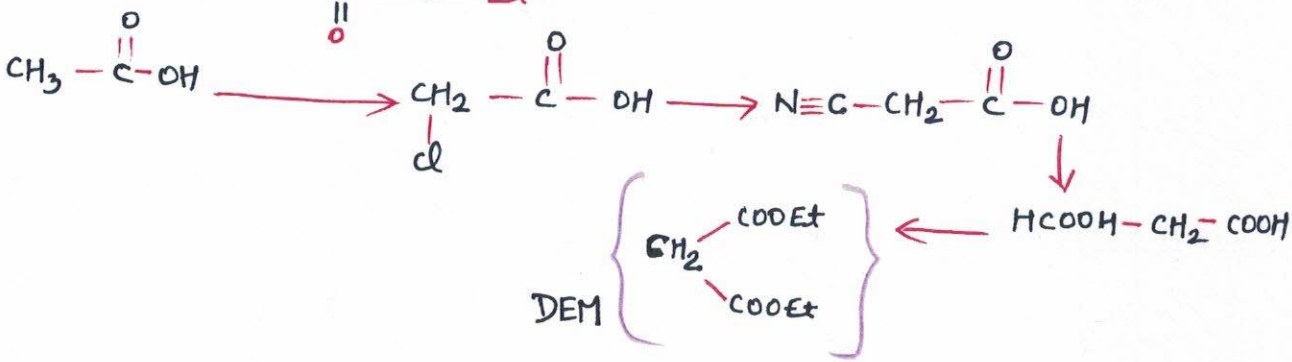


(EAA) → ?

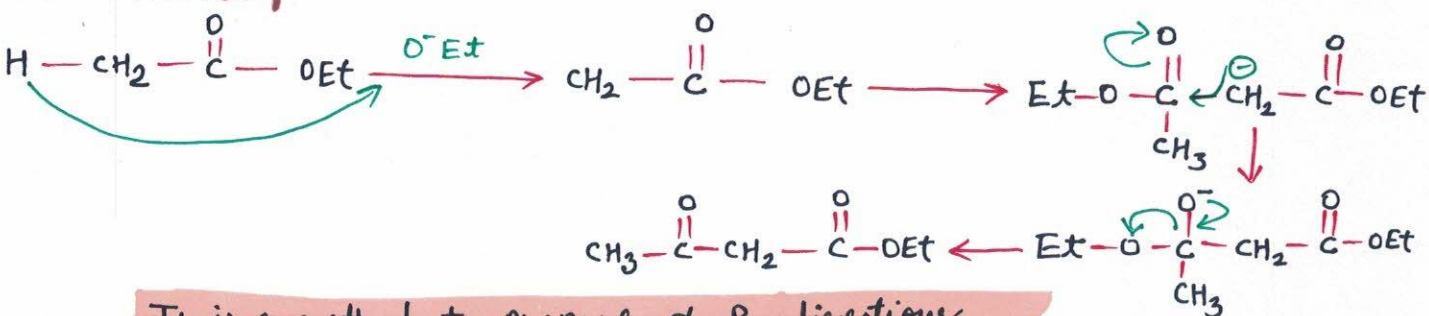
only esters having α hydrogen can undergo this reaction.

Convert CH_3COOH into DEM.

DEM = 

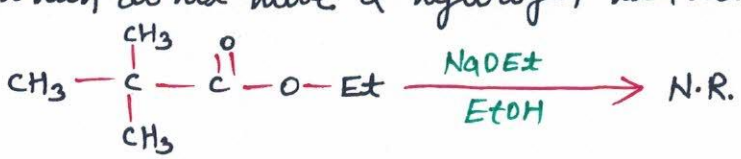


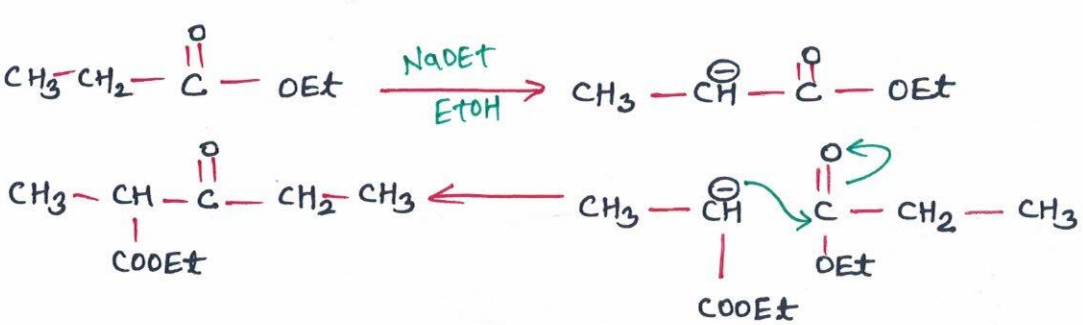
MECHANISM



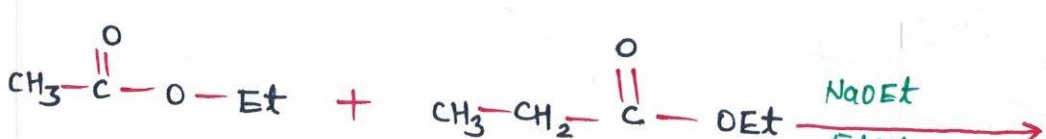
It is a method to prepare α, β directions.

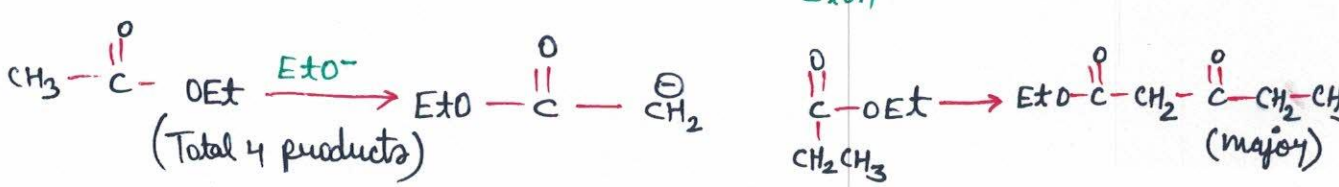
These esters which do not have α hydrogen will not undergo this reaction.

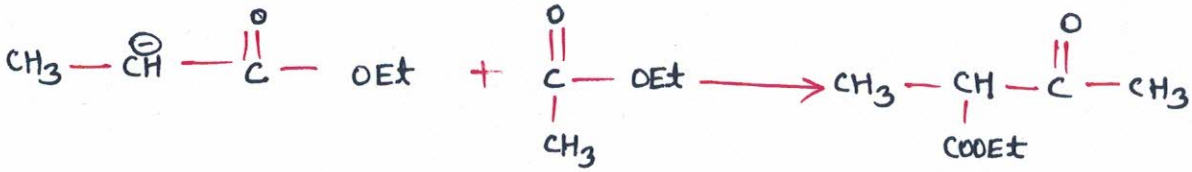


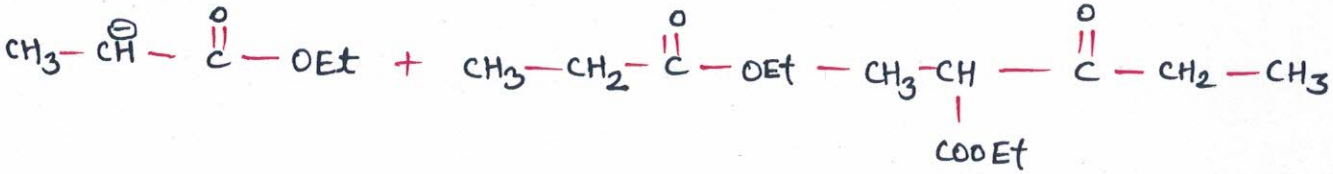


CROSSED CLAISEN'S CONDENSATION

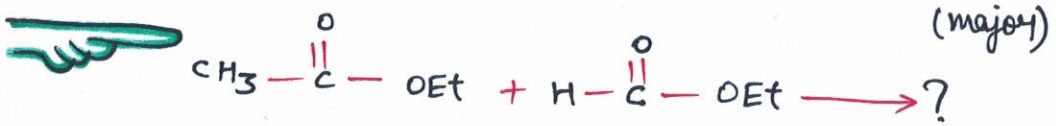


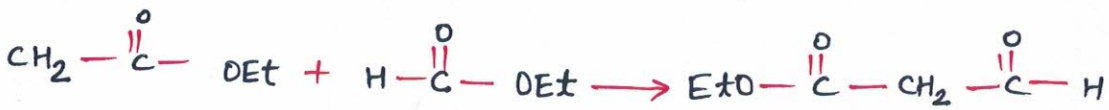


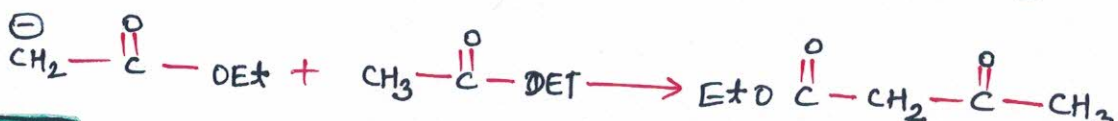


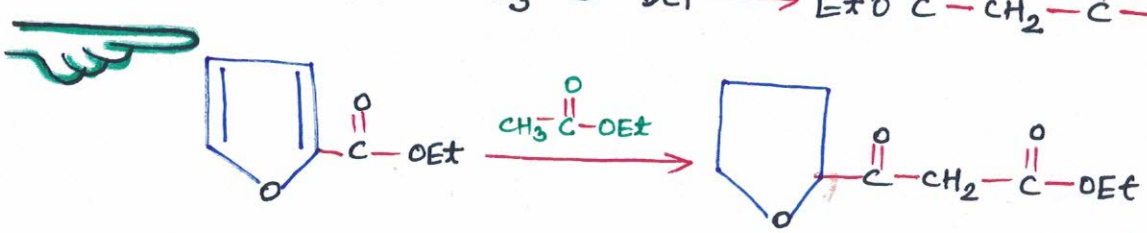




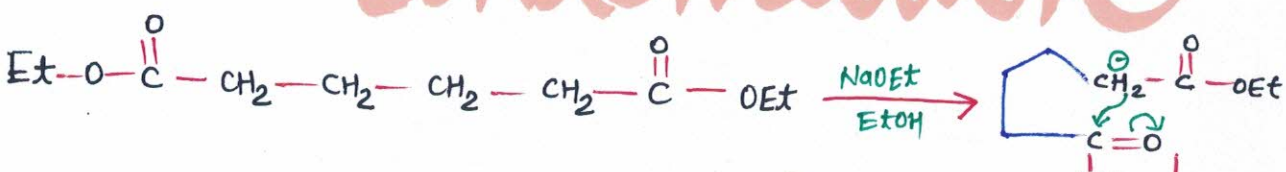


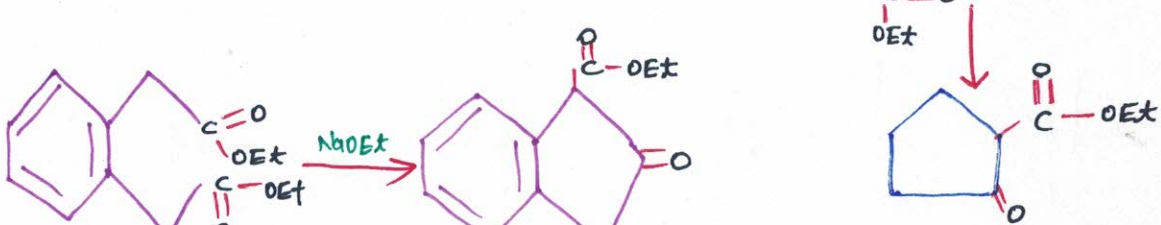




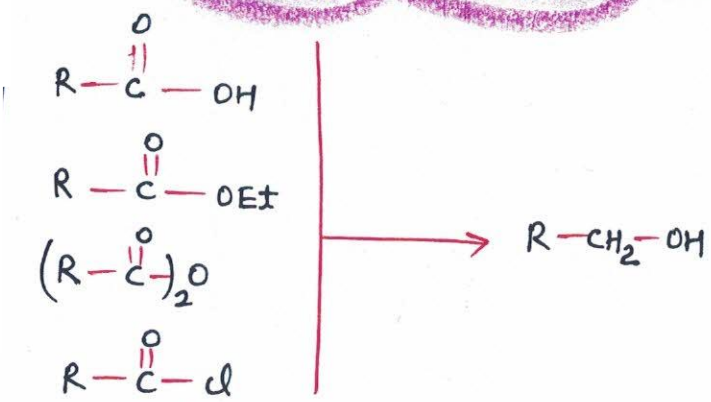


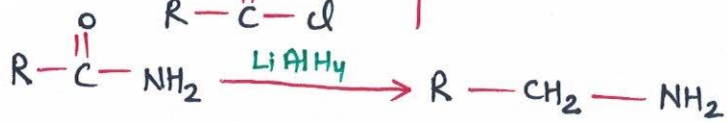
Intramolecular claisen's condensation

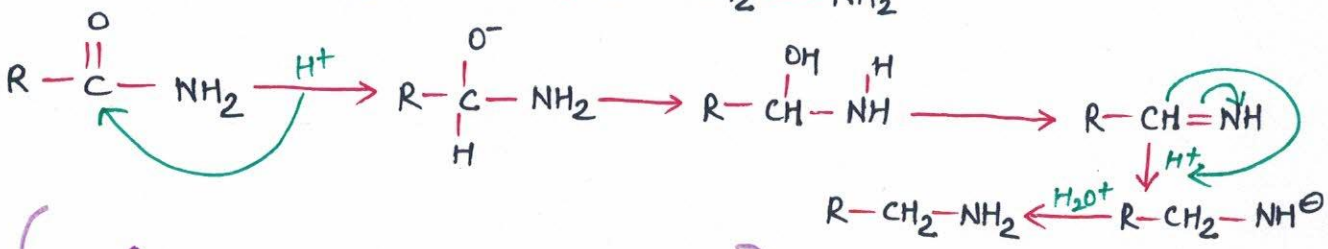


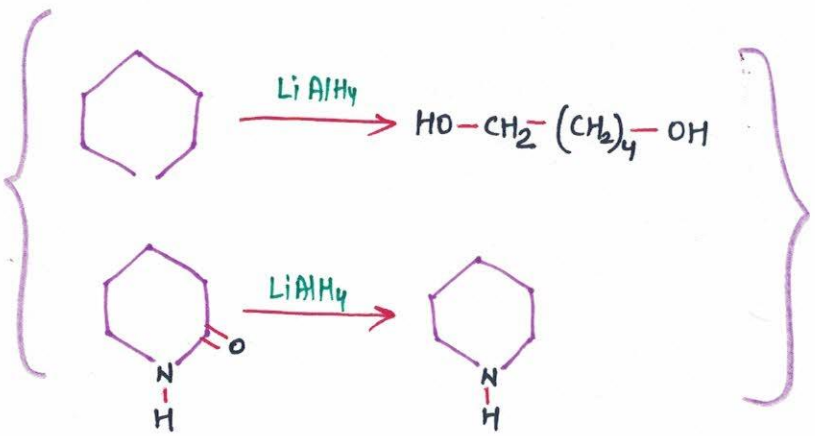


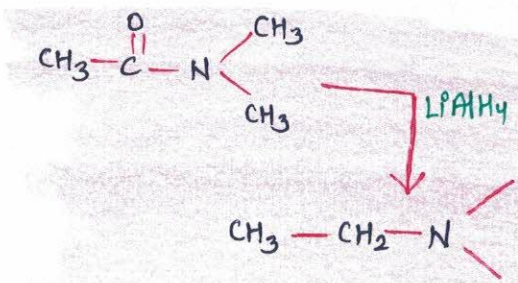
IMPORTANT POINT



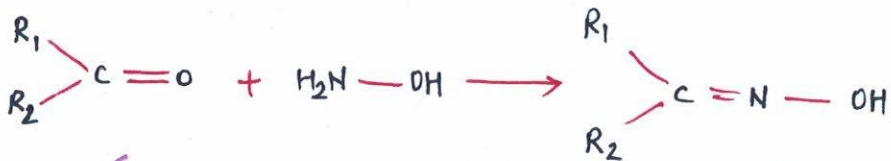
But 

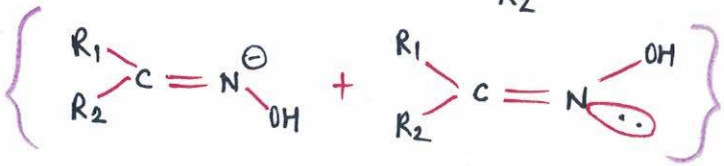


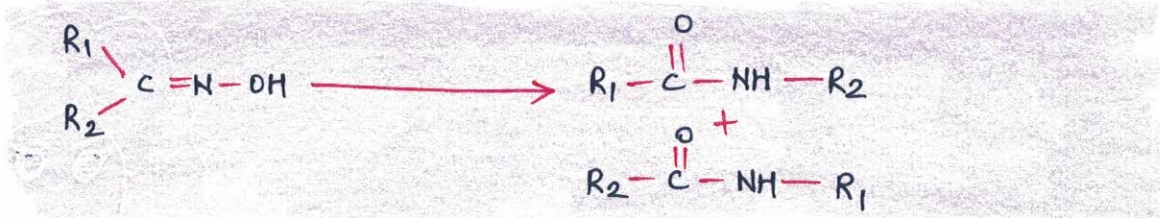




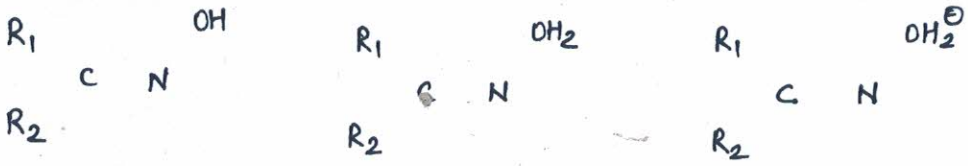
BACKMANN REARRANGEMENT

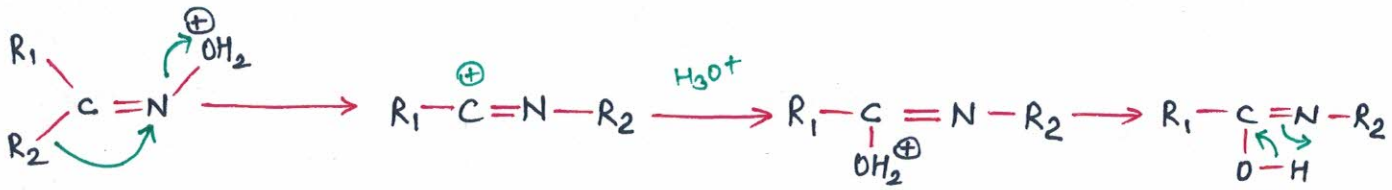




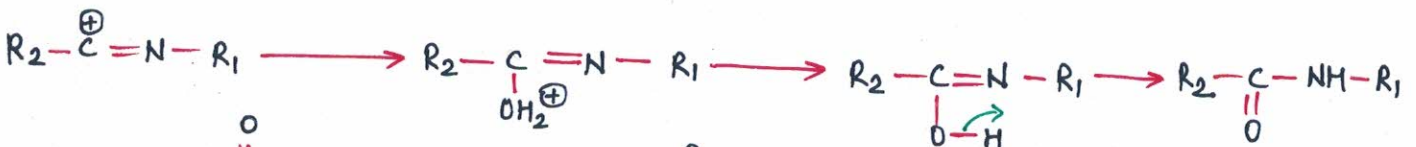


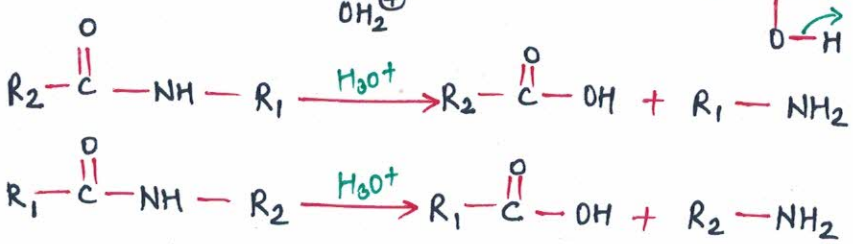
Always group trans to oxime will migrate.

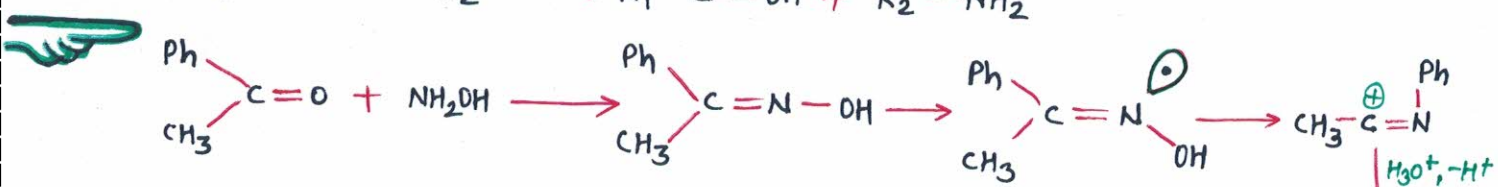


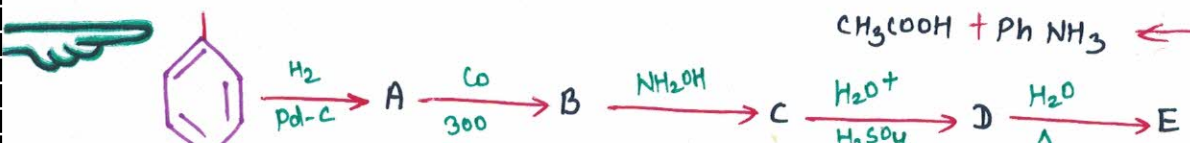


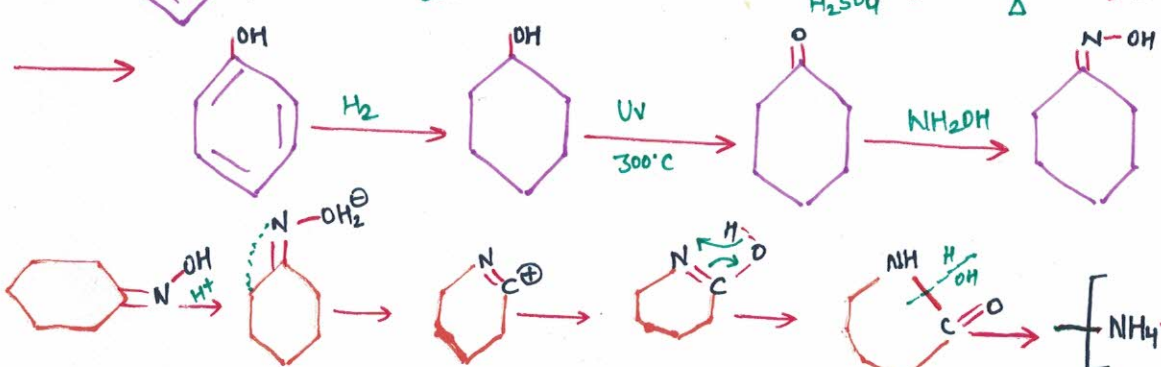




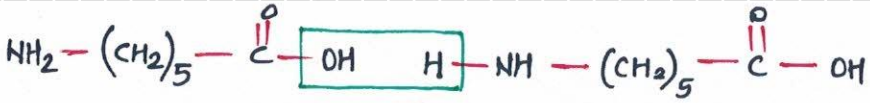








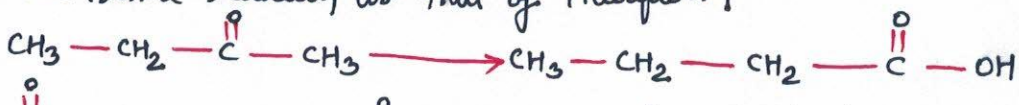
(Caprolactam) nylon 6 polyamide 6



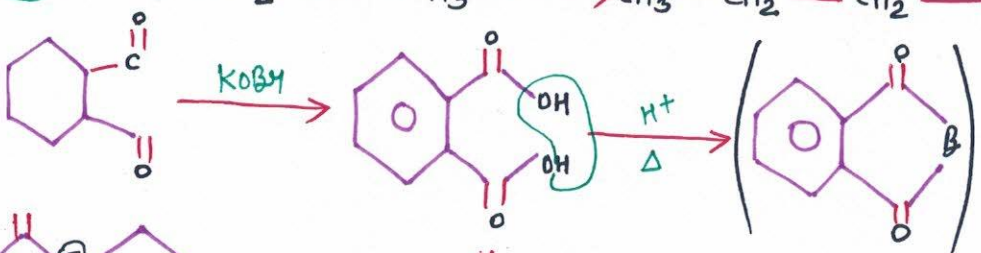
KOBr does the same reaction as that of Haloform

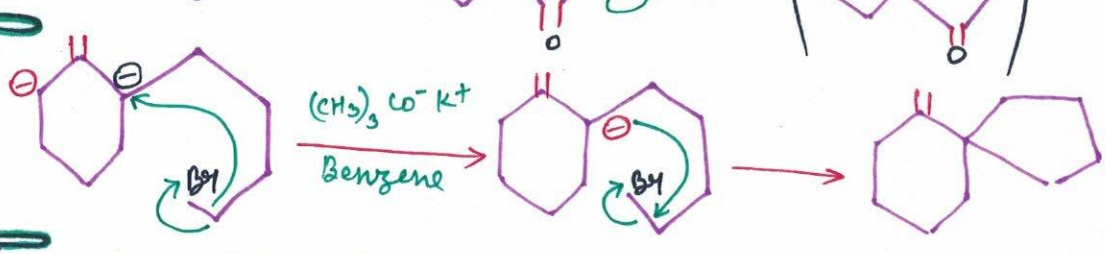
E.g.

1.



2.





Hydride shift is faster than methyl shift.